RS–MCA ENTROPY FRONTIERS

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Abstract. We study mutual correlated agreement (MCA) for sequences of Reed–Solomon codes: a slope is bad when its received-line word agrees with a low-degree polynomial on many coordinates although the two line generators are not both so explained on that same set. Unconditionally, we determine the high-agreement bad-slope count and translate exactly among shared leading coefficients of polynomials vanishing on the agreement supports, lists of nearby codewords, and intersections of a parity-check image line with column spans. Over polynomial-size evaluation fields in odd and characteristic two, exponentially large support families with few repeated differences disprove a uniform bound based only on the average fiber size of the leading-locator-coefficient map; the odd-characteristic example also transfers to circle evaluation codes. For supports formed from complete polynomial-map fibers and marked exceptional points, we prove an exact quotient/remainder normal form and quantify the enlargement caused by a smaller quotient field. Because maximum fiber size and same-fiber pair counts do not determine distinct slopes, our conditional upper theorem requires an exhaustive structural classification and explicit Fourier and line-incidence estimates, then bounds the bad-slope numerator by support counts divided by attained boundary-map values. The results give a target-aware threshold theorem and isolate what remains necessary for any unrestricted smooth- or circle-domain theorem, where "smooth" refers to complete power-map fibers on multiplicative cosets.

1. Introduction and main results

By a finite Reed–Solomon *row* we mean data $(\mathbb{F}, D, k)$, where $\mathbb{F}$ is a finite field, $D \subseteq \mathbb{F}$ is an *evaluation domain* of size n, and $C = \mathrm{RS}_{\mathbb{F}}(D, k) \subseteq \mathbb{F}^D$ consists of the evaluations on D of polynomials of degree less than k. Thus a support is simply a subset of the coordinate set D. A word $u \in \mathbb{F}^D$ is *explained on a support* $S \subseteq D$ if its restriction to S agrees with one such degree-less-than-k polynomial. When the evaluation points lie in a designated coefficient subfield, we write $D \subseteq \mathbb{B} \subseteq \mathbb{F}$; $\mathbb{B}$ controls locator coefficients and domain structure, while $\mathbb{F}$ controls codeword coefficients and slopes. Throughout, $h(x) = -x \log x - (1-x)\log(1-x)$ denotes binary entropy in natural units, and $H_2(x) = h(x)/\log 2$ denotes binary entropy in bits, with the usual convention $0 \log 0 = 0$.

A received pair $r = (r_0, r_1) \in (\mathbb{F}^D)^2$ determines the *received line* $\{r_0 + \gamma r_1 : \gamma \in \mathbb{F}\}$. The pair is simultaneously explained on S if each of the two received words has an explanation there, possibly by different polynomials. A finite slope γ is *support-wise MCA-bad at agreement a* if some support $S \subseteq D$, $|S| \geq a$, explains $r_0 + \gamma r_1$ but does not simultaneously explain the pair. The same support is used in both tests; this is the support-wise convention. A tuple (γ, S, h), where $\deg h < k$ and $h|_S = (r_0 + \gamma r_1)|_S$, is a *witness* to that bad slope when the pair is not simultaneously explained on S, or equivalently is *support-wise nontrivial* there. Finally, $B_C^{\mathrm{MCA}}(a)$ is the maximum, over all received pairs, of the number of distinct bad slopes. It counts slopes, not witnesses or supports.

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Here is the asymptotic formulation used throughout. If $\varnothing \neq \Gamma \subseteq \mathbb{F}$ is the *challenge set*, meaning the set of slopes allowed in the test, let $B^{\mathrm{MCA}}_{C,\Gamma}(a)$ denote the same maximum with $\gamma \in \Gamma$, and at the radius grid point $\delta = 1 - a/n$ put

$$e_{\mathrm{MCA},\Gamma}(C,\delta) = \frac{B^{\mathrm{MCA}}_{C,\Gamma}(a)}{|\Gamma|}. \tag{1.1}$$

For the standard full-field challenge we abbreviate

$$e_{\mathrm{MCA}}(C, 1 - a/n) = \frac{B^{\mathrm{MCA}}_{C}(a)}{|\mathbb{F}|}, \qquad \delta^*_C(\varepsilon) = \sup\{\delta < 1 - k/n : e_{\mathrm{MCA}}(C,\delta) \leq \varepsilon\},$$

with radii interpreted on the closed integer grid. Taking $\Gamma = \mathbb{F}$ in (1.1) gives exactly this customary formulation. By a *row sequence* we mean a family $C_n = \mathrm{RS}_{\mathbb{F}_n}(D_n, k_n)$ indexed so that $|D_n| = n$; the fields, domains, and dimensions may all vary with n. Every claimed upper bound is uniform over received pairs, whereas a lower construction need only exhibit the stated received pair. For such a sequence, challenge sets Γ_n, and targets $0 \leq \varepsilon_n \leq 1$, define

$$\begin{aligned} &B^*_n = \lfloor \varepsilon_n |\Gamma_n| \rfloor\,, \\ &a^*_n = \min\{a \in \{k_n + 1, \ldots, n\} : B^{\mathrm{MCA}}_{C_n,\Gamma_n}(a) \leq B^*_n\}, \qquad \delta^*_n = 1 - \frac{a^*_n}{n}, \end{aligned} \tag{1.2}$$

whenever the set defining a^*_n is nonempty. Thus the asymptotic RS–MCA problem is to determine δ^*_n, or equivalently the first *safe agreement*: agreement a is safe when its bad-slope numerator is at most B^*_n, and unsafe otherwise. Since the numerator is nonincreasing in a, the resulting safe/unsafe boundary is the *MCA frontier*. This formulation retains the target and challenge size; replacing them by a zero-exponent slogan loses real information when the target is small.

The upper argument organizes witnesses as follows. A *compiler* is a proved implication that converts one family of row certificates—for example support-fiber, Fourier, or incidence bounds—into the next distinct-slope or threshold bound. A *chart* is a coordinate presentation of part of the witness incidence, including any auxiliary quotient, planted, tangent, or rank parameters. A *cell* is a specified subincidence in such a chart; after the received line and auxiliary parameters are fixed, its *realized cell* is an actual finite witness subset. A *profile* records the cell type together with those realized parameter values; a boundary profile additionally records its *full support slice* and *boundary map*. The slice is the set of supports compatible with the fixed profile parameters before first-match deletion; the map records the coefficient or syndrome data whose equal fibers are counted. We use "full support slice" and "full profile slice" synonymously.

Here a quotient parameter records factorization through a finite folding map, meaning a map with complete uniform fibers; a planted parameter records a forced algebraic factor or support block, and tangent or rank parameters record loci where a projection loses its expected rank. An ordered *first-match atlas* is witness-exhaustive if its realized cells cover every witness; a slope occurring in several cells is assigned to the first cell whose actual slope projection contains it. A cell is *paid* only when a proved bound controls that distinct-slope projection at the cell's natural scale—for a boundary profile, its full support count divided by the size of its realized boundary image, and otherwise an explicitly stated direct scale.

A *ledger* is the atlas together with all such payments, the treatment of its primitive residual profiles, and the final *add-back*, meaning the summation of all first-match profile budgets into one row-level bound. It is *closed* when each required payment and ray bound is a theorem for the row. The *profile envelope* is the row-level sum of the natural scales of all realized profiles, maximized over received lines; it is the budget to which a closed ledger compiles.

Here *quotient descent* means factorization through a folding map, whereas *field descent* means that the data are defined over a proper subfield. *Saturation* identifies witnesses producing the same projective explanation direction, and the setwise stabilizer is taken in the folding-symmetry group acting on the evaluation domain.

A residual profile is called *primitive* only after the atlas has certified the removal of its named quotient, field-descent, planted, rank, and ray-saturation degeneracies; primitivity is a ledger certificate, not merely trivial setwise stabilizer or a topological openness condition. A *leaf* is one fixed realized profile remaining at the end of this first-match process.

The domain families in the title carry additional folding structure. A *smooth multiplicative domain* is a coset $D = \theta H$ of a subgroup $H \leq \mathbb{B}^\times$, equipped with those power maps $x \mapsto x^c$ for which every attained value has exactly c preimages in D.

Assume now that q is an odd prime power. A *norm-one torus* over $\mathbb{F}_q$ has rational points $\{u \in \mathbb{F}_{q^2}^\times : u^{q+1} = 1\}$. A *Chebyshev twin-coset domain* is obtained from a disjoint inverse pair $gH \sqcup g^{-1}H$ in this group by the projection $\chi(u) = (u + u^{-1})/2$; the polynomial T_c satisfying $T_c(\chi(u)) = \chi(u^c)$ supplies its complete-fiber folding maps. We call this projected set a *circle x-domain*.

A *standard circle code* instead evaluates bounded-total-degree bivariate polynomials on points of $x^2 + y^2 = 1$. The two presentations are *diagonally equivalent* only after a coordinate bijection and coordinatewise multiplication by a fixed nonzero vector have been proved; that equivalence preserves all agreement sets. The precise construction, including the exceptional points at which u and u^{-1} coincide, is Definitions 7.2 and 7.3.

Proximity Prize and recent work. The Ethereum Foundation's preliminary Proximity Prize asks for the largest radius at which smooth-domain Reed–Solomon codes of rates $1/2, 1/4, 1/8, 1/16$ have MCA error at most a cryptographic target such as 2^{-128} [PP26, ABF26]. The foundational Reed–Solomon proximity-gap theorem reaches the Johnson regime, namely the classical list-decoding radius $1 - \sqrt{\rho}$ at code rate ρ [BCIKS20]. Recent work sharpens the positive theory in that range and proves substantial lower obstructions beyond it [BCHKS25, CS25, Hab25]; general reductions from list decoding and polynomial-generator MCA supply further routes to proximity and correlated agreement [GCXK25, BCGM25]. Near capacity, Goyal and Guruswami prove MCA and proximity gaps for folded Reed–Solomon, multiplicity, and subspace-design codes and transfer them to random linear and random-evaluation Reed–Solomon ensembles [GG25]. The random-ensemble parameters were subsequently improved through row-span-constrained local properties [GGSW26], while a direct syndrome-space approach gives near-capacity results for random linear codes [YZ26]. Explicit constant-alphabet Alon–Edmonds–Luby (AEL) and subspace-design constructions inherit broad random-linear-code local properties [GGH26]; these are concatenated or additive code families, not asymptotic ordinary Reed–Solomon codes over $\mathbb{F}_2$. Jo proves that affine-line MCA is exactly invariant under row-wise interleaving; for a general generator with s coefficients over a field of size q, the loss is at most $1 + q^{-1} + \cdots + q^{-(s-1)}$ [Jo26]. The Reed–Solomon proximity testing protocol WHIR is an important application and terminology source for MCA [ACFY25].

These results do not automatically cover ordinary Reed–Solomon codes on a fixed multiplicative coset or a fixed circle domain: random evaluation points, folding, multiplicity, and subspace-design hypotheses change the code family. Conversely, multiplicative-subgroup domains over prime fields already admit near-capacity counterexamples [KKH26]. We therefore do not claim the first MCA formalism, the first MCA counterexample, or an unrestricted capacity theorem.

Our contribution is complementary. We first give exact finite-row compilers through syndromes and auxiliary simple poles: the former replaces a word by its coset modulo the code, while the latter divides by $X-\alpha$, $\alpha \notin D$, to place a list on one received line. We then construct a target-aware profile envelope and an exact quotient–remainder normal form separating whole folding fibers from the *support remainder*, the exceptional selected points outside those complete fibers. Odd- and characteristic-two field-drop examples show how quotient coefficients in a proper subfield enlarge every *live prefix slot*, meaning a prefix coordinate not forced by lacunarity (prescribed gaps among the highest locator coefficients), planted data, or previously fixed quotient data and therefore still ranging over its recorded coefficient field. Finally, the first-match formalism isolates the residual Sidon and distinct-ray hypotheses required by any valid upper theorem. The finite ledgers and reductions used by this formulation are developed in [Cho26a, Cho26b].

For an a-point support $S \subseteq D$, its monic *locator* is $Q_S(X) = \prod_{t\in S}(X-t) = X^a + c_1(S)X^{a-1} + \cdots + c_a(S)$. The *depth-$w$ locator prefix* is $\Phi_w(S) = (c_1(S), \ldots, c_w(S))$, and a *prefix fiber* is a set $\Phi_w^{-1}(z)$ of supports with the same prescribed leading coefficients. The natural near-capacity lower construction takes $w = a-k-1$. Its *identity profile* is the unprofiled family of all a-subsets of D, with no quotient, planted, or field-descent restriction. Averaging its $\binom{n}{a}$ supports over the ambient prefix space $\mathbb{B}^w$ gives the ambient *identity scale*

$$\overline{N}_1(a) = \binom{n}{a}|\mathbb{B}|^{-w}, \qquad D \subseteq \mathbb{B} \subseteq \mathbb{F}. \tag{1.3}$$

Here $\mathbb{B}$ is the coefficient field containing the domain; the code and challenge field $\mathbb{F}$ may be a *scalar extension*, meaning that the same evaluation points are retained while polynomial coefficients and slopes are allowed in a larger field $\mathbb{F} \supseteq \mathbb{B}$. It is tempting to conjecture that $\overline{N}_1$, up to $e^{o(n)}$, also bounds every received line. The *identity entropy crossing* is the agreement at which $\log \overline{N}_1(a)$ meets the logarithm of the target budget; for a subexponential target this is the zero of the identity exponent $n^{-1}\log \overline{N}_1(a)$. The countertheorem shows that the identity scale is not a universal bound without a quotient envelope. A large-depth identity prefix can contain a bounded-degree *quotient profile*, meaning supports assembled from complete fibers of a folding map, whose live coefficients are defined over a much smaller subfield. At the identity entropy crossing that profile is exponentially larger than $\overline{N}_1$. The same construction survives on the circle after the coordinate bijection and nonzero diagonal rescaling of Proposition 8.15.

The second point is analytic. We identify supports with their incidence vectors in the torsion-free group $\mathbb{Z}^D$. For a support family F, its *additive energy* is $E(F) = |\{(x_1,x_2,x_3,x_4) \in F^4 : x_1 - x_2 = x_3 - x_4\}|$, normalized as $\Delta(F) = E(F)/|F|^3$. A large prefix fiber need not have high energy. We call a positive-rate contribution from fibers with exponentially small Δ the *Sidon-heavy branch*; the name records scarcity of repeated differences, not an assertion that the fiber is a literal Sidon set.

More precisely, for a support map $\Phi : \Omega^0 \to \Sigma$, write $F_s = \Phi^{-1}(s)$, $f_s = |F_s|$, $L = |\Phi(\Omega^0)|$, and $\overline{N} = |\Omega^0|/L$. Its normalized order-$q$ moment is $L^{-1}\sum_s (f_s/\overline{N})^q$; the Sidon-heavy part restricts this sum to $\Delta(F_s) \le e^{-\sigma N}$, where N is the number of active coordinates and $\sigma > 0$ is fixed. A *logarithmic moment order* is an order $q = q_N \to \infty$ growing at most polylogarithmically in N. We split that moment mass into the low-energy branch and a high-energy branch. Here *positive rate* means growth $e^{cN-o(N)}$ for a fiber, or $e^{cNq-o(Nq)}$ for an order-q moment, with some fixed $c > 0$.

If the Sidon branch is paid by an explicit Fourier estimate, the Balog–Szemeredi–Gowers (BSG) theorem and the Boolean-cube difference-growth theorem, namely $|U-U| \ge |U|^{3/2}$ for $U \subseteq \{0,1\}^d$, rule out a positive-rate high-energy fiber. This is a genuine theorem, but it does not pay the Sidon branch at the deployed *deep prefix*—the large locator-prefix depth used at the entropy crossing, unrelated to the high-agreement syndrome range called the deep regime below—merely by naming that branch a cell.

The Li–Wan distinct- coordinate sieve combined with Weil square-root cancellation supplies the payment only under its displayed pointwise inequality for *minor characters*. The *rational phase* is the rational function inside the corresponding additive-character sum; it is *nondegenerate* here when it is nonconstant, is not Artin–Schreier, and has not already been routed through a quotient, field-descent, or rank-loss profile. We denote that numerical condition by (PF). The remaining exceptional *major characters*, whose phases are constant, Artin–Schreier degenerate (a Frobenius coboundary $G^p - G$ plus a constant for the chosen additive character, where p is the coefficient-field characteristic), or factor through an earlier algebraic profile, require the separate aggregate estimate (MA).

The third point concerns the MCA numerator. The *Q statistic* is the largest support fiber of a profile boundary map, and *SP* is its second moment, equivalently the number of ordered support pairs with the same boundary value. These quantities count supports or support pairs, whereas MCA counts distinct *slope rays*: the actual projective explanation directions after repeated witness representatives have been identified. A *ray compiler* is a theorem converting the support pair count into a distinct-slope bound, either directly or through a proved lower multiplicity for every slope. The one-parameter moving-root theorem supplies such a count after a chart has genuinely been reduced to one projective locator pencil $Q_0 + \lambda Q_1$. It does not exhaust the high-dimensional *balanced-core* kernels, meaning residual locator families with a factored common core (a shared locator factor), equal-degree monic moving locators with a common prefix, and more than one live projective parameter, occurring at the frontier. The exact direct-bound-or-multiplicity hypothesis used by the compiler is denoted (RC).

For the finite theorem below, a *syndrome* is the parity-check image of a received word. A syndrome line meets the span V_E of the parity-check columns indexed by a possible error set E. The meeting is *transverse* when the whole syndrome line is not contained in V_E, so that this support supplies at most one slope. Thus a "transverse syndrome-line incidence" is a distinct line parameter γ for which that meeting occurs.

Theorem 1.1 (Unconditional finite-row theorem package)**.** *The following assertions hold for every finite field row in their displayed ranges.*

(i) *In the* deep range $3(n-a) \le n-k$, *where three permitted error supports are still too small to support a nonzero RS codeword, the full-field MCA numerator is exactly $n-a+1$; for a challenge subset $\Gamma \subseteq \mathbb{F}$ it is* $\min\{|\Gamma|, n-a+1\}$.

(ii) *The bad-slope numerator is exactly the transverse syndrome-line incidence* (3.4)*; in particular one fixed* noncommon support*—a support on which the two received words are not both explained—carries at most one finite bad slope.*

(iii) *Writing $L = \lceil \binom{n}{a} |\mathbb{B}|^{-(a-k-1)} \rceil$, locator prefixes give an exact list of size at least L in the dimension-$k+1$ code; when $|\mathbb{F}| > n$, the collision-aware pole transform gives at least*

$$\left\lceil \frac{L(|\mathbb{F}| - n)}{|\mathbb{F}| - n + k(L-1)} \right\rceil$$

distinct MCA-bad slopes in the dimension-k code.

(iv) *Every depth-w prefix fiber of N many m-sets, with $0 \le w \le m-2$, embeds after a scalar extension satisfying $|\mathbb{F}| - n > (m-w-1)\binom{N}{2}$ into one received line with at least N distinct MCA-bad slopes.*

(v) *For every complete degree-c folding map, a depth-w complete-fiber-plus-remainder prefix with remainder degree $r < c$ has the exact finite-row normal form of Theorem 8.3. For fixed c, r, the natural quotient-profile scale at the identity crossing has field-sensitive exponent* (QR9).

(vi) *Q and SP support statistics do not determine the syndrome-line incidence. The precise multiplicity condition under which a pair count compiles to distinct rays is* (12.3).

The assertions are proved in Sections 3, 4, 8 and 18 and Theorem 4.5.

Theorem 1.2 (Countertheorem to unrestricted identity-only bounds)**.** *Fix $0 < \alpha < 1/2$.*

(a) *There are odd-characteristic smooth multiplicative rows with $\log|\mathbb{B}_n| = O(\log n)$, $a_n/n \to \alpha$, and $\overline{N}_1(a_n) = e^{o(n)}$, but a square-quotient prefix fiber—that is, a profile of complete fibers of $x \mapsto x^2$—has size $\exp((h(\alpha)/4 + o(1))n)$, exponentially small normalized additive energy, and, after scalar extension, that many distinct MCA-bad slopes. The same obstruction transfers to standard bivariate circle codes.*

(b) *There are characteristic-two smooth multiplicative rows with the same ambient identity-scale behavior but a cubic-quotient fiber of complete $x \mapsto x^3$ fibers of size $\exp((h(\alpha)/6+o(1))n)$, again giving exponentially many distinct MCA-bad slopes after scalar extension.*

Consequently smoothness and $\log|\mathbb{B}_n| = o(n)$ do not imply a uniform ambient identity-normalized Sidon payment or an identity-scale MCA numerator bound. They also invalidate any unrestricted identity-frontier formula derived sharply from those assertions; threshold counterexamples result for target budgets below the constructed distinct-slope count.

Proof. Part (a) is Theorem 8.9 and Corollary 8.16; part (b) is Theorem 8.11. The MCA statements use the exact prefix-to-line compiler of Theorem 4.5. □

Remark 1.3 (Threshold and field-size scope)*.* Here $w_n/n = \Theta(1/\log n) = o(1)$. The countertheorem therefore refutes ambient identity-normalized Sidon payment, identity-only numerator bounds, and *row-sharp* threshold derivations (derivations retaining the row's actual sublinear displacement rather than absorbing it into an unspecified $o(1)$ radius error), but it does not by itself refute a coarse radius identity with an $o(1)$ error. The domain field $\mathbb{B}_n$ is polynomial-size, whereas the scalar extension used to separate all exponentially many slopes has $\log|\mathbb{F}_n| = \Theta(n)$. A claim for a prescribed Proximity-Prize target must still use the exact target-aware bracket and verify its field and reserve hypotheses.

For the upper theorem, a realized first-match profile λ has a *full support slice* Ω^0_λ, namely all supports compatible with its fixed profile parameters before first-match deletion, a coefficient field $\mathbb{B}_\lambda$, and R_λ effective prefix equations. It comes with a proved *boundary map* $\Phi_\lambda : \Omega^0_\lambda \to \mathbb{B}_\lambda^{R_\lambda}$. For a received line (r_0, r_1) and agreement a, let $\Lambda(r_0, r_1; a)$ be the finite set of its realized profiles whose first-match cells are nonempty at that agreement. Put

$$A_\lambda = |\mathbb{B}_\lambda|^{R_\lambda}, \qquad L_\lambda = |\Phi_\lambda(\Omega^0_\lambda)|, \qquad \overline{N}_\lambda = \frac{|\Omega^0_\lambda|}{L_\lambda},$$

and define the image-normalized profile envelope

$$\mathfrak{E}_n(a) = 1 + (n - a + 1) + \sup_{(r_0,r_1)} \sum_{\lambda\in\Lambda(r_0,r_1;a)} (1 + \overline{N}_\lambda). \tag{1.4}$$

The formal codomain has A_λ points, the realized image has L_λ points, and $\overline{N}_\lambda$ is therefore the average full-slice fiber size at the realized image scale. For each received line the sum includes the identity profile and all quotient, Chebyshev, planted, and remainder profiles that meet that line; the supremum makes the envelope a row-level quantity. With subexponentially many profiles, the sum and maximum have the same exponential scale. The ambient scale $\overline{N}_\lambda^{\mathrm{amb}} = |\Omega_\lambda^0|/A_\lambda$ may replace $\overline{N}_\lambda$ only after a full-image certificate

$$L_\lambda \geq e^{-o(n)} A_\lambda \tag{FI}$$

has been proved uniformly. Otherwise effective-image collapse is *routed*—assigned by the first-match rule to an earlier profile so that its slopes and all witnesses above them are removed from the later residual—as an effective-image rank-collapse profile, and the image scale is retained.

Here *effective-image collapse* means that the realized image is exponentially smaller than the formal codomain. This is distinct from a ray-saturation profile, which records many raw witness representatives producing the same explanation or slope ray.

The countertheorem is exactly a row for which a quotient profile in $\mathfrak{E}_n$ is exponentially larger than the ambient identity term. A *shift-pair chart* records two residual supports with the same boundary value; a balanced-core chart records the higher-dimensional locator kernel that may remain after their common core is factored.

Definition 1.4 (Ledger-admissible smooth/circle sequence). *A sequence (C_n, a_n), with $C_n = \mathrm{RS}_{\mathbb{F}_n}(D_n, k_n)$, is ledger-admissible if the following conditions hold uniformly in every received line.*

(A1) *The domains are among the structured classes of Definitions 7.1 and 7.3, or are circle codes diagonally equivalent to such RS rows, and the folding maps have complete uniform fibers.*

(A2) *A first-match atlas covers every bad-slope witness and has $e^{o(n)}$ profiles. The total* distinct-slope *contribution of its algebraic cells is at most $e^{o(n)}\mathfrak{E}_n(a_n)$.*

(A3) *Every primitive profile uses the image normalization above. Its leaf has an* active coordinate set *of size N, namely the evaluation positions still free to enter or leave the support after the profile parameters have been fixed, and has fixed support size m_N, and density $m_N/N \in [\alpha, 1-\alpha]$ for one row-uniform constant $\alpha > 0$. Writing L_N for the size of its realized boundary image, there is a logarithmic order $q_N \to \infty$, $q_N \leq (\log N)^C$, satisfying $\log L_N/q_N = o(N)$, unless it carries a separately proved primitive-Q bound. Any use of the ambient scale is accompanied by* (FI).

(A4) *Every primitive prefix leaf satisfies both the pointwise minor-arc inequality* (PF) *of Definition 7.12 and the major-arc aggregate* (MA) *of Definition 7.11, or has a separately proved image-normalized Sidon/Fourier moment payment of the strength in Definition 12.3.*

(A5) *Primitive columns are genuine weighted Vandermonde columns, meaning columns of the form $\rho(t)(1, t, \ldots, t^{R_N-1})^\mathsf{T}$ at distinct t's with $\rho(t) \neq 0$. Every use of power-sum coordinates satisfies $R_N < \operatorname{char} \mathbb{B}_N$; small-characteristic leaves instead retain elementary coordinates or carry a direct Sidon/Q theorem.*

(A6) *Every residual shift-pair and balanced-core chart satisfies the ray compiler* (RC) *of Hypothesis 18.10, or has a direct distinct-slope bound at most its contribution to $e^{o(n)}\mathfrak{E}_n(a_n)$.*

(A7) *The profile envelope in (1.4), including all quotient subfields $\mathbb{B}_\phi$ with $\phi(D) \subseteq \eta\mathbb{B}_\phi$ for some scalar η (the* scaled quotient coefficient fields*) and all profile fields of definition, is used in the final budget; it is not replaced by the identity term unless that comparison is proved for the row.*

Theorem 1.5 (Conditional profile-envelope compiler). *Every ledger-admissible smooth/circle sequence satisfies*

$$B_{C_n}^{\mathrm{MCA}}(a_n) \leq e^{o(n)}\mathfrak{E}_n(a_n). \tag{1.5}$$

Proof. Condition (A2) supplies a witness-exhaustive first-match atlas and pays its algebraic slope projections. On every remaining primitive profile, conditions (A3)–(A5) and Theorem 17.2 give image-normalized Q. Condition (A6) and Proposition 18.11 convert those bounds to distinct slopes, and (A7) places every profile at its term in $\mathfrak{E}_n(a_n)$. Summing the $e^{o(n)}$ first-match payments and applying Lemma 2.10 proves (1.5). ☐

For the lower side, a *certified profile-list construction* is an explicit list inside one identity, quotient, Chebyshev, or remainder profile, together with the pole and challenge-intersection estimates that prove its bad-slope count exceeds the target. A profile-envelope budget is *safe* only when the closed ledger proves its total distinct-slope count is at most the target. The next theorem brackets the frontier between one certified unsafe agreement and one certified safe agreement.

Theorem 1.6 (Target-aware threshold bracket)**.** *Let $a_{-,n} < a_{+,n}$ be agreements for which a certified profile-list construction is unsafe at $a_{-,n}$ and the closed-ledger profile-envelope budget is safe at $a_{+,n}$. Then the threshold in (1.2) exists and*

$$a_{-,n} < a_n^* \le a_{+,n}, \qquad 1 - \frac{a_{+,n}}{n} \le \delta_n^* < 1 - \frac{a_{-,n}}{n}. \tag{1.6}$$

In particular, if $a_{+,n} - a_{-,n} = o(n)$, $k_n/n \to \rho$, and the common crossing is $a_n = k_n + 1 + g_n n + o(n)$, then $\delta_n^* = 1 - \rho - g_n + o(1)$.

We call a range of consecutive agreement values, or an *agreement window*, *identity-dominant* when the entire profile envelope is, up to $e^{o(n)}$, bounded by the identity scale together with the universal constant and deep terms. Only in such a window can the profile compiler reduce to the familiar one-variable identity frontier.

Corollary 1.7 (Identity-frontier corollary)**.** *Assume the full-field challenge $\Gamma_n = \mathbb{F}_n$, and suppose in a window that*

$$\mathfrak{E}_n(a) \le e^{o(n)}\Big(1 + n - a + \overline{N}_1(a)\Big).$$

Put $\rho_n = k_n/n$, $\beta_n = \log_2 |\mathbb{B}_n|$, $\tau_n = n^{-1}\log_2(1 + B_n^*)$, and define

$$F_n(g) = H_2(\rho_n + g) - \beta_n g, \qquad g_{T,n} = \sup\Big(\{g \in [0, 1-\rho_n] : F_n(g) \ge \tau_n\} \cup \{0\}\Big). \tag{1.7}$$

The right crossing *is this rightmost point of the superlevel set $F_n(g) \ge \tau_n$. It is* interior *when $0 < g_{T,n} < 1 - \rho_n$ and the function passes from the superlevel side to the sublevel side there. By continuity, at an interior right crossing,*

$$H_2(\rho_n + g_{T,n}) - \beta_n g_{T,n} = \tau_n.$$

If the exact safe and unsafe tests furnish agreements $a_{\pm,n} = k_n + 1 + g_{T,n} n + o(n)$, then

$$\delta_n^* = 1 - \rho_n - g_{T,n} + o(1).$$

If additionally $\log(1 + B_n^*) = o(n)$, put

$$g^*(r, b) = \sup\{g \in [0, 1-r] : H_2(r+g) \ge bg\}.$$

Then $g_{T,n} = g^*(\rho_n, \beta_n) + o(1)$ and

$$\delta_n^* = 1 - \rho_n - g^*(\rho_n, \beta_n) + o(1). \tag{1.8}$$

For a proper challenge subset the exact bracket remains valid, but this corollary additionally requires a certified challenge-intersection lower bound; $|\Gamma_n|$ alone does not determine it.

The proof is not a change of notation: the upper implication uses the distinct-ray ledger, the lower implication uses an actual profile list and the collision-aware pole line, and monotonicity then gives (1.6). The complete argument, including the Stirling error and the *target reserve*—the quantitative margin by which the safe and unsafe integer bounds clear B_n^*—is given in Section 19.

The profile-envelope compiler is conditional only on the explicit enumerative statements in (A1)–(A7), not on an unnamed closed ledger. The analytic implication

$$\text{Sidon payment} + \text{positive-rate moment excess} \implies \text{high energy},$$
$$\text{high energy} \implies \text{BSG small doubling} \implies \text{contradiction}$$

is proved in full.

The polynomial-field construction below proves that a uniform ambient identity-normalized Sidon payment on the unrestricted deep prefix is false, even after exact stabilizers have been destroyed by a bounded planted remainder.

The exact bounded-remainder comparison identifies a missing exponential profile term rather than removing it. Arbitrary positive-density *partial-fiber occupancy*—supports selecting neither zero nor all points in a positive-density family of folding fibers—is not exhausted by that normal form. What remains genuinely open is a Sidon payment only on the residual obtained after every quotient, subfield, planted, and remainder profile has been removed, together with a complete partial-occupancy comparison, an exhaustive profile atlas, and a higher-dimensional transverse-secant ray bound. These are logically coupled conditions, not independent lemmas whose conjunction could prove the false unrestricted identity numerator theorem.

Definition 1.8 (Closed asymptotic ledger)**.** *A ledger-admissible sequence has a* closed asymptotic ledger *when, uniformly in every received line, all of the following are supplied as theorems rather than left as hypotheses:*

(L1) *its* exact-agreement witness incidence*, meaning the witnesses restricted to supports S with $|S| = a_n$, has a witness-exhaustive first-match atlas in the sense of Definition 2.7, with $e^{o(n)}$ realized profiles;*

(L2) *every algebraic or directly counted profile has a certified bound for its actual first-match slope set Z_i°, meaning the slopes assigned to profile i but to no earlier profile, and the sum of those bounds is at most its stated contribution to $e^{o(n)}\mathfrak{E}_n(a_n)$;*

(L3) *every primitive first-match residual of Definition 2.9 has its image-normalized scale, Sidon/Fourier or direct primitive-Q estimate, and distinct-ray compiler proved at its stated profile scale; and*

(L4) *effective-image collapse is either routed to an earlier profile or the full-image certificate* (FI) *is proved before an ambient scale is used, and all remaining profile budgets add back to at most $e^{o(n)}\mathfrak{E}_n(a_n)$.*

Thus closure is an enumerative statement about distinct slope projections. Constructibility, a raw support count, or a support-pair moment alone does not close a ledger.

Theorem 1.9 (Compiler for closed ledgers)**.** *A closed asymptotic ledger satisfies* (1.5) *and therefore supplies the safe side of the target problem. Whenever a certified profile-list construction supplies the unsafe side, the threshold bracket of Theorem 1.6 follows. Under the additional identity-dominance, target-reserve, and exact lower-side hypotheses, this specializes to Corollary 1.7.*

Proof. The primitive analytic theorem is proved in Sections 12, 14 and 17. The ray step under (RC) is Proposition 18.11, first-match summation is Lemma 2.10, and the target-aware entropy comparison is in Section 19. Adding the algebraic, primitive, and direct-ray budgets gives (1.5). ∎

Theorem 1.10 (Primitive Sidon-heavy implication). *For each N, let $\Omega_N^0 \subseteq \{0,1\}^{T_N}$ be a full profile slice with $|T_N| = N$ and fixed support weight m_N, where $m_N/N \in [\alpha, 1-\alpha]$ for some constant $\alpha > 0$. Let $\Phi_N : \Omega_N^0 \to \mathbb{B}_N^{R_N}$, and put*

$$\mathcal{S}_N = \Phi_N(\Omega_N^0), \qquad L_N = |\mathcal{S}_N|, \qquad \overline{N}_N = |\Omega_N^0|/L_N.$$

Let $\Omega_N^\circ \subseteq \Omega_N^0$ be the primitive first-match residual after the algebraic profiles have been removed. For $s \in \mathcal{S}_N$, put $F_{N,s} = \Omega_N^\circ \cap \Phi_N^{-1}(s)$ and $f_{N,s} = |F_{N,s}|$. Assume there is a logarithmic order $q_N \to \infty$, $q_N \le (\log N)^C$, such that

$$\frac{\log L_N}{q_N} = o(N). \tag{1.9}$$

If the image-normalized Sidon-heavy moment at this order is $e^{o(Nq_N)}$ for every fixed energy cutoff $\Delta(F_{N,s}) \le e^{-\sigma N}$, $\sigma > 0$, then the leaf satisfies primitive Q:

$$\max_{s\in\mathcal{S}_N} |\Omega_N^\circ \cap \Phi_N^{-1}(s)| \le e^{o(N)}\overline{N}_N.$$

The same conclusion follows from an ambient Fourier payment only when that payment is proved directly; such a bound also certifies $|\mathcal{S}_N| \ge e^{-o(N)}|\mathbb{B}_N|^{R_N}$ by Remark 10.2. Consequently the Sidon-heavy branch is not an additional inverse theorem: once its Fourier/effective-image payment is verified, any remaining positive-rate Rényi excess—exponential excess of the normalized high moment $L_N^{-1}\sum_s (f_{N,s}/\overline{N}_N)^{q_N}$—is high-energy and is impossible in a Boolean primitive leaf.

Proof. This is Theorem 17.2. For completeness, if primitive Q failed at positive exponential rate, the elementary lower moment inequality $L_N^{-1}\sum_s R_s^{q_N} \ge L_N^{-1}(\max_s R_s)^{q_N}$, with $R_s = f_{N,s}/\overline{N}_N$, and (1.9) would make the ordinary normalized moment at the stipulated order have positive-rate excess. The Sidon payment removes the low-energy contribution. The ordinary-moment split of Proposition 12.2, which separates low- and high-energy fiber contributions, then produces a fiber of size $e^{\Omega(N)-o(N)}\overline{N}_N$, hence $e^{\Omega(N)-o(N)}$, with normalized energy $e^{-o(N)}$. By Balog–Szemeredi–Gowers and Boolean-cube difference growth, Proposition 17.1, such a fiber has only $e^{o(N)}$ points, a contradiction. ∎

Remark 1.11 (Scope of the primitive implication). The theorem is the precise primitive implication "pay structure, flatten Sidon, inverse high energy." It does not prove the source estimate (PF) or (MA), window-uniformity in the agreement (uniform constants for every integer agreement in the stated interval), the effective-image certificate, profile add-back, the lower-side reserve, or the conversion from support pairs to distinct MCA rays. Those remain the separate normalization, profile, Fourier, and ray interfaces recorded in Definition 1.4.

Terminology map. The following summary fixes the meaning of the recurring paper-specific terms; the later definitions supply their scheme-theoretic and finite-row forms.

Row and domain.: A row is $(\mathbb{F}, D, k)$, with coordinate/evaluation domain D; a row sequence allows all three data to vary with $|D| = n$.

Explanation, bad slope, and witness.: Explanation means agreement with a degree-less-than-k polynomial on the named support. A bad slope has a witness (γ, S, h) for which the line point is explained on S but the received pair is not simultaneously explained on that same S.

Locator, prefix, and identity profile.: The locator Q_S is the monic polynomial with root set S; a prefix fiber fixes its leading nontrivial coefficients. The identity profile is the full unprofiled support family, and $\overline{N}_1$ is its ambient average scale.

Folding, quotient, and circle domain.: A folding map has complete uniform fibers; a quotient support is a union of those fibers. A smooth multiplicative domain uses power foldings on a multiplicative coset, while a Chebyshev twin-coset domain projects two inverse torus cosets and uses the descended Chebyshev foldings.
Chart, cell, profile, and atlas.: A chart gives witness coordinates, a cell specifies a subincidence, and its realization is the corresponding finite witness subset after all parameters are fixed. A realized profile records the cell type and parameter values; a boundary profile also records its full support slice and boundary map. An ordered atlas is witness-exhaustive and uses actual first-match slope projections.
Payment and ledger.: A payment is a proved distinct-slope bound at the relevant profile scale. A ledger is the atlas, all its payments, the primitive residual analysis, and their add-back; it is closed only when every required bound is verified for the row.
Image scale and envelope.: For a full profile slice, the image scale is support count divided by the size of the realized boundary image. The ambient codomain scale may replace it only after (FI). The profile envelope sums these realized scales, together with the universal finite terms, uniformly over received lines.
Q, SP, and rays.: Q is a profile max-fiber bound, SP is the same-boundary support-pair count, and a ray bound counts distinct slopes. The passage from Q/SP to rays is precisely the separate (RC) interface.
Syndrome, energy, and Fourier payment.: A syndrome is a parity-check image, and the secant compiler counts where its received line meets error- support spans. Additive energy counts repeated support differences. The Sidon-heavy moment is the low-energy part of a normalized fiber moment; (PF) pays minor Fourier characters pointwise and (MA) pays the exceptional characters in aggregate.
Frontier.: The target-aware frontier is the first safe agreement, or equivalently its radius $1 - a/n$; the identity frontier is its conditional specialization when the identity profile dominates the full envelope. A target reserve is the explicit margin by which a certified numerator lies on the required side of the target budget.

2. MCA witnesses and first-match ledgers

Throughout this section $\mathbb{F}$ is a finite field, $D \subseteq \mathbb{F}$ has size n, and $C = \mathrm{RS}_{\mathbb{F}}(D, k)$. Write $\mathbb{F}[X]_{<k} = \{f \in \mathbb{F}[X] : \deg f < k\}$, and let $\mathrm{ev}_S : \mathbb{F}[X]_{<k} \to \mathbb{F}^S$ be restriction of the evaluation map.

The terminology in this section turns the informal statement "a point on the received affine line looks like a codeword on many coordinates" into a finite incidence problem. A *witness* records all data certifying that statement: the line parameter γ, the agreement support S, and the explaining polynomial h. A *profile* records a specified structural reason why witnesses may occur in abundance. An *atlas* is an ordered collection of such profiles covering all witnesses, and a *ledger* attaches a proved distinct-slope budget to every entry. These words have no additional geometric meaning beyond the exact definitions below.

Definition 2.1 (Explanation and support-wise nontriviality). *For $u \in \mathbb{F}^D$ and $S \subseteq D$, an* explanation of u on S *is a polynomial $f \in \mathbb{F}[X]_{<k}$ with $\mathrm{ev}_S(f) = u|_S$. We write $\mathrm{Expl}_C(u; S)$ when such an f exists. A pair $r = (r_0, r_1)$ is* simultaneously explained on S *when $\mathrm{Expl}_C(r_0; S)$ and $\mathrm{Expl}_C(r_1; S)$ both hold, with no requirement that the two explaining polynomials coincide. It is* support-wise nontrivial on S *when it is not simultaneously explained there. We denote this last condition by $\mathrm{NT}_C(r; S)$.*

Thus support-wise nontriviality is a condition on the same support that explains the affine combination. It does not mean that both coordinates are individually unexplained; failure of either explanation is enough.

Definition 2.2 (MCA-bad slope)**.** *Let $r = (r_0, r_1) \in (\mathbb{F}^D)^2$ and let a be an agreement threshold. A finite slope $\gamma \in \mathbb{F}$ is support-wise MCA-bad at agreement a if there are $S \subseteq D$, $|S| \geq a$, and $h \in \mathbb{F}[X]_{<k}$ such that $\mathrm{ev}_S(h) = (r_0 + \gamma r_1)|_S$ and $\mathrm{NT}_C(r; S)$ holds. Define*

$$B_C^{\mathrm{MCA}}(a) = \max_{r \in (\mathbb{F}^D)^2} |\{\gamma \in \mathbb{F} : \gamma \text{ is MCA-bad for } r \text{ at } a\}|.$$

The finite-slope restriction costs at most the point at infinity, which may also be treated by exchanging the two coordinates. The support-wise condition is essential: explanation and nontriviality are tested on the same set.

Definition 2.3 (Exact-agreement witness incidence)**.** *Assume $k + 1 \leq a \leq n$. For a received line $r = (r_0, r_1)$, its exact-a witness incidence is*

$$\mathcal{W}_a(r) = \left\{ (\gamma, S, h) : \begin{array}{ll} \gamma \in \mathbb{F}, \quad S \in \binom{D}{a}, & h \in \mathbb{F}[X]_{<k}, \\ \mathrm{ev}_S(h) = (r_0 + \gamma r_1)|_S, & \mathrm{NT}_C(r; S) \end{array} \right\}. \tag{2.1}$$

Its slope projection is $\pi_\gamma(\gamma, S, h) = \gamma$, and its exact bad-slope set is $Z_a(r) = \pi_\gamma(\mathcal{W}_a(r)) \subseteq \mathbb{F}$.

Lemma 2.4 (Reduction to exact agreement)**.** *For $a \geq k + 1$, the set $Z_a(r)$ is precisely the set of slopes that are MCA-bad at threshold a in Definition 2.2.*

Proof. An exact-a witness is a threshold-a witness. Conversely, let (γ, S, h) witness badness with $|S| \geq a$. At least one of $r_0|_S, r_1|_S$ is outside $\mathrm{ev}_S(\mathbb{F}[X]_{<k})$; call it $u|_S$. Interpolate u on any fixed k-subset $K \subseteq S$. If the resulting polynomial agreed with u at every point of $S \setminus K$, it would explain u on S, a contradiction. Hence some $x \in S \setminus K$ makes u unexplained on $K \cup \{x\}$. Extend this set to an a-subset $S' \subseteq S$. Non-explanation persists on supersets, while the same h explains $r_0 + \gamma r_1$ on S'. □

Definition 2.5 (Universal witness incidence)**.** *Let $\mathfrak{S}_a(D) = \coprod_{S \in \binom{D}{a}} \operatorname{Spec} \mathbb{F}$ be the finite reduced scheme of exact-a supports. The* universal witness incidence *is the row-wise locally closed set*

$$\mathfrak{W}_a \subseteq (\mathbb{A}^n)^2 \times \mathbb{A}^1_\gamma \times \mathfrak{S}_a(D) \times \mathbb{A}^k_h$$

whose fiber over $r \in (\mathbb{F}^D)^2$ is $\mathcal{W}_a(r)$. On the component indexed by S, it is cut out by $\mathrm{ev}_S(h) = (r_0 + \gamma r_1)|_S$ and by the open condition $(r_0|_S, r_1|_S) \notin \mathrm{im}(\mathrm{ev}_S)^2$.

Here $\mathfrak{S}_a(D)$ is simply the disjoint union of one point for each a-element support. The scheme notation packages all supports into one algebraic object; readers may equivalently read $\mathfrak{W}_a$ as the finite family of polynomial equation-and-inequation systems indexed by those supports. "Locally closed" means an intersection of a Zariski-open and a Zariski-closed set, and "constructible" means a finite union of such sets.

The evaluation equations are polynomial and the simultaneous-explanation locus is a closed linear subspace. Thus $\mathfrak{W}_a$ is locally closed on each support component, and hence locally constructible. It is generally not a closed subvariety: support-wise nontriviality is an open condition. Eliminating h, or adjoining locators, quotient data, and *rank pivots*, produces constructible images by Chevalley's theorem. A rank pivot is a specified matrix minor required to be nonzero, thereby selecting a chart on which a chosen rank is visible. We use Chevalley's theorem only in its standard form that the algebraic projection of a constructible set is constructible.

Definition 2.6 (Profile cell and realized profile)**.** *A* profile type *λ consists of a parameter space P_λ and a locally constructible incidence $\mathfrak{C}_\lambda \subseteq \mathfrak{W}_a \times P_\lambda$. For a received line r and a parameter $\xi \in P_\lambda(\mathbb{F})$, the corresponding* realized profile cell *is the actual finite witness set*

$$\mathcal{C}_{\lambda, r, \xi} = \{w \in \mathcal{W}_a(r) : (w, \xi) \in \mathfrak{C}_\lambda(\mathbb{F})\}.$$

It is a realized profile when this set is nonempty. A boundary profile also specifies a finite full support slice $\Omega^0_{\lambda,r,\xi}$ *and a proved boundary map* $\Phi_{\lambda,r,\xi} : \Omega^0_{\lambda,r,\xi} \to \Sigma_{\lambda,r,\xi}$. *Its image-normalized scale is*

$$L_{\lambda,r,\xi} = |\Phi_{\lambda,r,\xi}(\Omega^0_{\lambda,r,\xi})|, \qquad \overline{N}_{\lambda,r,\xi} = \frac{|\Omega^0_{\lambda,r,\xi}|}{L_{\lambda,r,\xi}}. \tag{2.2}$$

The ambient codomain size is not substituted for $L_{\lambda,r,\xi}$ *without the full-image certificate* (FI).

The parameters ξ are part of the profile, not hidden existential choices. Consequently an assertion that there are $e^{o(n)}$ profiles means that the number of nonempty pairs (λ, ξ) occurring for a received line is $e^{o(n)}$, uniformly in that line.

Definition 2.7 (Witness-exhaustive first-match atlas)**.** *Fix* r *and order its realized profile cells as* $\mathcal{C}_1, \dots, \mathcal{C}_s \subseteq \mathcal{W}_a(r)$. *Define their* actual *slope projections and first-match parts by*

$$Z_i = \pi_\gamma(\mathcal{C}_i), \qquad Z_i^\circ = Z_i \setminus \bigcup_{j<i} Z_j, \qquad \mathcal{C}_i^\circ = \mathcal{C}_i \cap \pi_\gamma^{-1}(Z_i^\circ). \tag{2.3}$$

The ordered family is a witness-exhaustive first-match atlas *if* $\mathcal{W}_a(r) = \bigcup_{i=1}^s \mathcal{C}_i$. *A certified upper ledger for this atlas consists of supersets* $E_i \supseteq Z_i^\circ$ *and numbers* U_i *satisfying* $|E_i| \le U_i$.

Witness exhaustivity is stronger than merely covering the set of slopes, and is the property needed to assert that no unclassified witness remains. The sets Z_i°, rather than arbitrary slope supersets, define the first-match assignment. In particular, $Z_a(r) = \coprod_i Z_i^\circ$.

Definition 2.8 (Profile payment)**.** *For a row sequence* $C_n = \mathrm{RS}_{\mathbb{F}_n}(D_n, k_n)$, *with* $|D_n| = n \to \infty$, *a* scaled realized profile cell *is a realized cell equipped with either its image-normalized scale* $\overline{N}_i$ *or a declared direct scale* M_i. *Such a cell is* paid *in a given first-match atlas if, uniformly in the received line and its realized parameter, its certified budget satisfies*

$$|Z_i^\circ| \le U_i \le \exp(o(n))\Big(1 + \overline{N}_i\Big). \tag{2.4}$$

More generally a direct profile scale M_i *is paid when* $U_i \le e^{o(n)} M_i$. *An atlas is paid at an envelope* E_n *when* $\sum_i U_i \le e^{o(n)} E_n$. *All* $o(n)$ *terms are uniform over the profiles and received lines under consideration.*

At image scale one always has $\overline{N}_i \ge 1$, so the additive 1 is harmless. It keeps the same payment notation valid for separately certified ambient or direct profile scales that can lie below one. Payment is an enumerative assertion about an actual slope projection; constructibility alone is not payment.

Definition 2.9 (Primitive first-match residual)**.** *After the first* j *cells of an ordered atlas have been routed, define the remaining witness incidence by*

$$\mathcal{W}_{>j}(r) = \mathcal{W}_a(r) \cap \pi_\gamma^{-1}\left(\mathbb{F} \setminus \bigcup_{i \le j} Z_i\right). \tag{2.5}$$

A later realized boundary profile is a primitive first-match residual *when its witnesses lie in* $\mathcal{W}_{>j}(r)$, *every algebraic degeneracy named by the atlas has been routed among the first* j *cells, and the remaining boundary map carries the row-specific primitivity certificate used by the analytic and ray arguments. Its residual support set* Ω°_λ *is the subset of* Ω^0_λ *induced by* (2.5).

The residual in Definition 2.9 is a first-match excision, not a topological assertion. It is constructible in the row-wise finite incidence after taking the displayed projections, but need not be Zariski open. Nor is "primitive" synonymous with having trivial stabilizer: it records the specific quotient, field-descent, rank, planted, and ray-saturation exclusions certified by the atlas.

Lemma 2.10 (First-match bound)**.** *If every received line admits a witness-exhaustive first-match atlas and a certified upper ledger with $\sum_i U_i \le U$, then $B_C^{\mathrm{MCA}}(a) \le U$.*

Proof. For a fixed received line, exact-agreement reduction and witness exhaustivity give the disjoint union $Z_a(r) = \coprod_i Z_i^\circ$. Since $Z_i^\circ \subseteq E_i$, its size is at most U_i. Summing and then maximizing over received lines proves the claim. □

The first-match rule is the combinatorial form of excision in the witness incidence. Profile cells need not be disjoint. Once a slope is charged to the first profile in whose actual projection it occurs, all of its later witnesses are removed from the residual by (2.5).

3. Exact syndrome geometry and the deep regime

The *syndrome* of a word is the result of applying a parity-check matrix; it vanishes exactly on codewords. Thus syndrome space forgets the part of a received word already lying in the code and retains only its error class. This description separates possible error supports from distinct line parameters. Put $R = n - k$, the redundancy (and syndrome-space dimension), and choose the generalized Vandermonde parity-check matrix H with columns

$$h_x = \lambda_x(1, x, \dots, x^{R-1})^\mathsf{T}, \qquad \lambda_x = \left(\prod_{y \in D\setminus\{x\}} (x-y)\right)^{-1}. \tag{3.1}$$

Lagrange interpolation gives $\sum_{x\in D} \lambda_x x^j = 0$ for $0 \le j \le n-2$, so this matrix annihilates every degree-less-than-k evaluation. Every set of at most R columns is independent. For $E \subseteq D$, write $V_E = \mathrm{Span}\{h_x : x \in E\}$; it is exactly the set of syndromes of words supported on E. We write $s(u) = Hu^\mathsf{T}$ for the syndrome of u.

Proposition 3.1 (Syndrome-line normal form)**.** *Let $S = D \setminus E$. A word u is explained on S if and only if $s(u) \in V_E$. Consequently γ is MCA-bad on S for a pair (u_0, u_1) if and only if*

$$s(u_0) + \gamma s(u_1) \in V_E, \qquad \{s(u_0), s(u_1)\} \not\subseteq V_E. \tag{3.2}$$

For fixed E, there is at most one such finite slope.

Proof. If $u = c + e$, where $c \in C$ and e is supported on E, then $s(u) = s(e) \in V_E$. Conversely, if $s(u) = \sum_{x\in E} e_x h_x$, subtract the word supported on E with values e_x; the result has zero syndrome and belongs to C. Apply this equivalence to the line point and to both coordinates. Passing (3.2) to $\mathbb{F}^R/V_E$ shows that two distinct solutions would put both projected syndromes at zero, contradicting badness. □

Since $\lambda_x \neq 0$, one has $[h_x] = [1 : x : \cdots : x^{R-1}]$; these are the points of the rational normal curve in projective syndrome space, while the weights λ_x merely choose vector representatives. A *secant span* is the linear span V_E of a selected set of those points. We call the meeting of a syndrome line with V_E *transverse* when the entire line is not contained in V_E, equivalently when its two generators are not both in V_E. The next result is a *compiler* in the sense that it translates the bad-slope problem, exactly and in both directions, into this line–secant incidence problem. A *ray* means a one-dimensional direction in syndrome or coefficient space; different raw witnesses may represent the same ray and hence the same slope.

Theorem 3.2 (Exact syndrome–secant compiler)**.** *Put $t = n - a$, and for a syndrome line $\ell_{y_0,y_1} = \{y_0 + \gamma y_1 : \gamma \in \mathbb{F}\}$ define*

$$\Theta_t(y_0, y_1) = |\{\gamma \in \mathbb{F} : \exists E \subseteq D,\ |E| \le t, y_0 + \gamma y_1 \in V_E,\ \{y_0, y_1\} \not\subseteq V_E\}|. \tag{3.3}$$

Then

$$B_C^{\mathrm{MCA}}(a) = \max_{y_0,y_1 \in \mathbb{F}^{n-k}} \Theta_t(y_0, y_1). \tag{3.4}$$

For each fixed E, the affine line has at most one transverse intersection parameter with V_E. For a Reed–Solomon code the spaces V_E are the spans of at most t points of the weighted rational normal curve in (3.1). Thus the higher-dimensional ray problem is exactly a transverse line–secant incidence problem, with repeated intersection parameters deduplicated.

Proof. Apply Proposition 3.1 with $S = D \setminus E$, and take the union over $|E| \le t$. The syndrome map is surjective, so maximizing over received pairs is the same as maximizing over (y_0, y_1). In the quotient $\mathbb{F}^{n-k}/V_E$, a transverse intersection solves $\overline{y}_0 + \gamma\overline{y}_1 = 0$ with the two projected vectors not both zero, and therefore has at most one finite solution. The last assertion is the Vandermonde description of the parity columns. □

A *parity-column circuit* is a minimally linearly dependent set of columns of H. The next lemma says that many distinct transverse rays cannot all be supported inside an independent union of columns.

Lemma 3.3 (Independent-union ray bound)**.** *Let every set of at most R parity columns be independent, put $t < R$, and fix a syndrome line. For each slope in a transverse set Z, choose an error set E_γ, $|E_\gamma| \le t$, witnessing (3.3). If $U = \bigcup_{\gamma\in Z} E_\gamma$ has size at most R, then $|Z| \le t + 1$. Consequently, more than $t + 1$ transverse slopes force the witness union to contain a parity-column circuit.*

Proof. If $|Z| \le 1$, there is nothing to prove; assume that Z contains two distinct slopes. Then Two distinct line points lie in V_U, hence $y_0, y_1 \in V_U$. Column independence gives unique coefficient lifts $e_0, e_1 \in \mathbb{F}^U$, and every chosen lift is $e_\gamma = e_0 + \gamma e_1$. Let U' be the set of s coordinates where the pair (e_0, e_1) is nonzero. If $s \le t$ and E_γ contained all of U', then $y_0, y_1 \in V_{U'} \subseteq V_{E_\gamma}$, contradicting the transversality of this particular witness. Hence some $x \in U' \setminus E_\gamma$ exists, and uniqueness of the lift gives $e_0(x) + \gamma e_1(x) = 0$. Each nonzero affine coordinate function has at most one root, so $|Z| \le s \le t$. If $s > t$, every chosen lift of weight at most t has at least $s - t$ zero coordinates. Double counting the pairs (γ, x) with $e_0(x) + \gamma e_1(x) = 0$ gives $|Z|(s - t) \le s$, whence $|Z| \le \lfloor s/(s - t)\rfloor \le t + 1$. If $|U| \le R$, independence holds; the contrapositive proves the circuit assertion. □

For a fixed received line, the exact bad-slope set is therefore the set of unique cancelling line parameters contributed by the transverse quotient equations. For each E, $|E| \le n - a$, project the two syndromes to $\mathbb{F}^R/V_E$. If the second projection is nonzero and the first is a scalar multiple of it, record the unique cancelling scalar; otherwise record nothing. Deduplicating these scalars gives the numerator. In particular, the number of supports, the number of pairs of supports, and the number of distinct slope rays are three different quantities.

We call $3(n - a) \le d - 1$ the *deep regime*: the allowed error radius $n - a$ is so small relative to the minimum codeword weight d that the union of three permitted error supports still cannot contain a nonzero codeword support.

Theorem 3.4 (Deep-regime upper bound)**.** *Let $C \subseteq \mathbb{F}^n$ be linear with minimum nonzero weight d, let $0 \leq a \leq n$, put $r = n - a$, and assume $3r \leq d - 1$. Then*

$$B_C^{\mathrm{MCA}}(a) \leq r + 1. \tag{3.5}$$

Proof. Fix (u_0, u_1), and let Z be the slopes for which $u_0 + \gamma u_1$ is within distance r of C. If $|Z| \leq r + 1$, there is nothing to prove. Otherwise choose distinct $\gamma_1, \gamma_2 \in Z$, codewords p_1, p_2, and error sets E_1, E_2, each of size at most r, such that $u_0 + \gamma_i u_1 = p_i$ off E_i. Solving the two affine equations gives $c_0, c_1 \in C$ for which the error pair $e_i = u_i - c_i$ is supported on $T = E_1 \cup E_2$, $|T| \leq 2r$.

For any $\gamma \in Z$, let p be a closest explaining codeword. The codeword $p - (c_0 + \gamma c_1)$ is supported on at most $3r \leq d - 1$ positions and is zero. Thus $\mathrm{wt}(e_0 + \gamma e_1) \leq r$. Let T' be the support of the error pair. If $|T'| \geq r + 1$, every $\gamma \in Z$ makes at least $|T'| - r$ nonzero affine coordinate functions $e_{0,x} + \gamma e_{1,x}$ vanish. Each coordinate has at most one root, so $|Z|(|T'| - r) \leq |T'|$ and $|Z| \leq r + 1$.

It remains that $|T'| \leq r$. If γ is MCA-bad on a set S, $|S| \geq n - r$, its explaining codeword equals $c_0 + \gamma c_1$, since their difference would otherwise have weight at most $2r \leq d - 1$. The set S must meet T', and at a point of $S \cap T'$ the nonzero pair satisfies $e_{0,x} + \gamma e_{1,x} = 0$. Such a point determines γ, giving at most $|T'| \leq r$ bad slopes. $\square$

Recall that for a nonempty challenge set $\Gamma \subseteq \mathbb{F}$, $B_{C,\Gamma}^{\mathrm{MCA}}(a)$ denotes the same maximum with slopes restricted to Γ.

The following "tangent floor" is an elementary family of neighboring coordinate-span intersections. The name does not assert a hidden differentiability or multiplicity condition.

Proposition 3.5 (Universal tangent floor)**.** *For every Reed–Solomon code, every $a \geq k+1$, and every nonempty $\Gamma \subseteq \mathbb{F}$,*

$$B_{C,\Gamma}^{\mathrm{MCA}}(a) \geq \min\{|\Gamma|, n - a + 1\}. \tag{3.6}$$

Proof. Put $j = n - a$, $t = \min\{|\Gamma|, j + 1\}$, and choose distinct $\gamma_1, \ldots, \gamma_t \in \Gamma$ and independent parity columns $h_{x_1}, \ldots, h_{x_t}$. Define $s_1 = \sum_i h_{x_i}$ and $s_0 = -\sum_i \gamma_i h_{x_i}$, and lift them to received words. For $E_i = \{x_\ell : \ell \neq i\}$, $s_0 + \gamma_i s_1 \in V_{E_i}$, whereas $s_1 \notin V_{E_i}$. Proposition 3.1 makes every γ_i bad on $D \setminus E_i$, which has size at least a. $\square$

Corollary 3.6 (Exact deep numerator)**.** *If $C = \mathrm{RS}_{\mathbb{F}}(D, k)$, $1 \leq k < n$, $k + 1 \leq a \leq n$, $\varnothing \neq \Gamma \subseteq \mathbb{F}$, and $3(n - a) \leq n - k$, then*

$$B_{C,\Gamma}^{\mathrm{MCA}}(a) = \min\{|\Gamma|, n - a + 1\}.$$

Proof. The upper bound is Theorem 3.4, intersected with Γ. The hypotheses imply $a \geq k + 1$, so Proposition 3.5 gives equality. $\square$

Theorem 3.7 (Subfield confinement)**.** *Let $\mathbb{B} \subseteq \mathbb{F}$, $D \subseteq \mathbb{B}$, and let $u_0, u_1 \in \mathbb{B}^D$. If $\gamma \in \mathbb{F} \setminus \mathbb{B}$ and $u_0 + \gamma u_1$ is explained on S, then the pair is explained on the same S. Hence every MCA-bad slope of a $\mathbb{B}$-valued received line lies in $\mathbb{B}$.*

Proof. Choose a $\mathbb{B}$-basis $1, \beta_1, \ldots$ of $\mathbb{F}$. Decompose the coefficients of the explaining polynomial and of γ in this basis. Because $D \subseteq \mathbb{B}$, comparison of basis components on S gives a codeword explaining u_1 from any nonzero nonbase component of γ, and then a codeword explaining u_0 from the base component. $\square$

4. Exact locator prefixes and collision-aware pole lines

Assume $D \subseteq \mathbb{B} \subseteq \mathbb{F}$. For an m-set $S \subseteq D$, write

$$Q_S(X) = \prod_{x \in S} (X - x) = X^m + c_1(S) X^{m-1} + \cdots + c_m(S), \qquad \Phi_w(S) = (c_1(S), \ldots, c_w(S)).$$

The monic polynomial Q_S, whose simple roots are precisely the points of S, is the *support locator*. Its *depth-w locator prefix* is the vector of the first w coefficients below the leading coefficient. For $z \in \mathbb{B}^w$, the inverse image $\Phi_w^{-1}(z)$ is the *prefix fiber over z*: the family of all m-element supports having that same prefix. Thus "fiber" in this paper always refers to the inverse image of a displayed map.

Proposition 4.1 (Exact prefix-list bijection)**.** *Let $1 \le K \le m \le n$, put $w = m - K$, and set $U_z(X) = X^m + \sum_{i=1}^{w} z_i X^{m-i}$ for $z \in \mathbb{B}^w$. The codewords of $\mathrm{RS}_{\mathbb{F}}(D, K)$ agreeing with $U_z|_D$ on at least m points are exactly*

$$(U_z - Q_S)|_D, \qquad S \in \Phi_w^{-1}(z). \tag{4.1}$$

No such codeword agrees on more than m points. In particular, some received word has list size at least $\lceil \binom{n}{m} |\mathbb{B}|^{-w} \rceil$.

Proof. For $S \in \Phi_w^{-1}(z)$, the leading term and next w coefficients cancel, so $\deg(U_z - Q_S) < m - w = K$. Conversely, if P, $\deg P < K$, agrees with U_z at m points, then $U_z - P$ is monic of degree m with those m roots and equals their locator. Coefficient comparison gives the prefix. A nonzero degree-m polynomial has no additional root. Pigeonhole the $\binom{n}{m}$ supports over $|\mathbb{B}|^w$ prefixes. □

The *simple-pole conversion* chooses an auxiliary point $\alpha \notin D$ and divides coordinatewise by $X - \alpha$. This turns one list of nearby polynomials into points on a received affine line for the dimension-k code. A *collision* is an equality $P_i(\alpha) = P_j(\alpha)$ for two distinct listed polynomials; the next formula pays such equalities rather than assuming all slopes are distinct.

Theorem 4.2 (Collision-aware simple-pole conversion)**.** *Let $C = \mathrm{RS}_{\mathbb{F}}(D, k)$, $C^+ = \mathrm{RS}_{\mathbb{F}}(D, k+1)$, $1 \le k < n$, and $q = |\mathbb{F}| > n$. If a word has L distinct codewords of C^+ agreeing on at least $m \ge k+1$ points, then one received line for C has at least*

$$M(L) = \left\lceil \frac{L(q-n)}{q - n + k(L-1)} \right\rceil \tag{4.2}$$

distinct MCA-bad slopes at agreement m.

Proof. Let the received word be $U \in \mathbb{F}^D$, and let the listed polynomials be $P_1, \ldots, P_L$, each of degree at most k. For $\alpha \in \mathbb{F} \setminus D$, define the word coordinatewise by $f_\alpha(x) = U(x)/(x - \alpha)$ and set $g_\alpha(x) = -1/(x - \alpha)$. On an agreement set for P_i, the line point of slope $P_i(\alpha)$ equals $(P_i(x) - P_i(\alpha))/(x - \alpha)$, a degree-less-than-$k$ evaluation. The pair is not simultaneously explained on $k+1$ points: an explanation of g_α by G, $\deg G < k$, would make $(X - \alpha)G(X) + 1$ vanish at $k+1$ points although its degree is at most k and its value at α is 1.

For $i \ne j$, the equality $P_i(\alpha) = P_j(\alpha)$ holds at at most k poles. Some pole has at most $k\binom{L}{2}/(q-n)$ colliding pairs. If its value multiplicities are $m_1, \ldots, m_M$, then $\sum m_i^2 \le L + kL(L-1)/(q-n)$. Cauchy–Schwarz gives $L^2 \le M \sum m_i^2$, which yields (4.2). □

Corollary 4.3 (Identity-prefix MCA floor)**.** *For $m \ge k+1$, put $w = m - k - 1$ and $L_m = \lceil \binom{n}{m} |\mathbb{B}|^{-w} \rceil$. If $q > n$, then*

$$B_C^{\mathrm{MCA}}(m) \ge M(L_m). \tag{4.3}$$

For equal-size sets, the *Johnson distance* is $d_J(S, T) = |S \setminus T| = |T \setminus S|$. It counts the number of point replacements required to turn one support into the other.

Proposition 4.4 (Prefix rigidity and its packing limit)**.** *Two distinct m-sets in one depth-w prefix fiber have Johnson distance at least $w+1$. Consequently, with $t = \lfloor w/2 \rfloor$, every fiber has size at most*

$$\frac{\binom{n}{m}}{\sum_{i=0}^{t} \binom{m}{i}\binom{n-m}{i}}. \tag{4.4}$$

At $w \asymp n/\log|\mathbb{B}|$, this packing loss is only $e^{o(n)}$ and does not supply the random factor $|\mathbb{B}|^{-w}$.

Prefix rigidity says that sharing many leading locator coefficients forces distinct supports to be far apart in this metric. Here the *polynomial-field regime* means $|\mathbb{B}| = n^{O(1)}$, equivalently $\log|\mathbb{B}| = O(\log n)$.

Proof. If S, T have Johnson distance $j \le w$, factor the locator of their intersection, of degree $m-j$, from $Q_S - Q_T$. The residual factor is nonzero, so the difference has degree at least $m-j \ge m-w$, whereas the common prefix makes its degree at most $m-w-1$, a contradiction. Radius t Johnson balls about the fiber members are disjoint and each has the denominator in (4.4). For $w = n/\log|\mathbb{B}|$, the logarithm of that ball volume is $O(w\log(n/w)) = o(n)$ in the polynomial-field regime. □

Theorem 4.5 (Prefix fibers embed in received lines)**.** *Let $\mathcal{G} \subseteq \Phi_w^{-1}(z)$ be a fiber of N many m-sets, with $0 \le w \le m-2$, and put $k = m-w-1$. Every finite extension $\mathbb{F}/\mathbb{B}$ satisfying*

$$|\mathbb{F}| - n > k\binom{N}{2} \tag{4.5}$$

admits one received line for $\mathrm{RS}_{\mathbb{F}}(D,k)$ with at least N distinct MCA-bad slopes furnished by $\mathcal{G}$.

Proof. For distinct $S, T \in \mathcal{G}$, the polynomial $Q_S - Q_T$ is nonzero of degree at most k. Condition (4.5) leaves a pole $\alpha \in \mathbb{F} \setminus D$ avoiding the roots of all these differences. Thus the values $Q_S(\alpha)$ are pairwise distinct. By Proposition 4.1, the polynomials $U_z - Q_S$ form one list for the dimension-$k+1$ code; the pole line makes their values, and hence the resulting MCA slopes, pairwise distinct. □

Definition 4.6 (Profiled asymptotic row datum)**.** *A* profiled asymptotic row datum *consists of a row sequence $C_n = \mathrm{RS}_{\mathbb{F}_n}(D_n, k_n)$, $|D_n| = n$, with coefficient fields $D_n \subseteq \mathbb{B}_n \subseteq \mathbb{F}_n$, together with agreements a_n and, for every received line, a witness-exhaustive first-match atlas of realized profiles with their fields, full slices, boundary maps, and actual slope projections recorded. The identity term is $\overline{N}_1 = \binom{n}{a_n}|\mathbb{B}_n|^{-(a_n-k_n-1)}$; the correct global scale is the profile envelope $\mathfrak{E}_n(a_n)$ of (1.4).*

The identity term is the random size of one unprofiled prefix fiber. It is an exact lower scale by Proposition 4.1, but it is an upper scale only after every quotient/remainder term has been compared with it and the primitive max-fiber and ray conditions have been proved.

5. Cells as moduli strata

The word *cell* does not mean a topological cell decomposition. The ambient object is the locally constructible universal witness incidence $\mathfrak{W}_a$ of Definition 2.5; a profile cell is the parameterized incidence of Definition 2.6. The qualification "algebraic" below records how such a profile is presented, not whether its slope projection has been paid.

A *moduli stratum* here simply means a parameterized family of witnesses having the same declared structural features. "Description complexity $e^{o(n)}$" means that the number and degrees of equations, inequations, auxiliary parameters, and Boolean pieces together contribute only a subexponential factor to the final count. It is a bookkeeping condition, not a claim that every family has bounded dimension.

The *structure maps* of a profile are its displayed projections, boundary maps, and auxiliary-coordinate maps. A *planted divisor* is a fixed factor common to the support locators in the profile; a *tangent datum* records a rank-defective slope projection; and a *locator pencil* is a projective polynomial family $Q_0 + \lambda Q_1$. The *field of definition* is the smallest recorded subfield over which these parameter spaces, equations, and maps are defined.

Definition 5.1 (Algebraic profile cell)**.** *An* algebraic profile cell *is a profile type $\mathfrak{C}_\lambda \subseteq \mathfrak{W}_a \times P_\lambda$ for which P_λ, the incidence, and all structure maps are locally constructible of uniform subexponential description complexity. A description may use polynomial equalities, inequations, rank conditions, finite Boolean operations, and projections. Quotient maps, planted divisors, tangent points, fields of definition, and locator-pencil parameters are recorded in P_λ; they are not suppressed when counting realized profiles.*

Definition 5.2 (Paid cell)**.** *Fix a witness-exhaustive ordered atlas. A scaled realized cell $\mathcal{C}_i$ is* paid at its profile scale *when its actual first-match projection Z_i° has a certified bound*

$$|Z_i^\circ| \le U_i \le e^{o(n)}(1 + \overline{N}_i), \tag{5.1}$$

uniformly in the received line and the realized profile parameter. A directly counted cell is paid at a stated scale M_i when $U_i \le e^{o(n)} M_i$. An ordered family is paid at an envelope when the sum of its certified U_i's is at most that envelope up to $e^{o(n)}$. This is the specialization of Definition 2.8 to the moduli cells used below.

Payment depends on the chosen order because it bounds Z_i°, not a formal cell name. It is an enumerative certificate, not a topological property. A cell may be Zariski dense in an ambient chart but paid because its actual slope projection is small after first-match saturation. Conversely, a proper subvariety is not automatically paid: its degree, components, parameter census, and projection fibers must fit the profile budget. The term $1 + \overline{N}_i$ in (5.1) also prevents a nonempty cell from being assigned a budget below one.

In the next lemma an *extension locus* records data that factor only after enlarging the coefficient field; a *differential-locator locus* is a rank-loss locus of the locator-prefix derivative; a *saturation locus* identifies raw witnesses with the same projective explanation direction; and a *fixed-dimensional pencil locus* is a bounded-dimensional projective family of locators. The catalogue below then explains how these loci are used.

Lemma 5.3 (Constructibility of the algebraic cells)**.** *The quotient, Chebyshev, planted, tangent, extension, differential-locator, saturation, and fixed-dimensional pencil loci in Section 5.1 are locally constructible profile cells over $\mathfrak{W}_a$. The Sidon moment component is analytic and is not asserted to be a bounded-complexity constructible cell.*

Proof. Each listed algebraic condition is expressed by polynomial equations, rank conditions, and nonvanishing conditions. Common roots and planted blocks use divisibility or resultant conditions, where $\mathrm{Res}(f, g) = 0$ detects a common root; quotient pullbacks use coefficient identities after composition; tangency and differential defects use minors; field descent uses Frobenius-fixed coefficient equations $c^{|\mathbb{B}'|} = c$, which force c into the recorded subfield $\mathbb{B}'$; and saturation or fixed-dimensional pencils are projective incidence conditions. Chevalley's theorem and stability under finite Boolean operations and projections prove constructibility. The exceptional Fourier characters may themselves form a constructible subset of the character parameter space, but constructibility alone does not imply the aggregate estimate (MA). □

5.1. **Cell catalogue.** The catalogue below is a language for row-specific proofs, not a theorem that every displayed locus is automatically paid. In a concrete family, each item must be accompanied by a distinct-slope projection bound at its natural profile scale. The conditional compiler uses only those bounds, not constructibility itself.

Quotient and periodic cells. A quotient cell consists of pullbacks along a nontrivial map $\pi : D \to D'$: support membership is constant on the relevant fibers of π, so the support descends to the smaller domain D'. Such a support is called *periodic* with respect to π. Algebraically, the locator or boundary map factors through the quotient coordinate. The smaller support count is offset by the descended prefix depth and possibly a smaller coefficient field, so the cell is budgeted at its natural quotient term in $\mathfrak{E}_n$; no identity-scale payment is automatic.

Dihedral and Chebyshev cells. These are quotient cells with an additional involution, usually inversion $u \mapsto u^{-1}$, or with the dihedral symmetry generated by inversion and multiplication by a root of unity. The invariant coordinate is $x = (u + u^{-1})/2$, and the Chebyshev polynomial T_d is characterized by $T_d(x) = (u^d + u^{-d})/2$. On polynomial coordinates these cells are detected by composition with Dickson or Chebyshev maps; in the unnormalized coordinate the Dickson polynomial is characterized by $D_d(u+u^{-1}) = u^d+u^{-d}$. Equivalently they are detected by inversion invariance after a multiplicative change of variables. They must be separated because a naive quotient count may miss the two-to-one or dihedral fiber structure. The precise circle domains are defined in Section 7.

Planted-block cells. A *planted block* is a predetermined group of support positions forced by an algebraically controlled common divisor P. Here P is a polynomial factor common to every support locator Q_S, and its roots in D are the planted positions. Divisibility makes the locus constructible, but payment additionally requires a subexponential census of allowed P, the residual prefix estimate, and its slope projection; arbitrary planted subsets are not one profile.

Tangent, deep, and common-line cells. For a support S, the *Reed–Solomon evaluation variety* means the linear image $\mathrm{ev}_S(\mathbb{A}^k) \subseteq \mathbb{A}^S$. These cells record rank-defective contact between the received line and the corresponding support-restricted evaluation incidence. The deep regime is controlled exactly by Theorem 3.4. General tangent charts require the independent-minor and slope-projection criterion of Proposition 9.6; rank drop alone is not payment. A *common-line cell* is a locus on which several nominal witness charts project to the same received line, while a *tangent cell* is a determinantal locus where the differential of the slope projection has smaller than its expected rank.

Extension and field-descent cells. A witness lies in an extension or field-descent cell if its data is defined over a proper subfield (field descent) or factors only after enlarging the coefficient field (extension). Frobenius invariance is constructible. Its natural subfield profile can be larger than the identity profile, so a direct field-sensitive slope count is required.

Differential-locator cells. The derivative of the locator-prefix map may be expressed by a Vandermonde or Hasse-derivative Jacobian. The Hasse derivatives are defined by $f(X+T) = \sum_{j\geq 0} D^{(j)} f(X)T^j$; unlike division by $j!$, this convention remains valid in small characteristic. A *differential-locator cell* is the locus where that matrix loses its expected rank. It is determinantal, but its rank loss need not equal its codimension. Payment requires independent minors and a slope-projection estimate as in Proposition 9.8.

Saturation and effective-image-collapse cells. A *saturated line ray* is the projective class of the residual one-dimensional explanation direction, represented on the finite affine chart by its slope. Two witnesses are *saturation-equivalent* when they give the same direction after their common codeword, quotient, and planted factors have been removed. Thus *saturation* means passing from raw witnesses to the image of this equivalence relation. MCA counts these rays, not raw supports, and the image–fiber count for the saturation map must replace an uncorrected support count; otherwise split-pencil or balanced-core charts can be exponentially overcounted. The cell is constructible in the projective locator and codeword-ray incidence, where a *codeword ray* is the projective class of a nonzero explaining codeword. *Effective-image collapse* is the related event that a boundary map reaches exponentially fewer boundary values than its ambient codomain contains.

Balanced-core and split-pencil cells. A *common core* is a locator factor shared by every member of a family. A *balanced core* is a pair of equal-degree monic residual locators with a common depth-w prefix after common and planted factors have been removed. A *balanced-core chart* is a parameter family of such cores whose projective dimension exceeds one. In projective dimension one, the residual family is a *split pencil* $Q_0 + \lambda Q_1$ whose members split into roots in the evaluation domain. The *moving-root incidence* of the pencil is defined on the projective parameter line $\mathbb{P}^1(\mathbb{F}) = \mathbb{F} \cup \{\infty\}$ by

$$\{(\lambda, t) \in \mathbb{P}^1 \times D : Q_0(t) + \lambda Q_1(t) = 0, (Q_0(t), Q_1(t)) \neq (0, 0)\}.$$

It is then bounded by Proposition 9.10. Higher-dimensional balanced-core charts require a proved decomposition or a direct ray estimate.

Fourier/Sidon-heavy analytic component. A primitive prefix fiber may be too large while having exponentially small additive energy (few additive quadruples relative to its size), so approximate-group inverse theory cannot handle it. Here that phrase means inverse results that extract a large subset with small sumset or difference set from a set with large additive energy; exponentially small energy gives no such input. The required input is the analytic moment bound of Definitions 12.1 and 12.3. Li–Wan and Weil control its minor characters only when (PF) holds, while the major characters require (MA) or a direct max-fiber theorem. These are additive Fourier characters of the boundary target: minor characters obey the pointwise estimate (PF), and the exceptional major characters are controlled only through the aggregate estimate (MA). This analytic component is deliberately kept separate from the constructible atlas and from the distinct-slope ray compiler.

Remark 5.4. The catalogue is intentionally geometric. A row proof should not merely state that an exceptional family is bounded; it should identify the moduli stratum, the projection being counted, and the entropy or degree reason why the projection is small. This is what makes the ledger compatible with limits.

6. Counting principles for paid cells

The closed-ledger hypothesis is verified in practice by a small collection of counting principles. We record them to make the role of payment precise. The statements are intentionally asymptotic and profile-uniform; finite rows require the same arguments with constants.

Recall that h is the natural-log binary entropy fixed in the Introduction. A map has *generic fiber size* d when all points outside a lower-dimensional exceptional set have exactly d preimages. "Transverse" equations are locally independent equations, so each removes the expected finite-field dimension. A locus has *codimension* h when its local dimension is h below that of the ambient witness chart. The *support divisor* of S is $\sum_{x\in S}[x]$, equivalently the root data encoded by its locator Q_S.

Lemma 6.1 (Entropy loss from a fixed planted block)**.** *Let $P \subseteq D$ have size b, and consider supports $S \subseteq D$ of size a with $P \subseteq S$. If $b = \theta n + o(n)$, then their number is*

$$\binom{n-b}{a-b} = \exp\left(n\Big((1-\theta)h((a/n-\theta)/(1-\theta)) + o(1)\Big)\right).$$

For fixed density a/n, this is exponentially smaller than $\binom{n}{a}$ whenever $\theta > 0$. If $b = o(n)$, the loss is absorbed into the profile factor $e^{o(n)}$.

Proof. This is Stirling's formula applied to $\binom{n-b}{a-b}$. Strict entropy loss for $\theta > 0$ follows from the strict concavity of the binary entropy function and from the fact that requiring a positive-density block removes independent support choices. $\square$

Lemma 6.2 (Quotient entropy principle)**.** *Suppose a quotient cell factors the moving support through a map $\pi : D \to D'$ of generic fiber size $d \geq 2$, and that the support is constant on a positive-density family of quotient fibers. Then the number of such supports is exponentially smaller than $\binom{n}{a}$, unless the number of constrained quotient fibers is $o(n)$. In the latter case the quotient data contributes only $e^{o(n)}$ profiles.*

Proof. On each constrained fiber, a support choice is replaced by a bounded list of invariant choices rather than arbitrary subsets of the fiber. For a positive-density number of fibers this reduces the entropy per fiber by a positive amount, uniformly away from the endpoints. Summing over quotient fibers gives an exponential entropy drop. If only $o(n)$ fibers are constrained, the number of choices for them is at most $C^{o(n)} = e^{o(n)}$ for a bounded constant C. $\square$

Lemma 6.3 (Rank-defect payment criterion)**.** *Let $M(x)$ be a matrix of polynomial functions on a witness chart. The locus* $\operatorname{rank} M \leq r - h$ *is cut out by minors. It has codimension at least h only if one exhibits h independent local equations (or proves the corresponding determinantal transversality), and a numerator payment also requires a bound for its projection to slope space. Under those two additional hypotheses a codimension-h finite-field estimate gives the claimed factor $|\mathbb{B}|^{-h+o(n)}$; generic rank alone gives no such conclusion.*

Proof. The rank locus is defined by the stated minors. If h independent equations cut it transversely, bounded-degree point counting gives at most $|\mathbb{B}|^{\dim X - h + o(n)}$ chart points. The assumed slope-projection bound converts this to a ray estimate. Without independence the loss can be only one even when the rank drops by h (for example $M(x) = xI_h$), and without projection control a point count need not be a distinct-slope bound. $\square$

Lemma 6.4 (One-pencil moving-root payment)**.** *Consider a primitive split-pencil chart in which the residual locator is $Q_\lambda = Q_0 + \lambda Q_1$, $\lambda \in \mathbb{P}^1$, after all common roots have been factored out. Earlier first-match cells have removed common-root, quotient, field-descent, and ray-saturation degeneracies. Assume operationally that $(Q_0(t), Q_1(t)) \neq (0, 0)$ for every moving root $t \in D$. Then any fixed moving root determines at most one parameter λ, except for the common-root locus already removed. Hence a one-pencil chart contributes at most a constant number of primitive slope rays per moving-root profile.*

Proof. For a point $t \in D$, the condition $Q_0(t) + \lambda Q_1(t) = 0$ is linear in λ. If $Q_1(t) \neq 0$, it determines a unique λ. If $Q_1(t) = 0$, then primitivity excludes $Q_0(t) = 0$ unless t belongs to the common-root divisor already paid; otherwise there is no solution. Thus a moving root supplies at most one parameter. Projective normalization adds only the point at infinity, another constant case. $\square$

Lemma 6.5 (Saturation principle)**.** *Let a cell be parameterized by support divisors and explaining polynomials, but let the MCA numerator count only slope rays. If every relevant fiber of the projection from supports to saturated codeword rays has size at least H, then a raw support count must be divided by H, or else the cell must be routed to an earlier ray-saturation cell in the first-match ledger. After saturation, first-match counting is performed in projective ray space and not in the raw support space.*

Proof. MCA counts slopes γ, not witnesses. If many witnesses produce the same slope, counting supports overestimates the numerator. The saturation map identifies witnesses with the same codeword ray or line ray. First-match assignment is then applied to saturated rays. The statement is just the fiber-counting identity $|\mathrm{im}\,\pi| = \sum_y 1 \le H^{-1}|\mathrm{domain}|$ when all fibers have size at least H, with the residual small fibers isolated as a separate constructible cell. □

These principles explain the asymptotic add-back condition in Definition 1.8. Each paid cell either has a genuine exponential loss, or it is a bounded-complexity profile with only subexponentially many choices. The primitive leaf is precisely the part where no such loss or profile explanation remains.

7. Smooth and circle evaluation domains

This section records the domain geometry used to close the ledger. The multiplicative and circle cases are treated simultaneously through uniform-fiber folding maps. The point is not merely that such maps exist, but that their fibers are complete inside the evaluation domain and their locators have a *lacunary* top part, meaning that prescribed gaps occur among the highest-degree coefficients. These two facts make quotient cells visible in the witness moduli and keep the bounded-prefix Fourier calculation stable under planted cores.

Definition 7.1 (Complete-fiber folding map and smooth multiplicative coset)**.** *Let $D \subseteq \mathbb{B}$ be finite. A polynomial map $\varphi : D \to Q := \varphi(D)$ is a* c-fold complete-fiber folding map *if every $y \in Q$ has exactly c preimages in D. A support is* complete at this folding scale *if it is a union of whole fibers, equivalently if it is $\varphi^{-1}(E)$ for some $E \subseteq Q$. Passing from D to Q, and from a complete support to E, is the corresponding* quotient *or* folding.

A multiplicative coset *is a set $D = \theta H \subseteq \mathbb{B}^\times$, where $H \le \mathbb{B}^\times$ is a subgroup and $\theta \in \mathbb{B}^\times$. For every divisor $c \mid |H|$, the restriction*

$$\pi_c : D \longrightarrow D^{(c)}, \qquad \pi_c(x) = x^c, \qquad D^{(c)} = \{x^c : x \in D\},$$

is a c-fold complete-fiber folding map. We call D smooth at scale c *when this power folding is among the quotients retained by the row, and* smooth *for a family of scales when it is smooth at every scale in that family. Thus the adjective* smooth *here is an explicit uniform-fiber property, not a claim about nonsingularity of an algebraic variety. The locator of a complete support is $V_E(X^c)$; its nonleading terms occur only in degree gaps divisible by c. This controlled pattern of missing top coefficients is what* lacunary *means in this section.*

Definition 7.2 (Circle domain and circle code)**.** *Let $\mathbb{B}_0$ have odd characteristic and let $\mathcal{C}/\mathbb{B}_0$ be the affine conic $\mathcal{C} : x^2 + y^2 = 1$. A* standard circle domain *is a finite set $E \subseteq \mathcal{C}(\mathbb{B}_0)$. If $\mathbb{F} \supseteq \mathbb{B}_0$, the degree-$w$* circle code *on E is*

$$\mathcal{C}_w(\mathbb{F}, E) = \{(f(P))_{P\in E} : f \in \mathrm{im}(\mathbb{F}[x, y]_{\le w} \to \mathbb{F}[x, y]/(x^2 + y^2 - 1))\}.$$

The degree is the total-degree filtration before passage to the quotient ring. A circle x-domain *is instead a set of first coordinates obtained from the torus construction below; an*

RS code on that x-domain and a bivariate circle code are different presentations until a diagonal equivalence such as Lemma 7.9 has been exhibited.

Definition 7.3 (Norm-one torus, twin coset, and Chebyshev domain)**.** *Let $p \equiv 3 \pmod 4$, choose $i \in \mathbb{F}_{p^2}$ with $i^2 = -1$, and write*

$$\mathbb{U} = \ker\Big(N_{\mathbb{F}_{p^2}/\mathbb{F}_p} : \mathbb{F}_{p^2}^{\times} \to \mathbb{F}_p^{\times}\Big) = \{u \in \mathbb{F}_{p^2}^{\times} : u^{p+1} = 1\}.$$

The algebraic kernel of the norm morphism is the norm-one torus*; $\mathbb{U}$ is its cyclic group of $\mathbb{F}_p$-rational points. Put $\chi(u) = (u + u^{-1})/2 \in \mathbb{F}_p$. If $H \leq \mathbb{U}$ and $g \in \mathbb{U}$ satisfies $g^2 \notin H$, then*

$$\mathcal{D}(g, H) = gH \sqcup g^{-1}H$$

is a disjoint union exchanged by $u \mapsto u^{-1}$; it is called a twin coset*. Its projected set*

$$D(g, H) = \chi(\mathcal{D}(g, H)) \subseteq \mathbb{F}_p$$

has cardinality $|H|$ and is the Chebyshev twin-coset x-domain*.*

For $c \geq 1$, let T_c be the Chebyshev polynomial normalized by

$$T_c\left(\frac{u + u^{-1}}{2}\right) = \frac{u^c + u^{-c}}{2}.$$

Thus $T_c \circ \chi = \chi \circ (u \mapsto u^c)$. Whenever $c \mid |H|$ and the two image cosets remain disjoint, the restriction of T_c to $D(g, H)$ is the Chebyshev folding map*; a Chebyshev twin-coset domain equipped with these complete-fiber maps is the circle analogue of a smooth multiplicative coset.*

Remark 7.4 (Characteristic two and binary-extension rows)*.* The entropy function h, the locator-prefix bijection, the syndrome–secant identity, the pole-line compiler, and the quotient/remainder normal form are characteristic-free. The use of power sums $\sum_{t \in S} t^j$ is not: Newton's identities, which triangularly relate elementary locator coefficients to power sums, give a change from the first R elementary locator coefficients to the first R power sums is invertible only when $R < \operatorname{char} \mathbb{B}$. In characteristic two the elementary locator prefix therefore remains the canonical exact coordinate system, while any power-sum statement must be proved directly.

A literal Reed–Solomon code over $\mathbb{F}_2$ has length at most two. Thus an asymptotic "binary-field" RS row means a tower $\mathbb{F}_{2^{s_n}}$ (or a scalar extension of such a field), with $s_n \geq \log_2 n$; the regime in this paper permits $s_n = o(n)$. This is distinct from taking a binary subfield subcode $C \cap \mathbb{F}_2^D$ or a trace code obtained by applying the field trace coordinatewise, operations that do not preserve the RS agreement problem used here. The notation H_2 denotes base-two entropy and has no connection with the coefficient-field characteristic. The characteristic-free interleaving comparison of [Jo26] also does not turn an ordinary length-n RS code over $\mathbb{F}_2$ into an asymptotic family.

The standard affine circle does not survive unchanged in characteristic two: $x^2 + y^2 - 1 = (x+y+1)^2$, so it is a doubled line rather than the smooth norm-one conic used in Lemma 7.9 and Proposition 8.15. Smooth characteristic-two norm conics can instead be written in a form such as $x^2 + xy + \delta y^2 = 1$, with the quadratic polynomial chosen irreducible, but transferring the circle ledger to that model requires a separate torus uniformization. Accordingly Theorem 8.11 below is a theorem for smooth multiplicative rows and makes no claim for the standard characteristic-two circle.

Definition 7.5 (Map-smooth domain)**.** *Let $\mathbb{B} \subseteq \mathbb{F}$ be finite fields, let $\varphi \in \mathbb{B}[X]$ have degree $c \geq 1$, and let $D \subseteq \mathbb{B}$ have size n. The set D is (φ, c)-smooth if every nonempty fiber $\varphi^{-1}(y) \cap D$, $y \in \varphi(D)$, has exactly c elements. We write $Q = \varphi(D)$ and $N = n/c$.*

A multiplicative coset $D = \theta H \subseteq \mathbb{B}^\times$ is (X^a, a)-smooth for every divisor $a \mid |H|$. The circle case uses Chebyshev maps rather than power maps; see Lemma 7.8. The following lemma is the common quotient-support construction. It is included because it is the lower-side source of the entropy frontier and because the same lacunarity is used by the upper ledger to identify quotient cells.

Lemma 7.6 (Map-smooth fiber construction)**.** *Let D be (φ, a)-smooth, $|D| = n$, and $C^+ = \mathrm{RS}_{\mathbb{F}}(D, k+1)$. Put $\ell = \lfloor k/a \rfloor + 2$ and $A = a\ell$. If $\ell \le N - 1$, then there is $z \in \mathbb{B}$ and a word $u_z \in \mathbb{B}^D$ such that u_z has at least $\binom{N}{\ell}/|\mathbb{B}|$ distinct codewords of C^+ agreeing with it on at least A positions. Moreover $k + a + 1 \le A \le k + 2a$, with equality $A = k + 2a$ if $a \mid k$.*

Proof. For each ℓ-subset $E \subseteq Q$, let $P_E(Y) = \prod_{b \in E}(Y - b) = Y^\ell - e_1(E)Y^{\ell-1} + R_E(Y)$, with $\deg R_E \le \ell - 2$. The composed locator $P_E(\varphi(X))$ vanishes on the union of the a-point fibers above E, hence on $A = a\ell$ points of D. Set $z_E = -e_1(E)$ and $c_E = -R_E(\varphi)$. Since $\deg c_E \le a(\ell-2) \le k$, the word c_E lies in C^+, and it agrees on that fiber union with $u_{z_E}(x) = \varphi(x)^\ell + z_E\varphi(x)^{\ell-1}$. Among the $\binom{N}{\ell}$ subsets E, some value of $z_E \in \mathbb{B}$ occurs at least $\binom{N}{\ell}/|\mathbb{B}|$ times. If two such subsets with the same z gave the same codeword, then $R_E(\varphi) = R_{E'}(\varphi)$, hence $R_E = R_{E'}$ because composition with a nonconstant polynomial is injective, and then $P_E = P_{E'}$. Thus the codewords are distinct. The bounds on A follow from $\ell = \lfloor k/a \rfloor + 2$. □

Proposition 7.7 (Collision-aware lower cap for map-smooth domains)**.** *Let D be (φ, a)-smooth, $n < |\mathbb{F}| = q$, and let $C = \mathrm{RS}_{\mathbb{F}}(D, k)$. With ℓ, A as in Lemma 7.6, assume $\ell \le N - 1$, and put*

$$L = \left\lceil \binom{N}{\ell} |\mathbb{B}|^{-1} \right\rceil.$$

Then for every integer threshold $k + 1 \le m \le A$,

$$B_C^{\mathrm{MCA}}(m) \ge \left\lceil \frac{L(q-n)}{q - n + k(L-1)} \right\rceil.$$

Proof. The list in Lemma 7.6 has size at least L at agreement A. Apply the collision-aware pole conversion Theorem 4.2. Its line pair is not simultaneously explained on any $k+1$ points, so an agreement set of size A also witnesses badness for every threshold m in the displayed range. □

We now verify the complete-fiber property for the construction in Definition 7.3. Thus $\mathcal{D} = gH \sqcup g^{-1}H \subseteq \mathbb{U}$, $D = \chi(\mathcal{D})$, and $T_a(\chi(u)) = \chi(u^a)$.

Lemma 7.8 (Chebyshev smoothness)**.** *Let $\mathcal{D} = gH \cup g^{-1}H \subseteq \mathbb{U}$ be a twin coset, let $D = \chi(\mathcal{D})$, and let $a \mid |H|$. If $g^{2a} \notin H^{(a)} = \{h^a : h \in H\}$, then D is (T_a, a)-smooth. If $\mathrm{ord}(g) \nmid 2|H|$, this holds for all $a \mid |H|$.*

Proof. The map $u \mapsto u^a$ has kernel of size a on $\mathbb{U}$, and that kernel lies in H. Hence it maps gH and $g^{-1}H$ onto $g^aH^{(a)}$ and $g^{-a}H^{(a)}$, respectively, with fibers of size a. The hypothesis says these image cosets are disjoint, so the image is again a twin coset. Since χ is exactly two-to-one on both twin cosets and identifies only inverse points, the equation $T_a(x) = y$ on D has exactly a solutions for each $y \in T_a(D)$. Taking $a = |H|$ shows that the all-scale condition is equivalent to $g^{2|H|} \ne 1$, and the implication to every smaller divisor follows by raising to $|H|/a$. □

Lemma 7.9 (Diagonal uniformization of circle codes)**.** *Assume $\mathbb{F} \supseteq \mathbb{F}_{p^2}$. Let $E \subseteq \{(x, y) \in \mathbb{F}_p^2 : x^2 + y^2 = 1\}$, define $\iota(x, y) = x + iy$, put $E' = \iota(E) \subseteq \mathbb{U}$, and let $\mathcal{C}_w(\mathbb{F}, E)$*

be the circle code of Definition 7.2. Under the coordinate bijection ι and the diagonal multiplier $t_u = u^{-w}$, the code $\mathcal{C}_w(\mathbb{F}, E)$ is $t \circ \mathrm{RS}_{\mathbb{F}}(E', 2w+1)$. Consequently list sizes, ordinary correlated-agreement (CA) numerators, and mutual-correlated-agreement (MCA) numerators are preserved.

Proof. The map $u = x + iy$ identifies $\mathbb{F}[x, y]/(x^2 + y^2 - 1)$ with $\mathbb{F}[u, u^{-1}]$. The subspace of total degree at most w maps to $u^{-w}\mathbb{F}[u]_{\le 2w}$, and both spaces have dimension $2w+1$. Multiplication by the nonzero diagonal vector $(u^{-w})_{u \in E'}$ preserves coordinatewise equality and hence every agreement notion used here. Therefore it preserves list, CA, and MCA counts exactly. $\square$

Definition 7.10 (Artin–Schreier phase)**.** *Let $\mathbb{B} = \mathbb{F}_{p^s}$, let $X/\mathbb{B}$ be the affine line or the norm-one torus, and write a nontrivial additive character as $\psi_a(t) = \exp(2\pi i\, \mathrm{Tr}_{\mathbb{B}/\mathbb{F}_p}(at)/p)$ with $a \in \mathbb{B}^\times$. A rational phase $R \in \mathbb{B}(X)$ is* Artin–Schreier degenerate for ψ_a *if*

$$aR = G^p - G + c$$

for some $G \in \mathbb{B}(X)$ and $c \in \mathbb{B}$. Away from the poles, $\psi_a(R)$ then has constant trace phase, so the square-root cancellation asserted by the ordinary Weil estimate is unavailable. A phase not admitting such a representation is called non-Artin–Schreier*. This qualification is always relative to the chosen additive character.*

The Fourier/Sidon payment uses bounded-prefix flatness as a sufficient minor-arc estimate, but the ledger also records low-energy heavy fibers as a separate first-match cell. We state here the smooth and circle flatness forms and then prove the Sidon-resolved payment. The proof and the general weighted statement are given in Sections 14 and 24; the Sidon-resolved payment is Theorem 14.8. For a full profile slice write

$$\Omega^0 = \Omega_{T,m} = \{x \in \{0,1\}^T : \sum_{t \in T} x_t = m\}, \qquad \overline{N}_0 = |\mathbb{B}|^{-R}\binom{|T|}{m}.$$

Here $\overline{N}_0$ is the ambient profile scale. The theorems below prove an ambient max-fiber bound, which is stronger than image-normalized Q and, by Remark 10.2, simultaneously certifies that the realized image has full exponential size. The prefix functions in a primitive chart are weighted rational functions $g_1, \dots, g_R$ on T, and the full profile map is $\Psi(x) = \sum_{t \in T} x_t(g_1(t), \dots, g_R(t))$. For $\alpha \in \mathbb{B}^R$, put $R_\alpha(t) = \alpha \cdot (g_1(t), \dots, g_R(t))$. A nonzero α is *major* when this phase is constant on a positive-density quotient fiber, is Artin–Schreier degenerate, has a pole or zero on a positive-density moving divisor, or factors through a routed quotient, Chebyshev, field-descent, tangent rank-loss, differential-locator, or ray-saturation profile; every other nonzero character is *minor*. Write $\mathfrak{M}$ for the set of major characters. Finally,

$$e_m(a_t : t \in T) = \sum_{S \in \binom{T}{m}} \prod_{t \in S} a_t$$

is the m-th elementary symmetric sum. Naming the corresponding primal stratum does not remove these characters from Fourier inversion on the full slice, so their aggregate is a separate input.

Definition 7.11 (Major-arc aggregate)**.** *Let $\mathfrak{M} \subseteq \mathbb{B}^R \setminus \{0\}$ be the major characters of a full profile slice and let ψ be a nontrivial additive character of $\mathbb{B}$. The chart satisfies* (MA) *if*

$$|\mathbb{B}|^{-R} \sum_{\alpha \in \mathfrak{M}} \left| e_m\Big(\psi(\alpha \cdot g(t)) : t \in T\Big) \right| \le e^{o(|T|)}\overline{N}_0. \tag{MA}$$

This is vacuous when $\mathfrak{M} = \varnothing$. It can be proved by a recursive quotient census or replaced by a direct full-slice max-fiber bound; first-match terminology alone does not imply it.

Definition 7.12 (Pointwise minor-arc range)**.** *A primitive weighted prefix chart satisfies the pointwise minor-arc range if the characteristic is larger than m, all minor-arc rational phases have total pole-degree at most Δ_{pole}, and, with*

$$\Lambda = C_0(\Delta_{\mathrm{pole}} + 1)\sqrt{|\mathbb{B}|} + |P|$$

for the absolute Weil constant C_0 and planted exceptional set P, one has

$$R\log|\mathbb{B}| + \log\binom{\Lambda + m - 1}{m} - \log\binom{|T|}{m} \le o(|T|). \tag{PF}$$

For circle twin-cosets the same definition is used with $2C_0(\Delta_{\mathrm{pole}} + 1)\sqrt{|\mathbb{B}|}$ in place of $C_0(\Delta_{\mathrm{pole}} + 1)\sqrt{|\mathbb{B}|}$, accounting for the two inverse-coset lifts on the torus before the involution quotient.

Theorem 7.13 (Bounded-prefix flatness on smooth cosets)**.** *Let T be a multiplicative coset or a coset with $o(|T|)$ planted deletions over $\mathbb{B}$. Let $g_1, \dots, g_R$ be the weighted rational prefix functions of a primitive smooth-domain chart. If the chart satisfies the pointwise minor-arc range and* (MA)*, then for every target $z \in \mathbb{B}^R$,*

$$|\{x \in \Omega_{T,m} : \Psi(x) = z\}| \le e^{o(|T|)}\overline{N}_0.$$

The same bound holds for every first-match residual $\Omega^\circ \subseteq \Omega_{T,m}$.

Proof. For a minor character α, the subgroup indicator expands as an average of multiplicative characters. Weil's bound for a nonconstant rational phase gives $|\sum_{t\in T}\psi(jR_\alpha(t))| \le C_0(\Delta_{\mathrm{pole}} + 1)\sqrt{|\mathbb{B}|} + |P| = \Lambda$ for $1 \le j \le m$. The cycle-index estimate—the generating-function count of distinct-coordinate tuples grouped by permutation cycle lengths—used in Theorem 14.1 bounds the complete minor-character contribution by $\binom{\Lambda+m-1}{m}$. The zero character contributes $\overline{N}_0$, and (MA) controls the omitted major characters. Condition (PF) makes the minor contribution at most $e^{o(|T|)}\overline{N}_0$. Passing to a first-match residual can only decrease fibers. $\square$

Theorem 7.14 (Bounded-prefix flatness on circle domains)**.** *Let $T = \chi(gH \cup g^{-1}H)$ be a circle twin-coset x-domain, with the branch points $u = u^{-1}$ of χ and any repeated projected coordinates already isolated as planted exceptions. For the primitive weighted prefix functions on a circle or diagonally uniformized circle-code chart, assume the circle version of the pointwise minor-arc range and* (MA)*. Then every fiber $\Psi^{-1}(z)$ of the full profile slice and every first-match residual fiber is bounded by $e^{o(|T|)}\overline{N}_0$.*

Proof. Pull the phase back by $x = \chi(u) = (u + u^{-1})/2$ to the norm-one torus. A minor arc becomes a nonconstant rational function on the genus-zero torus, not of Artin–Schreier form. Weil's bound on the two torus branches gives the same estimate as above with $2C_0(\Delta_{\mathrm{pole}} + 1)\sqrt{|\mathbb{B}|}$. The rest is identical to the proof of Theorem 7.13. If the pullback is constant on a power fiber, the witness lies in a Chebyshev quotient cell by first match. If it also collapses under $u \mapsto u^{-1}$, it lies in the corresponding *dihedral quotient cell*, meaning the quotient by multiplication by roots of unity ζ satisfying $\zeta^c = 1$ for the relevant folding degree c, together with inversion. $\square$

Corollary 7.15 (Sidon moment bound from full Fourier control)**.** *On every primitive smooth or circle chart satisfying the pointwise minor-arc estimate* (PF) *and* (MA)*, the Fourier/Sidon-heavy cell has image-normalized logarithmic moment $e^{o(nq)}$ for every logarithmic $q \le (\log n)^A$.*

Proof. The full-profile max-fiber theorem gives the ambient estimate $f_s \le e^{o(N)}\overline{N}_0$ for every residual fiber. It therefore certifies (FI), and Lemma 10.1 gives the image-normalized Sidon moment. The estimate (MA), not the name of a first-match cell, controls the major-character contribution. □

8. Exact quotient–remainder profiles and the field-drop obstruction

The quotient and remainder variables can be separated exactly at the locator-prefix level. This removes a spurious factor coming from summing over marked remainder sets, but it also exposes a genuine scaled quotient- coefficient-field term which prevents an unrestricted comparison with the identity profile.

Definition 8.1 (Quotient/remainder profile and field drop). *Let $\phi : D \to Q$ be a c-fold complete-fiber folding map in the sense of Definition 7.1. A* complete-fiber-plus-remainder support *is*

$$S = \phi^{-1}(E) \sqcup R,$$

where $E \subseteq Q$ and $R \subseteq D \setminus \phi^{-1}(E)$. The set R is the support remainder*: it records the selected points in fibers that are not declared complete. A* quotient/remainder profile *fixes ϕ, its degree c, the size $r = |R|$, the relevant coefficient field, and any planted remainder data, while E and the unfixed part of R range subject to the displayed disjointness. A fixed-R profile is the subprofile in which the remainder support itself is part of the label. The adjective* Euclidean *used below refers to separating the locator factor of degree $r < c$ from the composed quotient locator; it does not mean that an arbitrary support remainder has size less than c.*

We call $\mathbb{B}_\phi \subseteq \mathbb{B}$ a scaled quotient coefficient field *when, for some $\eta \in \mathbb{B}^\times$,*

$$Q = \phi(D) \subseteq \eta\mathbb{B}_\phi.$$

When this field is recorded in a profile, it is taken minimal among the subfields with this property. Then the j-th elementary quotient coefficient lies in $\eta^j\mathbb{B}_\phi$. A field drop *is the use of a proper such subfield. It is a* positive-rate field drop *along a sequence when*

$$1 - \frac{\log|\mathbb{B}_\phi|}{\log|\mathbb{B}|}$$

stays bounded away from zero and a positive-density number on the entropy scale of quotient-prefix slots is live. The payment saved in each live slot is the factor $|\mathbb{B}|/|\mathbb{B}_\phi|$; this is the term quantified in Proposition 8.7.

Definition 8.2 (Balanced quotient core). *A* split locator *is a monic squarefree polynomial all of whose roots lie in the evaluation domain D. After common locator factors, fixed quotient data, and planted remainder points have been removed, let A and B be two monic split locators of the same degree e. They form a* depth-w balanced core *if they have the same first w nonleading coefficients, equivalently*

$$\deg(A - B) \le e - w - 1.$$

It is a balanced quotient core *when the remaining locator factors are pullbacks through the same folding map ϕ. The word* balanced *records equality of degree and leading normalization; the* core *consists of the roots still allowed to move after common and planted factors are removed. If the resulting projective locator family is one-dimensional it is a split pencil $Q_0 + \lambda Q_1$; in higher dimension this definition supplies no distinct-ray bound, and the separate ray compiler is required.*

For a monic polynomial

$$A(X) = X^d + a_1 X^{d-1} + \cdots + a_d$$

and an integer $0 \le u \le d$, write $\mathrm{pref}_u(A) = (a_1, \ldots, a_u)$. Equivalently, if $A^\vee(Z) := Z^d A(Z^{-1})$, then $\mathrm{pref}_u(A)$ is the coefficient vector of $A^\vee(Z) \bmod Z^{u+1}$, with its constant coefficient omitted.

Theorem 8.3 (Exact quotient–remainder prefix normal form)**.** *Let $\mathbb{B} \subseteq \mathbb{F}$ be finite fields, let $D \subseteq \mathbb{B}$ have n distinct elements, and let $\phi \in \mathbb{B}[X]$ be monic of degree $c \ge 2$. Assume that the restriction*

$$\phi : D \longrightarrow Q := \phi(D)$$

has complete fibers of size c; in particular, $N := |Q| = n/c$. Fix integers m, r with $0 \le r < c$ and $cm + r \le n$. For

$$E \in \binom{Q}{m}, \qquad R \in \binom{D \setminus \phi^{-1}(E)}{r},$$

put

$$\begin{aligned} V_E(Y) &:= \prod_{y \in E}(Y - y) = Y^m + \sum_{j=1}^{m} v_j(E) Y^{m-j}, \\ P_R(X) &:= \prod_{x \in R}(X - x) = X^r + \sum_{i=1}^{r} p_i(R) X^{r-i}, \\ L_{E,R}(X) &:= P_R(X) V_E(\phi(X)). \end{aligned} \tag{QR1}$$

Then $L_{E,R}$ is the monic locator of the support $\phi^{-1}(E) \sqcup R$, of degree $A = cm + r$.

Let $0 \le w \le A$ and put $d = \lfloor w/c \rfloor$.

(i) *If $w \ge r$, every nonempty fiber of $(E, R) \mapsto \mathrm{pref}_w(L_{E,R})$ determines R uniquely and, after R is recovered, is exactly one quotient-prefix fiber. More precisely, for every prefix value z there are unique data R_z and $\eta_z \in \mathbb{B}^d$, whenever the fiber is nonempty, such that*

$$\begin{aligned} &\{(E, R) : \mathrm{pref}_w(L_{E,R}) = z\} \\ &\quad = \{(E, R_z) : E \in \tbinom{Q \setminus \phi(R_z)}{m},\ \mathrm{pref}_d(V_E) = \eta_z\}. \end{aligned} \tag{QR2}$$

Thus no factor counting possible remainder sets occurs in a fixed global prefix fiber.

(ii) *If $w < r$, then necessarily $w < c$, and the quotient set E is invisible to the first w coefficients. There is a fixed unitriangular bijection $T_{\phi,m,w} : \mathbb{B}^w \to \mathbb{B}^w$ such that*

$$\mathrm{pref}_w(L_{E,R}) = T_{\phi,m,w}(\mathrm{pref}_w(P_R)) \tag{QR3}$$

for every admissible (E, R). Consequently, with the convention $\binom{u}{v} = 0$ for $v > u$,

$$\begin{aligned} &\#\{(E, R) : \mathrm{pref}_w(L_{E,R}) = z\} \\ &\quad = \sum_{\substack{R \in \binom{D}{r} \\ \mathrm{pref}_w(P_R) = T^{-1}_{\phi,m,w}(z)}} \binom{N - |\phi(R)|}{m}. \end{aligned} \tag{QR4}$$

Thus this case is a remainder-prefix problem, multiplied by the number of quotient sets still available after R is fixed.

Proof. For a monic degree-e polynomial A, set $A^\vee(Z) = Z^e A(Z^{-1})$. Also set

$$\Phi(Z) := Z^c \phi(Z^{-1}) = 1 + \phi_1 Z + \cdots + \phi_c Z^c.$$

Taking reciprocal polynomials in (QR1) gives the exact formal-power-series identity

$$L^{\vee}_{E,R}(Z) = P^{\vee}_{R}(Z)\left(\Phi(Z)^m + \sum_{j=1}^{m} v_j(E)Z^{cj}\Phi(Z)^{m-j}\right). \tag{QR5}$$

All factors on the right have constant coefficient one except for the displayed scalar coefficients $v_j(E)$.

Suppose first that $w \geq r$. Since $r < c$, no term containing a v_j occurs below degree c. For $1 \leq i \leq r$, the coefficient of Z^i on the right of (QR5) is $p_i(R)$ plus an expression depending only on $p_1(R), \dots, p_{i-1}(R)$, on m, and on Φ. The first r coefficients therefore recover $p_1(R), \dots, p_r(R)$ successively. They recover the monic polynomial P_R, hence its root set R, uniquely.

We may now divide $L^{\vee}_{E,R}$ by the unit $P^{\vee}_R$ modulo Z^{w+1}. In the remaining factor of (QR5), the coefficient of Z^{cj} is $v_j(E)$ plus an expression depending only on $v_1(E), \dots, v_{j-1}(E)$, on m, and on Φ. Hence the coefficients at degrees $c, 2c, \dots, dc$ recover $v_1(E), \dots, v_d(E)$ successively. Conversely, P_R and these d coefficients determine the right side of (QR5) modulo Z^{w+1}, because every term containing v_j with $j > d$ starts in degree $cj > w$. This proves (QR2); admissibility of E is exactly the condition $E \cap \phi(R) = \varnothing$.

If $w < r < c$, every term involving a v_j starts in degree at least $c > w$, so modulo Z^{w+1} the bracket in (QR5) is the fixed unit $\Phi(Z)^m$. Multiplication by this unit sends the first w coefficients of $P^{\vee}_R$ to the first w coefficients of $L^{\vee}_{E,R}$ by a unitriangular transformation. It is independent of E and invertible. For a fixed R, the admissible E are precisely the m-subsets of $Q \setminus \phi(R)$, of which there are $\binom{N-|\phi(R)|}{m}$. Summing over the indicated remainder-prefix fiber proves (QR4). □

Proposition 8.4 (Complete support factorization and its limitation)**.** *Under the complete-fiber hypotheses of Theorem 8.3, every support $S \subseteq D$ has the unique decomposition*

$$E(S) = \{y \in Q : \phi^{-1}(y) \subseteq S\}, \qquad R(S) = S \setminus \phi^{-1}(E(S)), \tag{QR5a}$$

and $Q_S(X) = P_{R(S)}(X)V_{E(S)}(\phi(X))$. The remainder meets every nonfull ϕ-fiber in at most $c-1$ points, but its total size may be $\Theta(n)$. The reciprocal identity (QR5) remains valid for this arbitrary remainder. When $|R(S)| \geq c$, however, the quotient coefficients and the remainder coefficients begin to interlace at degree c, so the triangular recovery theorem above does not give a comparison without an additional partial-occupancy atlas.

Proof. A fiber is put into $E(S)$ exactly when all of its points lie in S; all other selected points form $R(S)$. This proves existence, disjointness, the occupancy bound, and uniqueness. Multiplying the corresponding linear factors gives the locator identity. The reciprocal calculation used for (QR5) is purely multiplicative and does not require $\deg P_R < c$. If $\deg P_R \geq c$, both $p_c(R)$ and $v_1(E)$ occur in degree c, which proves the stated limitation of the triangular separation. □

Remark 8.5 (Exact dimension rounding). If the global locator degree is $A = cm + r$, the list-code dimension is K, and $w = A - K$, then the quotient depth in Theorem 8.3 is

$$d = \left\lfloor \frac{A-K}{c} \right\rfloor = m - \left\lceil \frac{K-r}{c} \right\rceil. \tag{QR5b}$$

Thus the descended list dimension is exactly $K_c = \lceil (K-r)/c \rceil$. The floor/ceiling discrepancy costs at most one factor $|\mathbb{B}_\phi|$ when the quotient coefficients lie in a field $\mathbb{B}_\phi$, namely $O(\log |\mathbb{B}_\phi|)$ bits, and is therefore $o(n)$ under the field-size hypothesis used below. It must nevertheless be retained in a finite adjacent-row certificate.

Remark 8.6 (Nonmonic maps and Chebyshev maps)*.* If the leading coefficient of ϕ is $b \neq 0$, replace ϕ by $b^{-1}\phi$ and Q by $b^{-1}Q$. This changes quotient coefficients by fixed nonzero scalar factors and leaves every support and every fiber cardinality unchanged. The theorem therefore applies to the Chebyshev folding map T_c after this normalization. On a circle twin-coset domain on which T_c has complete fibers of size c, it gives the same statement verbatim. The fixed points of the involution, coincident twin cosets, and ramified endpoints must be removed or recorded as a bounded planted set; for fixed c they change all logarithmic estimates below by only $o(n)$.

For a declared complete-fiber-plus-Euclidean-remainder profile, the exact theorem separates algebraic bookkeeping from prefix equidistribution; it does not supply the partial-occupancy exhaustion isolated in Proposition 8.4. Let $\mathbb{B}_\phi \subseteq \mathbb{B}$ be a subfield and suppose $Q \subseteq \eta\mathbb{B}_\phi$ for some $\eta \in \mathbb{B}^\times$. Then $v_j(E) \in \eta^j\mathbb{B}_\phi$. Hence, in case (i), the natural average scale of the quotient-prefix fiber with fixed R is

$$\overline{N}_{\phi,R}(w) := \binom{N - |\phi(R)|}{m} |\mathbb{B}_\phi|^{-\lfloor w/c \rfloor}. \tag{QR6}$$

Pigeonholing proves a fiber of size at least $\lceil \overline{N}_{\phi,R}(w) \rceil$. An upper bound of the same order is not a consequence of the normal form: it is precisely the uniform prefix-flatness, or Sidon/Fourier, input on the quotient row.

Proposition 8.7 (Identity–quotient exponent comparison)**.** *Consider a sequence to which Theorem 8.3 applies with fixed $c \geq 2$ and fixed $0 \leq r < c$. Suppose*

$$\frac{a}{n} \longrightarrow \alpha \in (0,1), \qquad \log|\mathbb{B}| = o(n), \qquad \log\binom{n}{a} - w\log|\mathbb{B}| = o(n), \tag{QR7}$$

where $a = cm + r$, and suppose $Q \subseteq \eta\mathbb{B}_c$ for a subfield $\mathbb{B}_c \subseteq \mathbb{B}$. Write $\overline{N}_{c,r}(w) = \overline{N}_{\phi,R}(w)$ for the natural scale in (QR6), with this fixed remainder profile. Then $w \to \infty$, so $w \geq r$ eventually, and

$$\log \overline{N}_{c,r}(w) = \frac{1}{c}\log\left(\binom{n}{a}|\mathbb{B}|^{-w}\right) + \frac{w}{c}\log\frac{|\mathbb{B}|}{|\mathbb{B}_c|} + o(n), \tag{QR8}$$

where replacing $N - |\phi(R)|$ by N changes only the $o(n)$ term. Equivalently, if $\lambda_c = \log|\mathbb{B}_c|/\log|\mathbb{B}|$ has a limit, then at the identity crossing

$$\frac{1}{n}\log \overline{N}_{c,r}(w) = \frac{h(\alpha)}{c}(1 - \lambda_c) + o(1), \tag{QR9}$$

using the entropy convention fixed in the Introduction. Thus the natural average scale of this bounded-degree quotient profile is comparable with the identity scale at exponential accuracy exactly when $\log|\mathbb{B}_c|/\log|\mathbb{B}| = 1 - o(1)$. A proper field drop of fixed relative degree produces a positive exponential quotient term.

Proof. Stirling's formula, $a/n \to \alpha$, and fixed c, r give

$$\log\binom{n/c + O(1)}{(a-r)/c} = \frac{1}{c}\log\binom{n}{a} + o(n).$$

Also $\lfloor w/c \rfloor \log|\mathbb{B}_c| = (w/c)\log|\mathbb{B}_c| + o(n)$, because the rounding error is at most $\log|\mathbb{B}_c| \leq \log|\mathbb{B}| = o(n)$. This proves (QR8). Since $\log\binom{n}{a} = nh(\alpha) + o(n)$, (QR7) gives $w\log|\mathbb{B}| = nh(\alpha) + o(n)$. In particular $w \to \infty$ because $\log|\mathbb{B}| = o(n)$. Substitution in (QR8) proves (QR9). $\square$

Remark 8.8 (Unbounded quotient degrees)*.* If $c = c_n \to \infty$, the complete-fiber choice contributes at most $2^{n/c} = e^{o(n)}$. When $w \geq r$, the remainder is recovered exactly and no further remainder multiplicity occurs. When $w < r < c$, the exact formula (QR4) reduces the profile to an ordinary prefix problem for the remainder support, multiplied by $e^{o(n)}$ quotient choices. Thus only bounded quotient degrees can create a new positive-rate field-drop term; large degrees either cost $e^{o(n)}$ or return to an identity/remainder prefix problem. This does not prove flatness of the remaining prefix problem.

We now show that the field-drop term is a real Sidon-heavy obstruction in the polynomial-field regime defined in Section 4, rather than an artifact of the comparison. A *complete square-fiber support* is a pullback $\{x \in D : x^2 \in E\}$ under the two-fold power map; the analogous complete cube-fiber support is defined using $x \mapsto x^3$. We identify a support with its incidence vector in $\{0,1\}^D$. Recall that for a support family $\mathcal{A}$, its additive energy and normalized energy are

$$E(\mathcal{A}) = |\{(A_1, A_2, A_3, A_4) \in \mathcal{A}^4 : A_1 - A_2 = A_3 - A_4\}|, \qquad \Delta(\mathcal{A}) = \frac{E(\mathcal{A})}{|\mathcal{A}|^3}.$$

At the subexponential identity scales used below, an exponentially large prefix fiber with $\Delta \leq e^{-\sigma n}$ is called a *Sidon-heavy obstructing fiber*: it contributes to the previously defined Sidon-heavy branch by carrying large ordinary moment mass while having exponentially little additive closure.

Theorem 8.9 (Polynomial-field Sidon and MCA obstruction)**.** *Fix $0 < \alpha < 1/2$. Let p tend to infinity through odd primes and put*

$$n = 2(p-1), \qquad \mathbb{B}_0 = \mathbb{F}_p, \qquad \mathbb{B} = \mathbb{F}_{p^2}. \tag{6.1}$$

Let $H \leq \mathbb{B}^\times$ be the subgroup of order n, choose a generator θ of $\mathbb{B}^\times$, and put $D = \theta H$. Choose even integers $a = a_p$ with $a/n \to \alpha$, let w be the largest even integer not exceeding $\log_{|\mathbb{B}|} \binom{n}{a}$, and put $k = a - w - 1$. Then

$$\log|\mathbb{B}| = O(\log n) = o(n), \qquad \operatorname{char} \mathbb{B} = p > a, \qquad w = \Theta(n/\log n), \tag{6.2}$$

and the identity scale satisfies

$$1 \leq \overline{N}_1 = \binom{n}{a} |\mathbb{B}|^{-w} < |\mathbb{B}|^2 = e^{o(n)}. \tag{6.3}$$

For $E \in \binom{D^2}{a/2}$, put $S_E = \{x \in D : x^2 \in E\}$. There is a depth-$w$ prefix z for which the complete-square-fiber subfamily

$$\mathcal{Q}_z = \{S_E : E \in \tbinom{D^2}{a/2},\ \Phi_w(S_E) = z\} \subseteq \mathcal{F}_z := \Phi_w^{-1}(z),$$

and the full prefix fiber obeys

$$|\mathcal{F}_z| \geq |\mathcal{Q}_z| \geq \binom{n/2}{a/2} p^{-w/2} = \exp\left(\left(\frac{1}{4}h(\alpha) + o(1)\right)n\right), \tag{6.4}$$

with h as fixed in the Introduction. For the complete-square profile $\Omega_{\mathrm{sq}} = \{S_E : E \in \binom{D^2}{a/2}\}$, one also has

$$L_{\mathrm{sq}} := |\Phi_w(\Omega_{\mathrm{sq}})| \leq p^{w/2}, \qquad \frac{|\Omega_{\mathrm{sq}}|}{L_{\mathrm{sq}}} \geq \exp\left(\left(\frac{1}{4}h(\alpha) + o(1)\right)n\right). \tag{6.4$'$}$$

Thus the square quotient term is exponentially large in image normalization as well.

Moreover there is a constant $\sigma = \sigma(\alpha) > 0$ such that

$$\max\{\Delta(\mathcal{F}_z), \Delta(\mathcal{Q}_z)\} \leq e^{-\sigma n}. \tag{6.5}$$

For the full a-subset slice define the ambient identity-normalized moment

$$\Gamma^{\mathrm{sid,amb}}_{r,\sigma} = |\mathbb{B}|^{-w} \sum_{\substack{z'\in\mathbb{B}^w \\ \Delta(\mathcal{F}_{z'})\le e^{-\sigma n}}} \left(\frac{|\mathcal{F}_{z'}|}{\overline{N}_1}\right)^r .$$

For every logarithmic moment order $r_n \to \infty$,

$$\liminf_{n\to\infty} \frac{1}{nr_n} \log \Gamma^{\mathrm{sid,amb}}_{r_n,\sigma} \ge \frac{1}{4}h(\alpha) > 0. \tag{6.6}$$

Thus no ambient identity-normalized Sidon payment can hold on an unrestricted deep prefix before quotient, subfield, planted, and remainder profiles have been routed, even when $\log|\mathbb{B}| = o(n)$.

Finally, every scalar extension $\mathbb{F}/\mathbb{B}$ satisfying

$$|\mathbb{F}| - n > k\binom{|\mathcal{F}_z|}{2} \tag{6.7}$$

has a received line for $C = \mathrm{RS}_{\mathbb{F}}(D,k)$ with at least

$$B^{\mathrm{MCA}}_C(a) \ge |\mathcal{F}_z| \ge \exp\left(\left(\frac{1}{4}h(\alpha) + o(1)\right)n\right) \tag{6.8}$$

distinct bad slopes. Such extensions exist with degree $[\mathbb{F}:\mathbb{B}] = O(n/\log n)$.

Proof. The cyclic group $\mathbb{B}^\times$ has order $(p-1)(p+1)$, divisible by $n = 2(p-1)$. Squaring H gives its subgroup of order $p-1$, which is the unique such subgroup and hence equals $\mathbb{B}_0^\times$. Therefore

$$D^2 = \theta^2\mathbb{B}_0^\times, \tag{6.9}$$

and the square map $D \to D^2$ is exactly two-to-one. The two roots of $1+X^2$ have order four and lie in H, while $D = \theta H \ne H$; thus D avoids the stereographic poles. Since $a/n \to \alpha < 1/2$, one has $a < p$ eventually. Stirling's formula and the definition of w give $w = nh(\alpha)/(2\log p) + o(n/\log n)$; the even rounding leaves a remainder in $[0,2)$, proving (6.2) and (6.3).

For $E \in \binom{D^2}{a/2}$, let S_E be its full square preimage. Then

$$Q_{S_E}(X) = \prod_{y\in E}(X^2 - y) = X^a - e_1(E)X^{a-2} + e_2(E)X^{a-4} - \cdots . \tag{6.10}$$

Among the first w nonleading coefficients, the odd ones vanish and the coefficient in gap $2j$ lies in $\theta^{2j}\mathbb{B}_0$. Thus at most $p^{w/2}$ prefixes occur. Pigeonholing the $\binom{n/2}{a/2}$ supports gives a value z with $|\mathcal{Q}_z| \ge \binom{n/2}{a/2}p^{-w/2}$. Since $\mathcal{Q}_z \subseteq \mathcal{F}_z$, this proves (6.4). Also $w\log p = \frac{1}{2}nh(\alpha) + o(n)$, so Stirling gives the stated exponent.

By Proposition 17.1, there is an absolute $A > 0$ such that

$$E(F) \ge |F|^3/K, \quad F \subseteq \{0,1\}^D \quad \Longrightarrow \quad |F| \le K^A. \tag{6.11}$$

Choose $0 < \sigma < h(\alpha)/(8A)$. If either $\mathcal{F}_z$ or $\mathcal{Q}_z$ had normalized energy greater than $e^{-\sigma n}$, apply (6.11) to that fiber with $K = e^{\sigma n}$. It gives $|F| \le e^{A\sigma n}$, contradicting (6.4) for large n. This proves (6.5).

The formal prefix target space $\mathbb{B}^w$ has size $|\mathbb{B}|^w$. Retaining only the summand of $\mathcal{F}_z$ in the Sidon-heavy moment and using $\log \overline{N}_1 = o(n)$ gives

$$\log \Gamma^{\mathrm{sid,amb}}_{r_n,\sigma} \ge -w\log|\mathbb{B}| + r_n\left(\log|\mathcal{F}_z| - \log\overline{N}_1\right) = -nh(\alpha) + o(n) + r_n\left(\frac{1}{4}h(\alpha)n + o(n)\right).$$

Division by nr_n proves (6.6). Here the moment really is the full unrestricted prefix moment: because $w < a < p$, Newton's identities give an invertible triangular change of coordinates between the first w locator coefficients and the first w power sums. Hence their fibers, and therefore their normalized Sidon moments, coincide.

Finally *Theorem* 4.5, with $m = a$, applies directly to $\mathcal{F}_z$ and gives (6.8). Since $\log|\mathcal{F}_z| = O(n)$ and $\log|\mathbb{B}| = \Theta(\log n)$, an extension satisfying (6.7) can be chosen of degree $O(n/\log n)$. $\square$

Remark 8.10 (Normalization of the square obstruction). Equation (6.6) is an ambient identity-normalized statement and does not use an unproved reverse ambient-to-image transfer. The corresponding image-normalized profile-envelope obstruction is the independent estimate (6.4′); the locator-prefix fiber itself supplies the MCA lower construction.

Theorem 8.11 (Characteristic-two cubic quotient obstruction). *Fix $0 < \alpha < 1/2$. Let $s \to \infty$ through odd integers, put*

$$q = 2^s, \qquad \mathbb{B}_0 = \mathbb{F}_q, \qquad \mathbb{B} = \mathbb{F}_{q^2}, \qquad n = 3(q-1), \tag{6.11a}$$

and let $H \le \mathbb{B}^\times$ be the subgroup of order n. Choose a generator θ of $\mathbb{B}^\times$ and put $D = \theta H$. Choose multiples of three $a = a_s$ with $a/n \to \alpha$, let w be the largest multiple of three not exceeding $\log_{|\mathbb{B}|}\binom{n}{a}$, and put $k = a - w - 1$. Then

(6.11b)
$$\log|\mathbb{B}| = O(\log n) = o(n), \qquad w = \Theta(n/\log n), \qquad 1 \le \bar{N}_1 := \binom{n}{a}|\mathbb{B}|^{-w} < |\mathbb{B}|^3 = e^{o(n)}.$$

For an a-set $S \subseteq D$, let

$$\Phi_w^{\mathrm{el}}(S) = (c_1(S), \ldots, c_w(S)), \qquad \Phi_w^{\mathrm{ps}}(S) = (p_1(S), \ldots, p_w(S)),$$

where $Q_S = X^a + c_1(S)X^{a-1} + \cdots$ and $p_j(S) = \sum_{x \in S} x^j$. Put

$$S_E := \{x \in D : x^3 \in E\}, \qquad \Omega_{\mathrm{cube}} = \{S_E : E \in \tbinom{D^3}{a/3}\}.$$

For each $\star \in \{\mathrm{el}, \mathrm{ps}\}$ there is a prefix $z_\star \in \mathbb{B}^w$ such that

$$\begin{aligned} &\mathcal{F}^\star_{z_\star} := \{S \in \tbinom{D}{a} : \Phi^\star_w(S) = z_\star\}, \\ &\mathcal{Q}^\star_{z_\star} := \mathcal{F}^\star_{z_\star} \cap \Omega_{\mathrm{cube}}, \\ &|\mathcal{F}^\star_{z_\star}| \ge |\mathcal{Q}^\star_{z_\star}| \ge \binom{n/3}{a/3} q^{-w/3} = \exp\left(\left(\frac{1}{6}h(\alpha) + o(1)\right) n\right). \end{aligned} \tag{6.11c}$$

These are two separately proved assertions; no Newton-coordinate equivalence between the elementary and power-sum fibers is asserted. For either coordinate system

$$L^\star_{\mathrm{cube}} := |\Phi^\star_w(\Omega_{\mathrm{cube}})| \le q^{w/3}, \qquad \frac{|\Omega_{\mathrm{cube}}|}{L^\star_{\mathrm{cube}}} \ge \exp\left(\left(\frac{1}{6}h(\alpha) + o(1)\right) n\right). \tag{6.11c′}$$

Thus the image-normalized cubic quotient profile is exponentially larger than the ambient identity term; this conclusion does not depend on ambient-to-image moment transfer.

There is $\sigma = \sigma(\alpha) > 0$ for which

$$\max\{\Delta(\mathcal{F}^\star_{z_\star}), \Delta(\mathcal{Q}^\star_{z_\star})\} \le e^{-\sigma n} \quad (\star \in \{\mathrm{el}, \mathrm{ps}\}). \tag{6.11d}$$

More explicitly, define the ambient identity-normalized Sidon moment by

$$\mathcal{G}^\star_{r,\sigma} = |\mathbb{B}|^{-w} \sum_{\substack{z \in \mathbb{B}^w \\ \Delta(\mathcal{F}^\star_z) \le e^{-\sigma n}}} \left(\frac{|\mathcal{F}^\star_z|}{\bar{N}_1}\right)^r. \tag{6.11e}$$

For every logarithmic order $r = r_n \to \infty$,

$$\liminf_{n \to \infty} \frac{1}{nr} \log \mathcal{G}^\star_{r,\sigma} \ge \frac{1}{6}h(\alpha) \quad (\star \in \{\mathrm{el}, \mathrm{ps}\}). \tag{6.11f}$$

Finally, for the elementary-prefix fiber, every finite extension $\mathbb{F}/\mathbb{B}$ *satisfying*

$$|\mathbb{F}| - n > k\binom{|\mathcal{F}^{\mathrm{el}}_{z_{\mathrm{el}}}|}{2} \tag{6.11g}$$

has a received line for $C = \mathrm{RS}_{\mathbb{F}}(D, k)$ *with*

$$B_C^{\mathrm{MCA}}(a) \geq |\mathcal{F}^{\mathrm{el}}_{z_{\mathrm{el}}}| \geq \exp\left(\left(\frac{1}{6}h(\alpha) + o(1)\right) n\right). \tag{6.11h}$$

Such extensions exist with degree $[\mathbb{F} : \mathbb{B}] = O(n/\log n)$. *This last claim uses only the elementary locator prefix and does not transfer to the degenerate standard characteristic-two circle.*

Proof. Since s is odd, $q \equiv -1 \pmod 3$, and hence $3(q-1) \mid (q^2-1)$. The group $\mathbb{B}^\times$ is cyclic, so H exists. Cubing H gives the unique subgroup of order $q-1$, namely $\mathbb{B}_0^\times$, and the kernel of cubing on H has order three. Consequently

$$D^3 = \theta^3\mathbb{B}_0^\times, \qquad x \longmapsto x^3 : D \longrightarrow D^3 \quad \text{is exactly three-to-one.} \tag{6.11i}$$

Stirling's formula gives $\log_{|\mathbb{B}|}\binom{n}{a} = nh(\alpha)/(2\log q) + o(n/\log n)$. The rounding error in w is less than three, which proves (6.11b); it also gives $w < a - 1$ for all large n.

For $E \in \binom{D^3}{a/3}$, let S_E be its complete inverse image under cubing. Its locator is

$$Q_{S_E}(X) = \prod_{y\in E}(X^3 - y) = X^a + c_3(E)X^{a-3} + c_6(E)X^{a-6} + \cdots. \tag{6.11j}$$

The coefficient in gap $3j$ belongs to the scalar copy $\theta^{3j}\mathbb{B}_0$, and all other coefficients vanish. Hence the complete-cube supports assume at most $q^{w/3}$ elementary prefixes.

The same count for power sums is direct rather than an application of Newton's identities. If $\mu_3 \leq H$ is the group of cube roots of unity, then, in characteristic two,

$$p_j(S_E) = \sum_{y\in E}\sum_{x^3=y} x^j = \begin{cases} 0, & 3 \nmid j, \\ \sum_{y\in E} y^{j/3}, & 3 \mid j, \end{cases} \tag{6.11k}$$

because $\sum_{\zeta\in\mu_3}\zeta^j$ is zero unless $3 \mid j$, and is $3 = 1$ in $\mathbb{B}$ otherwise. The coordinate in position $3j$ lies in $\theta^{3j}\mathbb{B}_0$, so there are at most $q^{w/3}$ power-sum prefixes as well. Pigeonholing $\binom{n/3}{a/3}$ complete-cube supports proves the displayed lower bound for $\mathcal{Q}^\star_{z_\star}$, and hence for $\mathcal{F}^\star_{z_\star}$, in each coordinate system. Moreover

$$\log\binom{n/3}{a/3} - \frac{w}{3}\log q = \frac{n}{3}h(\alpha) - \frac{n}{6}h(\alpha) + o(n),$$

which proves its exponent.

By Proposition 17.1, an absolute constant $A > 0$ satisfies

$$E(F) \geq |F|^3/K, \quad F \subseteq \{0,1\}^D \quad \Longrightarrow \quad |F| \leq K^A.$$

Choose $0 < \sigma < h(\alpha)/(12A)$. If any of the full or quotient fibers in (6.11c) had $\Delta > e^{-\sigma n}$, the preceding implication with $K = e^{\sigma n}$ would contradict its exponential lower bound. This proves (6.11d). Retaining its one summand in (6.11e) gives

$$\log\mathcal{G}^\star_{r,\sigma} \geq -w\log|\mathbb{B}| + r\left(\log|\mathcal{F}^\star_{z_\star}| - \log\bar{N}_1\right).$$

Here $w\log|\mathbb{B}| = nh(\alpha) + o(n)$ and $\log\bar{N}_1 = o(n)$. Divide by nr and use $r \to \infty$ to obtain (6.11f).

Finally Theorem 4.5, applied to the elementary fiber with $m = a$, gives (6.11h). Because its cardinality has logarithm $O(n)$, while $\log|\mathbb{B}| = \Theta(\log n)$, an extension satisfying (6.11g) may be chosen with degree $O(n/\log n)$. Neither this step nor Proposition 8.15 identifies the doubled characteristic-two affine circle with the present multiplicative row. $\square$

Remark 8.12 (Ambient moment versus cubic profile scale). The moments in (6.11e) are ambient identity-normalized statements on the full a-subset slice. They do not, without (FI), assert failure of an image-normalized primitive payment. In particular, $p_{2j} = p_j^2$ in characteristic two, so the power-sum map has a Frobenius image collapse. The robust profile-envelope obstruction is (6.11c′), and the exact MCA construction uses the elementary locator prefix only.

Definition 8.13 (Aperiodic support)**.** *For $S \subseteq D = \theta H$, put* $\mathrm{Stab}_H(S) = \{h \in H : hS = S\}$*. The support is* aperiodic *if* $\mathrm{Stab}_H(S) = \{1\}$*. This removes exact multiplicative periodicity; it does not by itself remove a planted quotient/remainder explanation.*

Proposition 8.14 (Aperiodic planted Sidon version)**.** *The obstruction in Theorem 8.9 persists after requiring every support to have trivial stabilizer under multiplication by H. More precisely, fix $x_0 \in D$, omit its square fiber, and use*

$$S_E = \{x_0\} \sqcup \{x \in D : x^2 \in E\}, \qquad E \in \binom{D^2 \setminus \{x_0^2\}}{a/2}. \tag{6.12}$$

At total support size $a+1$ and depth w, some prefix fiber in this aperiodic residual still has size $\exp((h(\alpha)/4 + o(1))n)$, has normalized energy at most $e^{-\sigma n}$ after reducing σ if necessary, and has identity scale $e^{o(n)}$. Every one of its supports has trivial H-stabilizer.

Proof. The locator is $(X - x_0)\prod_{y\in E}(X^2 - y)$. Its first w coefficients are determined by at most $w/2$ elements of scalar copies of $\mathbb{B}_0$; therefore some prefix z contains an exponential constructed subfamily $\mathcal{P}_z$ of the form (6.12), with only $e^{o(n)}$ changes caused by deleting one square fiber and increasing the support size by one. Let

$$\mathcal{F}_z^{\mathrm{ap}} = \{S \in \Phi_w^{-1}(z) : \mathrm{Stab}_H(S) = \{1\}\}.$$

Every member of $\mathcal{P}_z$ has exactly one square fiber on which the involution $x \mapsto -x$ is incomplete, namely $\{x_0, -x_0\}$. Multiplication by $h \in H$ commutes with this involution. If $hS_E = S_E$, it must preserve that unique incomplete fiber, so $h = \pm 1$; the value -1 does not preserve S_E, because $-x_0 \notin S_E$. Hence $h = 1$. Thus $\mathcal{P}_z \subseteq \mathcal{F}_z^{\mathrm{ap}}$, so the latter is exponential. Applying Proposition 17.1 to $\mathcal{F}_z^{\mathrm{ap}} \subseteq \{0,1\}^D$ gives its stated low energy. The identity scale changes by only a polynomial factor. The same one-summand calculation as in (6.6) therefore gives $\exp(\Omega(nr_n))$ Sidon moment for the stabilizer-deleted residual at every logarithmic order $r_n \to \infty$. $\square$

The theorem disproves two proposed unrestricted steps at once. The identity-scale quotient comparison fails by (QR9), and the large exceptional fiber is Sidon-heavy rather than an approximate group. Exact stabilizer deletion does not restore the claim: Proposition 8.14 is a planted quotient/remainder profile and must be charged at its own scale. The remaining possible Sidon theorem is therefore a statement only about the primitive residual after all such profiles have been routed.

Proposition 8.15 (Stereographic circle equivalence)**.** *Let $\mathbb{F}$ have odd characteristic, let $\mathcal{C} : x^2 + y^2 = 1$, and let $E \subseteq \mathcal{C}(\mathbb{F})$ avoid $(-1, 0)$. Put $t = y/(1+x)$, $T = t(E)$, and let $\mathcal{C}_u(E)$ be the evaluation code of the degree-at-most-u filtered residue-class space in $\mathbb{F}[x,y]/(x^2+y^2-1)$. Then $\mathcal{C}_u(E)$ is diagonally equivalent to* $\mathrm{RS}_{\mathbb{F}}(T, 2u+1)$*. The equivalence preserves every list, CA, and support-wise MCA bad-slope set.*

Proof. The inverse parametrization is $x = (1-t^2)/(1+t^2)$, $y = 2t/(1+t^2)$. On E, $1+t^2 = 2/(1+x) \neq 0$. Modulo the circle equation, the filtered degree-u space is

$$\{f_0(x) + yf_1(x) : \deg f_0 \le u,\ \deg f_1 \le u-1\},$$

with the second summand absent at $u = 0$. It has dimension $2u+1$. After substitution and multiplication by $(1+t^2)^u$, every element becomes a polynomial of degree at most $2u$. The induced map of filtered function spaces is injective before evaluation, because stereographic substitution is an isomorphism of the localized coordinate rings. The dimensions agree, so it is an isomorphism. Coordinatewise multiplication by the nonzero values $(1+t^2)^u$ preserves equality on each support and commutes with affine combinations. □

Corollary 8.16 (Circle quotient obstruction)**.** *In Theorem 8.9, put $u = (k-1)/2 = (a-w-2)/2$ and let $E \subseteq \mathcal{C}(\mathbb{F})$ be the inverse stereographic image of D. Then the standard bivariate circle code $\mathcal{C}_u(E)$ has the same bad-slope sets and satisfies (6.8). In particular the polynomial-field quotient obstruction transfers without loss to the circle.*

Proof. The integers a, w are even, so $k = a-w-1 = 2u+1$. The coset D avoids the zeros of $1+t^2$, and Proposition 8.15 transfers the received line coordinate by coordinate. □

The example is the first nontrivial member of a general quotient envelope.

Proposition 8.17 (Necessary power-quotient envelope)**.** *Let $D = \theta H$, $|H| = n$, write $D^{(c)} = \{x^c : x \in D\}$, let c divide both n and m, and let $0 \le w \le m$, and suppose the power quotient $D^{(c)}$ is contained in a scalar copy of a subfield $\mathbb{B}_c$. For global prefix depth w, some prefix fiber contains at least*

$$\left\lceil \binom{n/c}{m/c} |\mathbb{B}_c|^{-\lfloor w/c \rfloor} \right\rceil \tag{6.13}$$

complete-fiber supports.

Proof. The locator of the preimage of $E \subseteq D^{(c)}$ is $\prod_{y\in E}(X^c - y)$. Among its first w nonleading coefficients, only the gaps $c, 2c, \ldots, \lfloor w/c \rfloor c$ can be nonzero; they are the first $\lfloor w/c \rfloor$ elementary symmetric functions of E, up to fixed scalar factors. Pigeonhole the $\binom{n/c}{m/c}$ quotient supports over at most $|\mathbb{B}_c|^{\lfloor w/c \rfloor}$ prefixes. □

When $m = cM$ and $K = m - w$, the descended prefix depth is

$$M - \lceil K/c \rceil = \lfloor w/c \rfloor . \tag{6.14}$$

Thus an $O(1)$ rounding difference costs $O(\log|\mathbb{B}_c|)$ bits and is subexponential when every relevant coefficient field has logarithmic size $o(n)$. The positive-rate effect in Theorem 8.9 instead comes from the field drop $|\mathbb{B}|/|\mathbb{B}_c|$ repeated across $\lfloor w/c \rfloor$ live slots, as quantified in (QR8). Remainder supports and Chebyshev quotients contribute analogous profile terms and must be retained in $\mathfrak{E}_n$.

9. Algebraic profile criteria and balanced-core scope

Constructibility identifies where a witness lives; payment is a separate enumerative statement about the image in slope space. This section records the algebraic facts that are unconditional and then states the precise criteria required by the upper compiler. In particular, quotient classification does not imply payment at the identity scale, and a codimension-one support locus does not by itself give the required number of independent prefix equations.

Here a *balanced-core chart* means a parameter family of equal-degree monic residual locator pairs with a common prefix, obtained after all common factors, quotient pullbacks, planted blocks, and already certified rank degeneracies have been removed, whose remaining moving coefficient space has projective dimension greater than one. Thus "balanced core" is a description of what remains to be counted, not a smallness assertion. A *ray bound* counts the distinct projective line directions—equivalently, the distinct finite slopes in the chosen affine chart—in the slope projection of that residual incidence. The upper *compiler* is the collection of proved implications that turns profile-wise support, fiber, and ray bounds into a bound for the MCA numerator. A *saturation profile* records the projection that identifies raw support/explanation witnesses having the same residual projective direction; paying it requires either a lower bound on the fibers of that projection or a direct bound on its image.

Theorem 9.1 (Conditional algebraic-ledger criterion). *Suppose the quotient profiles are budgeted at their natural terms in $\mathfrak{E}_n$; the planted, tangent, extension, rank-pivot, and saturation profiles satisfy the direct projection criteria below; and every remaining balanced-core chart is either covered by the one-pencil theorem or satisfies the ray compiler* (RC)*. If the number of profiles is $e^{o(n)}$, then the algebraic part of the first-match ledger contributes at most $e^{o(n)}\mathfrak{E}_n$.*

Proof. Each criterion below supplies a bound for a set of distinct slopes, not only for its support incidence. The profile envelope contains every quotient and coefficient-field term. The first-match sets are disjoint after ordering, and there are $e^{o(n)}$ of them, so their certified bounds sum to $e^{o(n)}\mathfrak{E}_n$. □

Definition 9.2 (Periodicity scale). *Let a finite group G act on D. For $S \subseteq D$, the periodicity scale $c(S)$ is the order of the stabilizer of S in the quotient tower used by the row. In a multiplicative coset $D = \theta H$, this is the largest $c \mid |H|$ such that S is a union of cosets of the order-c subgroup. In a circle x-domain it is the largest Chebyshev scale for which S is a union of complete T_c-fibers.*

Proposition 9.3 (Stabilizer exhaustion and quotient budget). *In a smooth multiplicative or circle row, every support with $c(S) > 1$ lies in a power or Chebyshev quotient cell. A scale-c complete-fiber cell is budgeted at its natural descended prefix term, including the field of definition and the rounding depth $\lfloor w/c \rfloor$. Summing these terms over the quotient profiles gives the quotient part of $\mathfrak{E}_n$.*

Proof. For multiplicative cosets, the subgroup lattice of H partitions supports by stabilizer. If $c(S) = c$, then S is the pullback of a support $\bar{S}$ on D/H_c. If the received and explaining data are constant on the same fibers, invariant quotient descent applies. Otherwise the noninvariant residues require a separate first-match chart and direct certificate.

The number of divisors of $|H|$ is subexponential. The locator of a complete-fiber support is lacunary, so its live depth and coefficient field are those in Proposition 8.17. That proposition is a lower bound and shows why the quotient term cannot be replaced by the identity term. The corresponding upper budget is therefore recorded as a separate profile term. The Chebyshev case is identical after lifting to the torus and replacing X^c by T_c. Summation gives precisely the quotient portion of $\mathfrak{E}_n$, without asserting an identity-scale comparison. □

Proposition 9.4 (Exact invariant quotient descent). *Let $D = \theta H$, let $c \mid |H|$, and suppose $h \in \mathbb{F}[X]$, $\deg h < n$, satisfies $h(\zeta x) = h(x)$ for every $x \in D$ and every $\zeta \in \mu_c$. Then $h(X) = \bar{h}(X^c)$ with $\deg \bar{h} = \lfloor \deg h/c \rfloor$. The analogous statement on a circle twin coset holds for an invariant symmetric Laurent polynomial and the Chebyshev quotient.*

Proof. Write $h = \sum_{j=0}^{c-1} X^j h_j(X^c)$. For $\zeta \in \mu_c$, the difference $h(\zeta X) - h(X)$ has degree less than n and vanishes at all n points of D, so it is zero. Orthogonality of the characters $\zeta \mapsto \zeta^j$ forces $h_j = 0$ for $j > 0$. On the torus the same character decomposition is compatible with the involution $z \mapsto z^{-1}$, and invariant Laurent monomials descend to Chebyshev polynomials. Constancy only on a selected witness support is weaker and requires a separate quotient-cell certificate; it is not asserted here. ☐

Proposition 9.5 (Planted-block payment criterion)**.** *Let the allowed planted divisors of size ℓ form a family of $e^{o(n)}$ candidates, and suppose the residual prefix fiber for each candidate has size at most its profile term times $e^{o(n)}$. Then the planted cells are paid at the sum of those terms.*

Proof. After fixing P, the residual support is an $(a-\ell)$-subset of $D \setminus P$, and multiplication by Q_P gives a triangular shift of the prefix target. The asserted residual bound is therefore a direct fiber estimate. Summing it over the stipulated $e^{o(n)}$ algebraic candidates proves the criterion. Arbitrary choices among $\binom{n}{\ell}$ subsets are not planted profiles. ☐

Proposition 9.6 (Tangent and determinantal payment criterion)**.** *The* deep regime *is the high-agreement range $3(n-a) \le n-k$; in that range the deep cell has the exact bound of Theorem 3.4. A tangent, common-line, or rank-drop chart is paid if a nonzero collection of independent pivot minors cuts the* slope image*—the set of slopes actually attained by that chart—by the number of prefix equations charged to that chart, with degree and chart count $e^{o(n)}$.*

Proof. The deep assertion was proved exactly in Section 3. For a determinantal chart, the nonzero independent minors give the displayed codimension and a finite-field point bound of the expected order. The criterion deliberately requires independence and controls the projection to slopes. A matrix such as XI_r can lose rank r on a codimension-one locus, so rank loss r alone does not certify codimension r. ☐

Here *restriction of scalars* means choosing a basis of a bounded extension field over its recorded coefficient subfield and replacing every extension-valued variable by its tuple of coordinates over that smaller field, which we call the base field in this proposition.

Proposition 9.7 (Extension-cell criterion)**.** *Restriction of scalars converts a bounded-degree extension cell to a bounded number of base-field incidence charts. Such a cell is paid only after the resulting charts satisfy a direct slope-image bound at their coefficient-field profile scales.*

Proof. Choose a base-field basis. Coefficient variables become bounded tuples and the witness equations become bounded systems over the base field; the operation does not itself estimate their slope projection. Once the displayed direct estimate is provided, bounded extension degree and subexponential profile count preserve it under summation. ☐

Proposition 9.8 (Rank-pivot criterion)**.** *A differential-locator rank-pivot cell is paid when its selected minor is nonzero on the* ambient primitive chart*, meaning the full primitive chart before the rank conditions are imposed, the required independent minor conditions have the charged codimension, and the remaining fibers of the slope projection have the displayed profile bound.*

Proof. Under the hypotheses, standard determinantal incidence and finite-field point counting give the charged factor for the ambient chart. The stated fiber hypothesis transfers it to distinct slopes. Neither nonvanishing of small Vandermonde minors nor constructibility alone proves these conditions for the remaining high-dimensional balanced-core chart. ☐

Proposition 9.9 (Saturation criterion)**.** *A saturation cell is paid if the projection from raw witnesses to saturated line rays has a proved lower fiber bound, or if its image has a direct distinct-ray estimate at the profile scale.*

Proof. This is the elementary image–fiber identity. If every ray has at least H raw representatives, their number is at most the raw witness count divided by H. If no such lower bound is known, only a direct image estimate can certify payment. An upper bound on the number of profile lifts goes in the wrong direction. ☐

Proposition 9.10 (One-pencil moving-root bound). *Let $A, B \in \mathbb{F}[X]$, let $G = \gcd(A, B, \prod_{x \in D}(X - x))$, and put $g = \deg G$. If every parameter $[s : t]$ in a set $\mathcal{Z} \subseteq \mathbb{P}^1(\mathbb{F})$ makes $(sA + tB)/G$ have at least h distinct roots in $D \setminus Z(G)$, then*

$$|\mathcal{Z}| \leq \lfloor (n - g)/h \rfloor \, . \tag{7.1}$$

Proof. Count incidences $([s : t], x)$ with $x \in D \setminus Z(G)$ and $sA(x) + tB(x) = 0$. Each parameter contributes at least h incidences. For a fixed moving root, $(A(x), B(x)) \neq (0, 0)$, so exactly one projective parameter works. There are $n - g$ moving points, proving (7.1). ☐

Remark 9.11 (Balanced-core exhaustion remains separate). The preceding proposition applies only after a chart is genuinely a one-dimensional locator pencil and the slope injects into its projective parameter. It neither covers a higher-dimensional coefficient family nor proves that such a family decomposes into subexponentially many pencils. That structural statement is part of (RC) or must be established by a direct ray count.

Lemma 9.12 (Subexponential profile atlas). *For a ledger-admissible sequence, the number of first-match profiles is $e^{o(n)}$.*

Proof. This is condition (A2) in Definition 1.4. For cyclic quotient towers the divisor count is subexponential, and for dyadic Chebyshev towers it is logarithmic. The statement does not include arbitrary planted subsets or an unproved decomposition of a higher-dimensional pencil; including either could create exponentially many profiles. ☐

The criteria above are modular. A concrete row must supply their missing projection and exhaustion estimates; replacing them by the word "constructible" would not be a proof.

10. The primitive prefix leaf

After first-match removal of paid cells, a residual prefix-boundary chart has a fixed set of active coordinates T, $|T| = N$, and a fixed support size m. The full profile slice is

$$\Omega_{T,m} := \{x \in \{0,1\}^T : \sum_{t \in T} x_t = m\},$$

and the primitive residual is a subset $\Omega^\circ \subseteq \Omega_{T,m}$. The superscript circle records a first-match residual; no Zariski openness is asserted. The normalization is taken from the full profile slice, not from the possibly much smaller residual mass. It is nevertheless the *realized image*, rather than the formal codomain, that supplies the default number of target fibers.

The three words in *primitive first-match residual* have separate content. "First match" means that a slope is removed as soon as one of its witnesses occurs in an earlier ordered cell; "residual" means that only witnesses surviving those removals remain; and "primitive" means that the row-specific quotient, planted, field-descent, rank, and ray-saturation certificates named by the atlas have all been applied. Primitivity does not mean merely trivial setwise stabilizer, and it does not assume the Fourier, Sidon, max-fiber, or ray estimates that will be imposed below.

The prefix syndrome map is a linear map $\Phi : \mathbb{Z}^T \to \mathbb{B}^R$, restricted first to the full slice and then to Ω°. In the unprofiled locator chart, when $R < \operatorname{char} \mathbb{B}$, it is equivalent through Newton's triangular change of coordinates to

$$\Phi(x) = \sum_{t \in T} x_t (t, t^2, \ldots, t^R). \tag{9.1}$$

Equivalently, the augmented columns $\widetilde{v}_t = (1, t, \ldots, t^R)$ have constant first coordinate $\sum_t x_t = m$, and the other R coordinates are Φ. A residual quotient or rational chart is allowed to replace these columns by $v_t = \rho(t)(1, t, \ldots, t^{R-1})$ only when that genuine weighted Vandermonde representation, with $\rho(t) \neq 0$, has been proved for the chart. It is not a consequence merely of constructibility or of deleting a Jacobian locus. The number R is the number of independent prefix equations; in the unprofiled MCA boundary leaf $R = w = a - K$.

Put $M = |\Omega_{T,m}|$, and write

$$\mathcal{S} = \Phi(\Omega_{T,m}), \qquad L = |\mathcal{S}|, \qquad A = |\mathbb{B}|^R, \qquad \overline{N}^{\mathrm{img}} = M/L, \qquad \overline{N}^{\mathrm{amb}} = M/A, \tag{9.2}$$

and for $s \in \mathcal{S}$ put $F_s = \Omega^\circ \cap \Phi^{-1}(s)$ and $f_s = |F_s|$. Unless an ambient full-image certificate has been proved, we set $\overline{N} = \overline{N}^{\mathrm{img}}$. The certificate is

$$L \geq e^{-o(N)} A. \tag{FI}$$

The set $\mathcal{S}$ is the *effective image*: it consists of the prefix targets actually attained by the full profile slice. The formal ambient target space $\mathbb{B}^R$ can be much larger. Condition (FI), the *full-image certificate*, says that these two target counts differ by at most a subexponential factor. An *effective-image collapse* is the failure of this comparison; it must be retained as an effective-image rank-collapse profile, or handled throughout with image normalization. Under (FI), the two scales differ by only $e^{o(N)}$, so either may be used at exponential accuracy. If the certificate fails, effective-image collapse must be retained as an effective-image rank-collapse profile, or all estimates on this leaf must stay at image scale. Deleting earlier cells can only decrease every fiber.

The image-scale frontier condition is $\log M - \log L = o(N)$. The familiar ambient equation $\log \binom{N}{m} - R \log|\mathbb{B}| = o(N)$ is interchangeable with it only under (FI). This distinction is immaterial for a surjective prefix map but essential for a collapsed primitive image.

Lemma 10.1 (Exact image–ambient moment conversion)**.** *For $q \geq 1$, extend f_s by zero from $\mathcal{S}$ to $\mathbb{B}^R$ and define*

$$\Gamma_q^{\mathrm{img}} = \frac{1}{L} \sum_{s \in \mathcal{S}} \left(\frac{f_s}{\overline{N}^{\mathrm{img}}} \right)^q, \qquad \Gamma_q^{\mathrm{amb}} = \frac{1}{A} \sum_{s \in \mathbb{B}^R} \left(\frac{f_s}{\overline{N}^{\mathrm{amb}}} \right)^q.$$

Then

$$\Gamma_q^{\mathrm{amb}} = \left(\frac{A}{L} \right)^{q-1} \Gamma_q^{\mathrm{img}}. \tag{9.3}$$

Thus an ambient-normalized moment bound implies the corresponding image-normalized bound. The converse is valid with subexponential loss at logarithmic order only under (FI).

Proof. Since $\overline{N}^{\mathrm{amb}} = M/A$ and $\overline{N}^{\mathrm{img}} = M/L$, zero extension and one rearrangement give

$$\Gamma_q^{\mathrm{amb}} = \frac{1}{A} \sum_{s \in \mathcal{S}} \left(\frac{A f_s}{M} \right)^q = \left(\frac{A}{L} \right)^{q-1} \frac{1}{L} \sum_{s \in \mathcal{S}} \left(\frac{L f_s}{M} \right)^q.$$

Because $L \leq A$, the asserted one-way implication follows. Under (FI), $(q-1)\log(A/L) = o(Nq)$, which proves the reverse transfer at the scale used here. $\square$

Remark 10.2 (Full-slice flatness certifies the image size). If a Fourier argument proves $\max_s|\Omega_{T,m} \cap \Phi^{-1}(s)| \le e^{o(N)}\overline{N}^{\mathrm{amb}}$, then averaging over the nonempty image fibers gives $A/L \le e^{o(N)}$. Hence such an ambient flatness theorem itself proves (FI); image collapse cannot be silently assumed away before that estimate has been established.

Definition 10.3 (Primitive Q). *The* Q branch *is the primitive max-fiber problem: it asks whether one effective prefix target contains substantially more residual supports than the image-normalized average. A primitive prefix leaf satisfies Q with loss $e^{o(N)}$ if*

$$\max_{s \in \mathcal{S}} f_s \le e^{o(N)}\overline{N}^{\mathrm{img}}.$$

A sequence of primitive leaves satisfies logarithmic-moment Q if, at every logarithmic order $q = q_N \to \infty$, $q \le (\log N)^C$, used by the argument,

$$\Gamma_q^{\mathrm{ord}} := \Gamma_q^{\mathrm{img}} = L^{-1} \sum_{s \in \mathcal{S}} \left(\frac{f_s}{\overline{N}^{\mathrm{img}}} \right)^q \le e^{o(Nq)}.$$

Here a logarithmic moment order *means $q = q_N \to \infty$ with $q \le (\log N)^C$ for a fixed C. The displayed normalized q-th moment is a Rényi-type probe of the fiber distribution: letting q grow slowly makes it approximate the largest normalized fiber without introducing an exponential moment-order cost. An ambient formulation is permitted only when it is proved directly or when* (FI) *has already been certified.*

The next lemma is the primitive logarithmic moment equivalence used in the earlier RS–MCA reductions. It is included because it is the exact interface between moment estimates and max-fiber Q.

Lemma 10.4 (Logarithmic moments and primitive Q). *Assume $q = q_N \to \infty$ and $\log L/q = o(N)$. Then $\Gamma_q^{\mathrm{ord}} \le e^{o(Nq)}$ implies primitive Q. Conversely, primitive Q implies $\Gamma_q^{\mathrm{ord}} \le e^{o(Nq)}$.*

Proof. Let $M_* = \max_s f_s/\overline{N}^{\mathrm{img}}$. Since $M_*^q \le \sum_s (f_s/\overline{N}^{\mathrm{img}})^q = L\Gamma_q^{\mathrm{ord}}$, the moment bound gives $\log M_* \le (\log L + \log \Gamma_q^{\mathrm{ord}})/q = o(N)$. Conversely, $\sum_s f_s \le |\Omega_{T,m}| = L\overline{N}^{\mathrm{img}}$, so $\Gamma_q^{\mathrm{ord}} \le M_*^{q-1}$. Primitive Q gives $M_* = e^{o(N)}$, hence $\Gamma_q^{\mathrm{ord}} = e^{o(Nq)}$. ☐

The genuine Vandermonde condition removes the ordinary rank-defect alternative.

Lemma 10.5 (Augmented Vandermonde independence). *Let $t_1, \dots, t_r \in \mathbb{B}$ be distinct and $r \le R+1$. Then the columns $\tilde{v}_{t_i} = (1, t_i, \dots, t_i^R)$ are linearly independent. The same holds for $\rho(t_i)(1, t_i, \dots, t_i^{R-1})$ when $r \le R$ and every $\rho(t_i) \neq 0$.*

Proof. In either case the leading $r \times r$ minor, after removing the nonzero column weights when present, is $(t_i^j)_{0 \le j < r, 1 \le i \le r}$. Its determinant is $\prod_{i<j}(t_j - t_i) \neq 0$. ☐

In the high-energy argument below the Vandermonde lemma is not used: Boolean-cube growth gives the contradiction directly. The lemma certifies only the exact locator chart, or a residual chart for which the displayed weighted form has separately been verified; it does not classify arbitrary rank defects as paid cells.

11. Prefix coordinates and the Vandermonde map

This section explains why the primitive residual has the linear Vandermonde form used in Section 10. Let $S \subseteq D$ have size m, and write its monic locator as

$$Q_S(X) = \prod_{t \in S}(X - t) = X^m + q_1(S)X^{m-1} + \cdots + q_m(S).$$

The first R boundary coordinates may be taken either as elementary symmetric coefficients $q_1, \dots, q_R$ or, when $R < \operatorname{char} \mathbb{B}$, as power sums $p_j(S) = \sum_{t \in S} t^j$, $1 \le j \le R$. Newton identities give triangular polynomial changes of variables between these two coordinate systems; the diagonal entries are $1, 2, \dots, R$, so the change is invertible in characteristic greater than R.

For the unprofiled locator chart, fixing the support size makes the zeroth power sum constant and Newton's identities identify the first R locator coefficients with the power sums $\sum_t x_t t^j$, $1 \le j \le R$. Thus (9.1) is exact. For a general residual chart one first fixes the planted roots, quotient data, common factors, and all coordinates charged to earlier cells. The fixed part contributes a constant syndrome s_0. To use the primitive theorem one must then exhibit the remaining map explicitly, either as the same power-sum map, as a genuine common-weight Vandermonde map, or as a map for which the required Sidon estimate has been proved directly. Coordinate-dependent rational weights do not become a Vandermonde system merely because they are nonzero. Here "common-weight" means that one scalar $\rho(t)$ multiplies every monomial coordinate of the column at t, rather than using a different weight for each coordinate. This verification is condition (A5) of Definition 1.4.

The fixed-density condition is also intrinsic. In a prefix leaf the total support size is fixed by the agreement threshold, and all paid planted blocks have fixed size. Thus the number of remaining choices is fixed. If a profile leaves several colors of coordinates, one obtains a product of fixed-density slices; the proof applies to each color after encoding the colors in a bounded alphabet. For the main statement we present the Boolean case because it is the MCA support case and because the Boolean-cube difference bound applies directly.

Lemma 11.1 (Newton-coordinate fiber equivalence)**.** *Assume $R < \operatorname{char} \mathbb{B}$. For supports of fixed size, the fibers of the first R elementary locator coefficients are exactly the fibers of the first R power sums. Consequently the Q max-fiber problem may be stated using either coordinate system.*

Proof. Newton identities give, for $1 \le j \le R$,

$$p_j + q_1 p_{j-1} + \cdots + q_{j-1} p_1 + j q_j = 0.$$

Since j is invertible in $\mathbb{B}$, q_j is determined by $p_1, \dots, p_j$, and conversely p_j is determined by $q_1, \dots, q_j$. The transformations are triangular inverse polynomial maps on the first R coordinates. □

The *description entropy* of a profile family is the logarithm of the number of allowed auxiliary/profile descriptions. It measures the census of cells and is distinct from the algebraic description complexity of one cell, which measures the number and degrees of its defining equations.

Lemma 11.2 (Primitive profiles have subexponential multiplicity)**.** *If the cell ledger fixes only bounded-complexity quotient, planted, tangent, extension, and split-pencil profiles, and the total description entropy of all data outside the moving support is $o(n)$ bits, then the number of primitive profiles is $e^{o(n)}$.*

Proof. There are at most $2^{o(n)} = e^{o(n)}$ descriptions by hypothesis. In particular, a planted or common block of size $o(n)$ has $\sum_{j=o(n)} \binom{n}{j} = e^{o(n)}$ possible supports, but recording $s = o(n)$ freely chosen field values costs $s \log_2 |\mathbb{B}|$ bits and is subexponential only when that product is $o(n)$. Positive-density data requires its own profile census and entropy payment; it cannot be discarded as a bounded-complexity label. □

Remark 11.3. The subexponential profile lemma is the asymptotic counterpart of the finite ledger census. It is the reason a row proof must distinguish between a genuine cell payment and a mere change of variables. A change of variables with exponentially many charts would consume the whole frontier budget.

12. The Sidon split

The Q moment can be large for two opposite reasons. A fiber may contain many repeated additive differences, which is the high-energy situation seen by inverse additive combinatorics, or it may be large while its differences almost never repeat, which is the Sidon-like situation. The split below separates these mechanisms before either one is estimated.

Every support is identified with its incidence vector in the torsion-free group $\mathbb{Z}^T$. In particular, all sums and differences in this section are integer operations, even when the Reed–Solomon coefficient field has characteristic two. For a finite set $F \subseteq \{0,1\}^T$, define its additive energy

$$E(F) = |\{(a,b,c,d) \in F^4 : a-b=c-d\}|.$$

Let X, X' be independent uniform points of F. Then $E(F) = |F|^4 \sum_z \mathbb{P}(X - X' = z)^2$. It is convenient to normalize by

$$\Delta(F) = \frac{E(F)}{|F|^3}.$$

The quantity $\Delta(F)$ is an additive closure probability:

$$\Delta(F) = \mathbb{P}_{a,b,c\in F}(a-b+c \in F).$$

Indeed, once a,b,c are chosen, the fourth point in an energy quadruple is forced to be $d = a-b+c$.

Thus additive energy counts repeated differences, and $\Delta(F)$ is their normalized density. Values $\Delta(F) = e^{-o(N)}$ indicate substantial additive closure at exponential scale, whereas $\Delta(F) \le e^{-\sigma N}$ is the low-energy, Sidon-like regime. The logarithmic moments below are the slowly growing normalized moments from Definition 10.3; they measure how much of the Q obstruction is carried by fibers of either type. Throughout these sections, *positive rate* means a fixed positive exponential rate along a subsequence: for a fiber it means an excess of the form $e^{cN-o(N)}$, and for a logarithmic moment it means an excess of the form $e^{cNq-o(Nq)}$, for some constant $c > 0$.

For the fiber F_s, write $\Delta_s = E(F_s)/f_s^3$, with $\Delta_s = 0$ if $f_s = 0$.

Definition 12.1 (Sidon-heavy logarithmic moment)**.** *For $\sigma > 0$, the Sidon-heavy part of the normalized moment is*

$$\Gamma^{\mathrm{sid}}_{q,\sigma} = L^{-1} \sum_{\Delta_s \le e^{-\sigma N}} \left(\frac{f_s}{\overline{N}}\right)^q.$$

A sequence has no positive-rate Sidon-heavy obstruction if $\Gamma^{\mathrm{sid}}_{q,\sigma} \le e^{o(Nq)}$ for every fixed $\sigma > 0$ and every logarithmic q. Here "Sidon" refers to the exponentially small normalized energy Δ_s, while "heavy" refers to the positive-exponential contribution of those fibers to the normalized logarithmic moment; it does not assert that every low-energy fiber is individually large.

The following dichotomy is the correct replacement for the naive assertion that collision excess directly creates small doubling.

Proposition 12.2 (Ordinary moment split)**.** *Fix $\eta > 0$ and $\sigma > 0$. If $\Gamma^{\mathrm{ord}}_q \ge e^{\eta N q}$ and $\Gamma^{\mathrm{sid}}_{q,\sigma} \le \frac{1}{2} e^{\eta N q}$, then there is a syndrome s such that*

$$f_s \ge e^{\eta N - o(N)} \overline{N}, \qquad \Delta_s > e^{-\sigma N}.$$

If no positive-rate Sidon-heavy obstruction is present, then any positive-rate moment failure gives, after taking $\sigma \to 0$ slowly, a fiber with $f_s \ge e^{\eta N - o(N)}\overline{N}$ and $E(F_s) \ge f_s^3 e^{-o(N)}$.

Proof. The high-energy part of the moment is at least $\frac{1}{2}e^{\eta N q}$. Since there are L syndromes, some high-energy syndrome satisfies

$$\left(\frac{f_s}{\overline{N}}\right)^q \ge \frac{L}{2}\, e^{\eta N q} \cdot L^{-1} = \frac{1}{2}e^{\eta N q}.$$

Thus $f_s \ge e^{\eta N - o(N)}\overline{N}$, and by construction $\Delta_s > e^{-\sigma N}$. If $\Gamma^{\mathrm{sid}}_{q,\sigma} \le e^{o(Nq)}$ for every fixed σ, apply the first part along a sequence $\sigma_j \downarrow 0$ and diagonalize; the normalization $\Delta_s = E(F_s)/f_s^3$ gives the energy conclusion. □

The proof also explains the no-go point. Suppose a fiber is heavy but $\Delta_s \le e^{-\sigma N}$. A closure-rich diagram using tests $a - b + c \in F_s$ pays a factor Δ_s for each nonredundant closure. Such diagrams cannot account for the ordinary contribution f_s^q of q independent choices in F_s. Therefore Sidon-heavy fibers must be removed by a separate cell; they cannot be forced into the high-energy inverse branch.

Definition 12.3 (Sidon moment payment)**.** *A primitive prefix leaf has a Sidon moment payment if $\Gamma^{\mathrm{sid}}_{q,\sigma} = e^{o(Nq)}$ for every fixed $\sigma > 0$ and every logarithmic moment order used by the primitive-Q argument, uniformly over profiles. This is an analytic support-fiber statement. It is not equivalent to a first-match distinct-slope payment; the latter still requires* (RC) *or a direct ray bound.*

The equivalence in the definition follows from Lemma 10.4 restricted to the low-energy set of syndromes. Operationally, one proves payment either directly by bounding these fibers or indirectly by proving Fourier flatness on the primitive complement of algebraic major arcs.

13. Why the Sidon cell is necessary

It is tempting to try to prove primitive Q by expanding the logarithmic moment into diagrams and then showing that every primitive diagram contains many additive closure tests. That strategy is correct only for a closure-rich submoment. It cannot capture the ordinary moment on a heavy low-energy fiber. The obstruction is simple enough to record as a lemma.

Let $F = F_s$ be a fiber of size f. A closure test is an equation of the form $a - b + c \in F$, with $a, b, c \in F$ already chosen. Its success probability is $\Delta(F) = E(F)/f^3$. A closure-rich diagram with ℓ nonredundant closure tests and r free choices contributes, up to a subexponential diagram factor,

$$f^r \Delta(F)^\ell.$$

After the free coordinates have been chosen, in the image normalization of a primitive prefix leaf, the remaining $q - r$ coordinates are supplied only at the random scale $\overline{N}$. Thus its contribution is at most

$$e^{o(Nq)}\overline{N}^{q-r} f^r \Delta(F)^\ell.$$

Lemma 13.1 (No-go for ordinary moment capture)**.** *Assume that $f \ge e^{\eta N}\overline{N}$, $\Delta(F) \le e^{-\sigma N}$, and that a family of diagrams has $r = o(q)$ free fiber choices and $\ell \ge cq$ nonredundant closure tests. Then the total contribution of these diagrams is smaller than the ordinary fiber contribution f^q by a factor*

$$\exp\Big(-(\eta + c\sigma)Nq + o(Nq)\Big)$$

up to changing η by $o(1)$.

Proof. For one diagram the ratio to f^q is at most

$$e^{o(Nq)}\left(\frac{N}{f}\right)^{q-r}\Delta(F)^{\ell}\le e^{o(Nq)}\exp(-\eta N(q-r)-c\sigma Nq).$$

Since $r=o(q)$, this is $\exp(-(\eta+c\sigma)Nq+o(Nq))$. The number of bounded-complexity logarithmic diagrams is $e^{O(q\log q)}=e^{o(Nq)}$, so summing over them does not change the exponential rate. □

The lemma has an important conceptual consequence. A large Sidon-like fiber is not a hidden approximate group. It is a different phenomenon: many independent choices land in the same syndrome, but their pairwise differences almost never repeat. Additive-combinatorics inverse tools are designed for the opposite situation, where repeated differences are abundant. Therefore the first-match ledger must contain a Sidon/Fourier cell before the primitive inverse argument begins.

The valid implication is exactly Proposition 12.2. Ordinary moment mass splits into a low-energy part, which needs a separately proved Sidon moment estimate, and a high-energy part, which is dispatched by Proposition 17.1.

This mechanism does not replace a prefix-flatness or Sidon-payment theorem; it identifies exactly where one is needed. The first-match order is: algebraic major arcs first, then a separately certified Sidon/Fourier cell, and only then the high-energy primitive inverse step. Naming a large low-energy fiber a cell does not pay it.

14. Fourier flatness as a Sidon payment

This section records the general weighted form of the Fourier/Sidon payment. It is phrased for an arbitrary finite field $\mathbb{B}$, because the primitive residual charts that come from the moduli ledger need not have unweighted power-sum coordinates. Let T be a finite set, $|T|=N$, let $g:T\to\mathbb{B}^R$ be any function, and set

$$\Omega_{T,m}=\{x\in\{0,1\}^T:\sum_{t\in T}x_t=m\},\qquad \Psi(x)=\sum_{t\in T}x_tg(t).$$

In applications $g(t)=(\rho_0(t),\rho_1(t)t,\dots,\rho_{R-1}(t)t^{R-1})$ with nonzero rational weights ρ_i on the primitive chart.

Fourier inversion has three pieces. The zero character produces the average-fiber main term. A nonzero character is called a *major arc* when its phase has one of the algebraic degeneracies listed in Definition 14.4; every other nonzero character is a *minor arc*. The pointwise condition (PF) controls each minor character, while the aggregate condition (MA) controls the sum of all major characters. Neither estimate substitutes for the other.

The condition (PF') below is the raw power-sum flatness inequality when one has a uniform bound for every nonzero character. Condition (PF) is its row-specific Weil specialization for minor characters after substituting the pole-degree bound; the omitted major characters are then handled by (MA).

Theorem 14.1 (Prefix flatness from power-sum control)**.** *Fix a nontrivial additive character ψ of $\mathbb{B}$. For $\alpha\in\mathbb{B}^R\setminus\{0\}$ and $1\le j\le m$, put*

$$P_j(\alpha)=\sum_{t\in T}\psi\Big(j\alpha\cdot g(t)\Big).$$

Assume $|P_j(\alpha)|\le\Lambda$ for every $\alpha\ne0$ and every $1\le j\le m$. Then, for every $z\in\mathbb{B}^R$,

$$|\Psi^{-1}(z)|\le|\mathbb{B}|^{-R}\binom{N}{m}+\binom{\Lambda+m-1}{m}.$$

Consequently, if

$$R\log|\mathbb{B}| + \log\binom{\Lambda+m-1}{m} - \log\binom{N}{m} \le o(N), \tag{PF'}$$

then $\max_z|\Psi^{-1}(z)| \le e^{o(N)}|\mathbb{B}|^{-R}\binom{N}{m}$.

Proof. By orthogonality,

$$|\Psi^{-1}(z)| = |\mathbb{B}|^{-R}\sum_{\alpha\in\mathbb{B}^R}\psi(-\alpha\cdot z)e_m\Big(\psi(\alpha\cdot g(t)) : t\in T\Big),$$

where e_m denotes the m-th elementary symmetric coefficient. The term $\alpha = 0$ is the main term $|\mathbb{B}|^{-R}\binom{N}{m}$. For $\alpha \ne 0$, write $a_t = \psi(\alpha\cdot g(t))$. The generating identity $\sum_{r\ge0} e_r(a_t)u^r = \exp(\sum_{j\ge1}(-1)^{j-1}(\sum_t a_t^j)u^j/j)$ and the hypothesis $|\sum_t a_t^j| \le \Lambda$ for $j\le m$ give $|e_m(a_t : t\in T)| \le [u^m](1-u)^{-\Lambda} = \binom{\Lambda+m-1}{m}$. Summing the nontrivial characters and using the outside factor $|\mathbb{B}|^{-R}$ gives the first bound. The displayed condition (PF') makes the error term subexponential relative to the random-fiber main term. □

Remark 14.2 (Small characteristic cycles)*.* If some integers $j \le m$ vanish in the characteristic, the formal Li–Wan cycle index separates the cycles divisible by $p = \operatorname{char}\mathbb{B}$; the corresponding coefficient majorant is $[u^m](1-u)^{-\Lambda}(1-u^p)^{-(N-\Lambda)/p}$. This bookkeeping identity does not supply the missing character-sum bounds for those cycles, exclude Artin–Schreier phases, or prove a uniform Sidon payment. The pointwise theorem above is therefore deliberately stated in the large-characteristic range. A small-characteristic leaf needs a direct cycle-by-cycle certificate or a separate image-normalized Q/Sidon theorem.

Proposition 14.3 (Weighted Weil minor arcs)**.** *Let T be either a multiplicative coset in $\mathbb{B}^\times$ or the x-coordinate of a torus twin coset. Suppose that for every nonzero $\alpha\in\mathbb{B}^R$ not routed to an earlier major-arc cell, the phase $R_\alpha(t) = \alpha\cdot g(t)$ is a nonconstant rational function on the corresponding genus-zero curve, is not of Artin–Schreier form, and has total pole-degree at most $\Delta_{\rm pole}$. If $m < \operatorname{char}\mathbb{B}$, then the power-sum bound in Theorem 14.1 holds with*

$$\Lambda = C_0(\Delta_{\rm pole}+1)\sqrt{|\mathbb{B}|} + |P|$$

for multiplicative cosets, and with $2C_0(\Delta_{\rm pole}+1)\sqrt{|\mathbb{B}|}+|P|$ for circle twin-cosets. Here P is the planted exceptional set and C_0 is an absolute Weil constant.

Proof. For a multiplicative coset $\theta H \subseteq \mathbb{B}^\times$, write the coset indicator as the average of the multiplicative characters of $\mathbb{B}^\times/H$. Each nontrivial power sum becomes an average of mixed multiplicative-additive character sums $\sum_x \chi(x)\psi(jR_\alpha(\theta x))$. Since $1 \le j < m < \operatorname{char}\mathbb{B}$, multiplication by j does not make the rational function constant or Artin–Schreier. Weil's bound for one-variable mixed character sums gives $C_0(\Delta_{\rm pole}+1)\sqrt{|\mathbb{B}|}$. Removing planted exceptional points changes the sum by at most $|P|$. The circle case is the same after pulling back by $x = (u+u^{-1})/2$ to the norm-one torus; the two torus branches give the factor two. □

Definition 14.4 (Fourier major and minor arcs)**.** *Recall that a nonzero character α is a major arc for a primitive chart if the weighted phase R_α is constant on a positive-density quotient fiber, is Artin–Schreier degenerate, has a pole or zero supported on a positive-density moving divisor, or factors through a proper quotient, Chebyshev map, field-descent locus, tangent rank drop, differential-locator defect, or ray-saturation collapse. Every nonzero character not satisfying one of these routed algebraic conditions is a minor arc. This is a row-specific algebraic classification of Fourier frequencies, not a numerical* *estimate: major arcs still require* (MA)*, and minor arcs still require* (PF) *or another character-sum bound.*

Proposition 14.5 (Major-arc algebraic interface)**.** *Each condition in Definition 14.4 is a constructible condition on the dual phase parameters and points to a quotient, Chebyshev, field-descent, planted-divisor, differential, or rank-loss profile. This classification does not imply that the character vanishes after primal first-match deletion and does not prove* (MA)*. A flatness argument must also bound* (MA)*, recursively or directly.*

Proof. Constancy on a quotient fiber, Chebyshev factorization, Artin–Schreier degeneracy, a positive-density pole divisor, and a rank drop are respectively coefficient, Frobenius, resultant, or determinantal conditions. They therefore identify constructible dual loci on which a recursive estimate can be attempted. Fourier inversion sums characters, not primal witnesses, so no support-level payment follows from this fact. □

Corollary 14.6 (Sidon payment from prefix flatness)**.** *If a primitive smooth or circle chart satisfies the pointwise minor-arc range of Definition 7.12 and the major-arc aggregate* (MA)*, then its Fourier/Sidon-heavy logarithmic moment is $e^{o(Nq)}$, in image normalization, for every logarithmic $q \le (\log N)^C$, for a fixed C.*

Proof. Write $\overline{N}^{\mathrm{amb}} = |\mathbb{B}|^{-R}\binom{N}{m}$. The zero character gives this ambient random-fiber term. By Theorem 14.1 and Proposition 14.3, the minor characters contribute at most $e^{o(N)}\overline{N}^{\mathrm{amb}}$, and (MA) gives the same bound for the major characters. Hence every full-slice, and therefore every residual, fiber is at most $e^{o(N)}\overline{N}^{\mathrm{amb}}$. By Remark 10.2, this also proves the full-image certificate; the exact identity (9.3) then gives the stated image-normalized moment bound. □

Proposition 14.7 (Frontier Weil separation)**.** *Let $N = |T|$, let $m = \theta N + o(N)$ with $0 < \theta < 1$, and suppose a primitive smooth or circle chart has minor-arc power-sum parameter $\Lambda = o(N)$. If the ambient random-fiber scale $\overline{N} = |\mathbb{B}|^{-R}\binom{N}{m}$ satisfies $\log \overline{N} = o(N)$, or more generally $\log \overline{N} \ge -o(N)$, then the prefix-flat inequality* (PF') *holds. In particular, minor arcs alone cannot create a positive-rate max-fiber or Sidon-heavy moment at this near-zero ambient entropy scale.*

Proof. The elementary estimate $\binom{\Lambda+m-1}{m} = \binom{\Lambda+m-1}{\Lambda-1} \le (e(m+\Lambda)/\Lambda)^{\Lambda}$ gives $\log\binom{\Lambda+m-1}{m} = o(N)$, since $\Lambda = o(N)$. The relative minor-arc error exponent is

$$\log\binom{\Lambda+m-1}{m} - \log\overline{N} = R\log|\mathbb{B}| + \log\binom{\Lambda+m-1}{m} - \log\binom{N}{m}.$$

The assumed frontier inequality makes this at most $o(N)$; it may be negative by a linear amount, which is only stronger. Thus the minor-arc contribution to every full-profile fiber is at most $e^{o(N)}\overline{N}$. This statement concerns only minor characters; a full flatness conclusion also requires (MA). □

Theorem 14.8 (Sidon payment in the pointwise flat range)**.** *Let $\Omega^{\circ} \subseteq \Omega_{T,m}$ be a primitive first-match residual of a smooth or circle chart, with weighted prefix map Ψ and ambient scale $\overline{N}^{\mathrm{amb}} = |\mathbb{B}|^{-R}\binom{N}{m}$. Put $A_{\mathrm{amb}} = |\mathbb{B}|^{R}$ and $L = |\Psi(\Omega_{T,m})|$. Suppose the pointwise minor-arc condition* (PF) *and the aggregate condition* (MA) *hold. Then, for every fixed $\sigma > 0$ and every logarithmic $q \le (\log N)^C$, for a fixed C,*

$$\Gamma^{\mathrm{sid}}_{q,\sigma}(\Omega^{\circ}) \le e^{o(Nq)},$$

where Γ^{sid} uses the image normalization of Definition 12.1. Equivalently, no such pointwise-flat residual leaf supports positive-rate low-energy heavy fibers.

Proof. It is enough to bound every residual fiber, and a residual is contained in the full slice. Fourier inversion on that slice splits into the zero, minor, and major characters. The zero character contributes $\overline{N}^{\mathrm{amb}}$. The cycle-index estimate and Proposition 14.3, together with (PF), bound the complete minor contribution by $e^{o(N)}\overline{N}^{\mathrm{amb}}$. Condition (MA) bounds the complete major contribution by the same quantity. Thus every full-slice, and hence every residual, fiber is $e^{o(N)}\overline{N}^{\mathrm{amb}}$. This proves the full-image certificate by Remark 10.2, and Lemma 10.1 transfers the ambient moment bound to image normalization. □

Remark 14.9 (Role of the Sidon-heavy cell). The theorem implements "pay structure, flatten Sidon, inverse high energy" only in the range where both (PF) and (MA) are verified. At entropy depth a phase can have degree up to R, and the raw Weil parameter is of order $R\sqrt{|\mathbb{B}|}$; this is not automatically $o(N)$ when $R \asymp N/\log|\mathbb{B}|$. Outside the pointwise-flat range a different Sidon payment remains necessary.

15. External additive combinatorics

Only two external additive-combinatorics inputs are used in the primitive high-energy proof.

Theorem 15.1 (Balog–Szemeredi–Gowers, energy form)**.** *There is an absolute constant C such that if A is a finite subset of an abelian group and $E(A) \geq |A|^3/K$, then there is $A' \subseteq A$ with $|A'| \geq K^{-C}|A|$ and $|A' - A'| \leq K^C|A'|$.*

This is the usual energy form of the Balog–Szemeredi–Gowers theorem; polynomial dependence on K follows from Gowers's refinement of the Balog–Szemeredi theorem [BS94, Gow98]. The difference-set form follows from the sumset form by replacing A by $-A$ and using standard Ruzsa calculus, or directly by applying the graph form to differences.

Theorem 15.2 (Boolean-cube difference growth)**.** *If $U \subseteq \{0,1\}^d$, then*

$$|U - U| \geq |U|^{3/2}.$$

This is the Boolean-cube specialization of a broader sumset inequality of Green–Matolcsi–Ruzsa–Shakan–Zhelezov [GMR+22]. It also follows from the induced-doubling identity of Matolcsi–Ruzsa–Shakan–Zhelezov [MRSZ22]. Only the displayed Boolean statement is used below.

Remark 15.3. Entropy inverse theorems of Tao and the entropy-doubling theory of Green–Manners–Tao are compatible with this proof and are often useful after one has produced small-doubling trades [Tao10, GMT23]. They are not needed as hypotheses here. The Boolean-cube growth theorem gives a direct contradiction to a large small-doubling subset of a primitive Boolean support fiber.

16. Entropy inverse theory and its role

The terminology "Tao–Gowers method" can refer to two related but distinct steps. The first is BSG, used directly in Proposition 17.1. The second is Freiman-type structural theory, meaning inverse theorems that model a set with small sumset or difference set by a generalized arithmetic progression or a coset progression. It is not needed for the Boolean contradiction, but it can suggest candidate geometric strata in more general bounded-alphabet residuals; those strata still require direct payment proofs.

A *bounded integer trade* is a vector $Y \in \mathbb{Z}^T$ whose coordinates are bounded independently of N, with $\Phi(Y) = 0$ and, on a fixed-weight slice, $\sum_t Y_t = 0$. If Y is random, Y' is an independent copy, and $\mathrm{H}(Y - Y') \leq \mathrm{H}(Y) + o(N)$, then entropy BSG and entropy Freiman theory produce, after conditioning and losing $o(N)$ entropy, a set model with doubling $e^{o(N)}$. Tao's entropy inverse theorem gives this in *coset-progression* language—a subgroup coset plus a generalized arithmetic progression—for Shannon entropy [Tao10]; Green–Manners–Tao recast the theory in terms of entropic doubling and Ruzsa distance [GMT23]; Goh formulates an entropic analogue of additive energy and the entropic BSG connection [Goh24]. In a torsion model one may also use polynomial Freiman–Ruzsa technology after reducing bounded integer trades modulo a large prime with no wraparound.

In the present paper the Boolean-cube difference theorem makes this structural compression unnecessary for the primitive Q step. Once BSG finds $A' \subseteq F_s \subseteq \{0,1\}^T$ with $|A' - A'| \le e^{o(N)}|A'|$, Boolean-cube difference growth already forces $|A'| = e^{o(N)}$. This contradicts the exponential size supplied by moment excess. Therefore there is no separate entropy-Freiman hypothesis in the proof.

For non-Boolean residual trades, entropy inverse theory can suggest coset-progression incidence strata or rank-defect tests. This is a useful heuristic for constructing a row-specific atlas, not an exhaustive dichotomy: each suggested stratum still needs the profile census and slope-projection payment in (A2), and arbitrary weighted charts are not covered by Lemma 10.5. No such heuristic is used in the proof of the Boolean primitive theorem.

17. Primitive Q after the Sidon cut

The key primitive theorem is now short and self-contained modulo Theorems 15.1 and 15.2. It also addresses the main obstruction: ordinary Rényi mass can be Sidon-heavy, but after that cell is removed it must be high-energy, and high-energy is impossible in a large Boolean fiber.

Proposition 17.1 (High-energy Boolean fibers are small). *Let $F \subseteq \{0,1\}^T$, $|T| = N$. If $E(F) \ge |F|^3/K$, then $|F| \le K^{O(1)}$. Consequently an exponential set $|F| \ge e^{cN-o(N)}$ cannot have $E(F) \ge |F|^3/e^{o(N)}$.*

Proof. Apply Theorem 15.1. There is $F' \subseteq F$ with $|F'| \ge K^{-C}|F|$ and $|F' - F'| \le K^C|F'|$. Since $F' \subseteq \{0,1\}^T$, Theorem 15.2 gives $|F' - F'| \ge |F'|^{3/2}$. Hence $|F'|^{1/2} \le K^C$, so $|F| \le K^{3C}$ after increasing C. Taking $K = e^{o(N)}$ rules out $|F| \ge e^{cN-o(N)}$. ☐

Theorem 17.2 (Primitive Q after a Sidon moment payment). *Let $\Omega_N^0 \subseteq \{0,1\}^{T_N}$ be fixed-density full profile slices, with $|T_N| = N$ and $m_N/N \in [\alpha, 1-\alpha]$, and let $\Omega_N^\circ \subseteq \Omega_N^0$ be primitive first-match residuals. Suppose there is a logarithmic moment order $q_N \to \infty$ with $\log L_N/q_N = o(N)$, where $L_N = |\Phi_N(\Omega_N^0)|$ and $\overline{N}_N = |\Omega_N^0|/L_N$, for which $\Gamma^{\mathrm{sid}}_{q_N,\sigma} \le e^{o(Nq_N)}$ for every fixed $\sigma > 0$. Then the leaves satisfy primitive Q:*

$$\max_s |\Omega_N^\circ \cap \Phi_N^{-1}(s)| \le e^{o(N)}\overline{N}_N.$$

Proof. Assume primitive Q fails at positive exponential rate. The lower half of the moment sandwich gives $\Gamma^{\mathrm{ord}}_{q_N} \ge e^{\eta N q_N - o(Nq_N)}$ along a subsequence. The Sidon moment hypothesis and Proposition 12.2 then give a fiber F_s with $f_s \ge e^{\eta N - o(N)}\overline{N}_N$ and $E(F_s) \ge f_s^3 e^{-o(N)}$. Since $\overline{N}_N = |\Omega_N^0|/L_N \ge 1$, this fiber has size $e^{\eta N - o(N)}$, contradicting Proposition 17.1 with $K = e^{o(N)}$. Thus no positive-rate max-fiber failure occurs. ☐

18. Q, SP, and the MCA compiler

The *Q branch* is the maximum-prefix-fiber problem. The *SP branch* (support-pair, or shift-pair, branch) is its second-moment problem: it counts ordered pairs of supports in one prefix fiber. These names label two counting interfaces and carry no assertion that either bound has already been proved. Neither count is automatically the MCA numerator:

a bad slope begins with one support, while converting support pairs to projective slope directions requires a proved multiplicity or incidence bound. We first record the exact moment identities and then state that separate ray compiler.

Definition 18.1 (Q and SP ledgers)**.** *For a first-match residual profile λ, let Ω_λ^0 be its full profile slice, let $\Omega_\lambda^\circ \subseteq \Omega_\lambda^0$ be the residual, and put*

$$\mathcal{S}_\lambda = \Phi_\lambda(\Omega_\lambda^0), \qquad L_\lambda = |\mathcal{S}_\lambda|, \qquad \overline{N}_\lambda = |\Omega_\lambda^0|/L_\lambda, \qquad Q_\lambda(s) = |\Omega_\lambda^\circ \cap \Phi_\lambda^{-1}(s)|.$$

The Q ledger is discharged if $\max_s Q_\lambda(s) \le e^{o(n)}\overline{N}_\lambda$ for every primitive profile. The support-pair SP statistic is $\sum_s Q_\lambda(s)^2$. A distinct-ray ledger is discharged only by a direct slope bound or by (RC) *below. If the ambient target count $|\mathbb{B}_\lambda|^{R_\lambda}$ is used in place of L_λ, the ledger must also record the full-image certificate* (FI) *or a direct ambient theorem.*

The compiler uses two elementary inequalities repeatedly. They are included to make explicit how the logarithmic moment, max-fiber Q, and shift-pair second moment interact.

Lemma 18.2 (Moment sandwich)**.** *Let $N(s) \ge 0$ be fiber sizes over a finite syndrome set of size L, let $\overline{N} = L^{-1}\sum_s N(s)$, and put $R(s) = N(s)/\overline{N}$. For $r \ge 2$, define $\Gamma_r = L^{-1}\sum_s R(s)^r$. Then*

$$\max_s R(s) \le (L\Gamma_r)^{1/r}.$$

Conversely, if $\max_s R(s) \le M$, then $\Gamma_r \le M^{r-1}$.

Proof. The first inequality follows from $\max_s R(s)^r \le \sum_s R(s)^r = L\Gamma_r$. For the converse, $R(s)^r \le M^{r-1}R(s)$, and $L^{-1}\sum_s R(s) = 1$. ☐

Corollary 18.3 (Finite moment floor)**.** *Suppose a finite row has remaining Q bit margin Δ, meaning that one needs $\max_s R(s) \le 2^\Delta$. A moment proof of order r can certify this only if*

$$\log_2 \Gamma_r + \log_2 L \le r\Delta,$$

where L is the number of realized targets in the selected normalization. In an ambient prefix proof this specializes to $\log_2 \Gamma_r^{\mathrm{amb}} + R\log_2|\mathbb{B}| \le r\Delta$; transferring an image-normalized moment to that statement additionally costs $(r-1)\log_2(|\mathbb{B}|^R/L)$.

Proof. The moment sandwich gives $\log_2 \max_s R(s) \le (\log_2 L + \log_2 \Gamma_r)/r$. To make this at most Δ one needs the displayed inequality. ☐

The finite corollary explains why a small adjacent margin cannot be closed by a fixed-order moment when $R\log|\mathbb{B}|$ is large.

Definition 18.4 (Saturated line rays)**.** *Here a* ray *is the projective class of the residual one-dimensional line or explanation direction; on the finite affine chart it is represented by its slope γ. A raw witness (γ, S, h) determines a finite slope and an explanation ray. Two witnesses are saturation-equivalent if both agree after common codeword, quotient, and planted factors are removed. The saturated ray set is the image of this equivalence relation; its size is not obtained by dividing the raw witness count unless a lower fiber bound is proved.*

Distinct rays therefore mean distinct projective directions, and hence distinct finite slopes after the point at infinity is handled separately. *Saturation* is the passage from raw support/explanation witnesses to this projective image after common components have been factored out. Many raw witnesses may saturate to one ray. A *ray compiler* is a proved image–fiber or direct incidence estimate that converts the available support or support-pair count into a count of these distinct rays.

Definition 18.5 (Support interpolation functionals)**.** *Let $T \subseteq D$, $|T| = m \geq k$, put $w = m - k$, and let $Q_T(X) = \prod_{x \in T}(X - x)$. For a word u, define*

$$\mathcal{A}_T(u) = \left(\sum_{x \in T} \frac{u(x)x^r}{Q'_T(x)} \right)_{0 \leq r < w} \in \mathbb{F}^w. \tag{12.1}$$

Lemma 18.6 (Slope elimination)**.** *A word u is explained by $\mathrm{RS}_{\mathbb{F}}(D, k)$ on T if and only if $\mathcal{A}_T(u) = 0$. Hence $u_0 + \gamma u_1$ is explained on T if and only if*

$$\mathcal{A}_T(u_0) + \gamma \mathcal{A}_T(u_1) = 0. \tag{12.2}$$

If T is not a common-support explanation, it supports at most one finite slope.

Proof. The vectors $(Q'_T(x)^{-1}x^r)_{x \in T}$, $0 \leq r < w$, form a parity-check matrix for $\mathrm{RS}_{\mathbb{F}}(T, k)$, proving the first assertion. The second is linearity. If two distinct slopes solved (12.2), both functionals would vanish and the pair would be explained on T. □

Lemma 18.7 (Second moment identity for primitive pairs)**.** *For any prefix profile λ, the number of ordered pairs of residual supports with common prefix syndrome is exactly*

$$\sum_s Q_\lambda(s)^2, \qquad Q_\lambda(s) = |\Omega^\circ_\lambda \cap \Phi_\lambda^{-1}(s)|.$$

and it is bounded by

$$\left(\max_s Q_\lambda(s)\right) \sum_s Q_\lambda(s).$$

Proof. Partition the residual supports by their prefix value. Ordered pairs in one part give $Q_\lambda(s)^2$, and summing proves the identity. The final inequality is elementary. No assertion about MCA rays is made. □

Q therefore controls a support-pair statistic. It does not justify dividing that statistic by $|\Omega^0_\lambda|$: a ray can have a single representative at exact agreement.

Proposition 18.8 (Q and SP do not determine rays)**.** *Let $\Phi : \Omega \to \Sigma$ be any support-profile map with $\Omega \neq \varnothing$, put $Q(s) = |\Phi^{-1}(s)|$, and $\mathrm{SP} = \sum_s Q(s)^2$. Regard Ω as an abstract set of candidate noncommon supports, and suppose the only retained slope-incidence axiom is that each support carries at most one slope. For every nonempty finite slope set Γ and every $1 \leq M \leq \min\{|\Omega|, |\Gamma|\}$, within the class of abstract incidence structures obeying these retained data there is a partial support-to-slope map having exactly M distinct values. There is also a one-value map. Hence no ray bound can be deduced from Q, SP, and that axiom alone; an RS-specific bound must use additional algebraic incidence information.*

Proof. Choose M distinct supports and M distinct slopes and map them bijectively, leaving all other supports unlabeled. This changes neither Φ, its fibers, nor SP, and it respects one-slope-per-support. Mapping the same chosen supports to one slope gives identical Q and SP data with a one-ray image. The missing datum is the image incidence, not another moment of the prefix fibers. □

Proposition 18.9 (Exact pair-to-ray multiplicity requirement)**.** *Let $\mathcal{P} = \{(A, B) \in \Omega^2 : \Phi(A) = \Phi(B)\}$, so $|\mathcal{P}| = \mathrm{SP}$. Let Z be a bad-slope set and $I \subseteq Z \times \mathcal{P}$ an incidence relation, and let $H, J \geq 1$. If every $\gamma \in Z$ is incident with at least H pairs and every pair is incident with at most J slopes, then*

$$|Z| \leq \frac{J\,\mathrm{SP}}{H} \leq \frac{J(\max_s Q(s))|\Omega|}{H}. \tag{12.3}$$

Consequently this pair-count argument yields a ray estimate at the same profile scale provided $J|\Omega|/H = e^{o(n)}$; alternatively one may prove a direct transverse-secant estimate. Merely proving that every ray has one representing pair is insufficient.

Proof. Double counting gives $H|Z| \le |I| \le J|\mathcal{P}| = J\,\mathrm{SP}$. The second inequality follows from $\mathrm{SP} \le (\max_s Q(s)) \sum_s Q(s) = (\max_s Q(s))|\Omega|$. $\square$

By Theorem 3.2, the genuinely missing invariant can be taken to be $\Theta_t(y_0, y_1)$. Equivalently one may construct a higher-dimensional split-pencil *eliminant*, meaning a polynomial in the slope parameter obtained after eliminating locator and support variables, whose degree bounds that transverse secant incidence. By Lemma 3.3, only the dependent-union, or maximum-distance-separable (MDS) circuit, strata can support more than $t+1$ rays; this localizes the remaining geometry but does not bound it.

Here an *MDS-circuit stratum* is a chart on which the relevant union of generalized Vandermonde parity-check columns has a minimal linear dependence. Equivalently, deleting any one column restores independence. The name identifies the only dependence pattern left by the MDS property; it does not provide a count for that stratum.

The next hypothesis packages exactly this missing conversion. Its number H_λ is a lower bound on how many certified support pairs represent one bad ray, while J_λ is an upper bound on how many bad rays one pair can represent. The ratio in (RC) is what makes the resulting double count stay at the profile scale.

Hypothesis 18.10 (Ray compiler). For every received line and primitive residual profile λ, let Z_λ° be its actual first-match set of unpaid bad slopes and put

$$\mathcal{P}_\lambda = \{(A, B) \in (\Omega_\lambda^\circ)^2 : \Phi_\lambda(A) = \Phi_\lambda(B)\}.$$

The row satisfies (RC) on this profile if either it has the direct bound $|Z_\lambda^\circ| \le e^{o(n)}(1 + \overline{N}_\lambda)$, or there are an incidence $I_\lambda \subseteq Z_\lambda^\circ \times \mathcal{P}_\lambda$ and numbers $H_\lambda, J_\lambda \ge 1$ such that

$$\text{(RC)} \qquad \deg_{I_\lambda}(\gamma) \ge H_\lambda, \qquad \deg_{I_\lambda}(A, B) \le J_\lambda, \qquad \frac{J_\lambda |\Omega_\lambda^0|}{H_\lambda} = e^{o(n)}.$$

All bounds are uniform in the received line and realized profile. The incidence must arise from the RS witness geometry; its existence is not a consequence of Q or SP. In particular, one representing pair per ray does not supply the required lower degree.

Proposition 18.11 (Q plus the ray compiler)**.** *If primitive Q is discharged and* (RC) *holds on every residual profile, then all primitive ray ledgers are discharged at total cost* $e^{o(n)}\mathfrak{E}_n$*.*

Proof. The boundary prefix fibers are bounded by Q. For every remaining unpaid slope choose one noncommon support, as permitted by Lemma 18.6; (RC) bounds their distinct slope set by $e^{o(n)}\overline{N}_\lambda$. Sum over the subexponential profile set and use (1.4). $\square$

Proposition 18.12 (Primitive residual numerator)**.** *For a ledger-admissible sequence, the total primitive contribution is at most* $e^{o(n)}\mathfrak{E}_n$*.*

Proof. The certified Sidon payment and Theorem 17.2 give Q on the primitive leaves. The profile sum is part of $\mathfrak{E}_n$, and Proposition 18.11 applies using (RC). $\square$

Proposition 18.13 (Asymptotic MCA numerator bound)**.** *For a ledger-admissible sequence at agreement* a_n*,*

$$B_{C_n}^{\mathrm{MCA}}(a_n) \le e^{o(n)}\mathfrak{E}_n(a_n).$$

Proof. Apply the first-match ledger. The algebraic and direct-ray cells are paid by (A2), the Fourier/Sidon cell is certified by (A4), and the remaining primitive part is Proposition 18.12. The first-match bound adds the distinct-slope budgets. $\square$

This is also a detailed proof of Theorem 1.5; a closed ledger is the theorem-verified special case. The threshold statement uses the entropy calculation and the lower construction.

19. Entropy bookkeeping and the lower side

Let $\rho_n = k_n/n$, $K_n = k_n + 1$, $\beta_n = \log_2 |\mathbb{B}_n|$, and write $a_n = K_n + g_n n + O(1)$. Stirling's formula gives

$$\log_2 \overline{N}_1(a_n) = n\Big(H_2(\rho_n + g_n) - \beta_n g_n\Big) + O(\log n + \beta_n), \tag{13.1}$$

where H_2 is the bit-valued entropy fixed in the Introduction. Let $\varnothing \neq \Gamma_n \subseteq \mathbb{F}_n$, let $0 \le \varepsilon_n^* \le 1$, and put $B_n^* = \lfloor \varepsilon_n^* |\Gamma_n| \rfloor$. Write $\ell_n = o(n)$ for the ledger overhead in bits, so that its certified bound is $2^{\ell_n} \mathfrak{C}_n$. Agreement a is *safe* if $B^{\mathrm{MCA}}_{C_n,\Gamma_n}(a) \le B_n^*$ and unsafe otherwise. Since the numerator is nonincreasing in a, when a safe agreement exists its integer threshold is

$$a_n^* = \min\{a \in \{k_n + 1, \dots, n\} : B^{\mathrm{MCA}}_{C_n,\Gamma_n}(a) \le B_n^*\}.$$

A target crossing is a comparison with $\log_2 B_n^*$, not automatically the zero of the entropy expression. Safety at $a_n \ge k_n + 1$ is impossible when $B_n^* < \min\{|\Gamma_n|, n - a_n + 1\}$ by Proposition 3.5.

We call the target *viable at agreement* a when it is not already below this universal floor, namely when $B_n^* \ge \min\{|\Gamma_n|, n - a + 1\}$. Viability is only a necessary condition for safety; it is not an upper bound. A *target reserve* is the explicit bit slack in one of the two comparisons: on the safe side it is the amount by which $\log_2 B_n^*$ exceeds $\ell_n + \log_2 \mathfrak{C}_n(a)$, and on the unsafe side it is the amount by which the logarithm of a proved bad-slope construction exceeds $\log_2 B_n^*$. An *agreement reserve* s_n is a number of integer agreement steps taken past a candidate crossing; away from tangency (that is, when the derivative at the crossing is nonzero), the derivative of the entropy exponent converts those steps into the corresponding bit reserve.

For a concave entropy level function, the superlevel set can have two endpoints. Its *right crossing* is the larger, equivalently the supremum of the superlevel set. It is an *interior right crossing* when it lies strictly inside $[0, 1 - \rho_n]$ and satisfies the level equation there. If its derivative is nonzero, the crossing is transverse; if the derivative vanishes, it is tangential and a linear agreement reserve need not produce a linear bit gain.

Lemma 19.1 (Target-aware safe side)**.** *Under a closed ledger, agreement a_n is safe at target B_n^* as soon as*

$$2^{\ell_n} \mathfrak{C}_n(a_n) \le B_n^*. \tag{13.2}$$

If $\mathfrak{C}_n(a) = e^{o(n)} \overline{N}_1(a)$ in a window and $\log(1 + B_n^*) = o(n)$, the leading crossing in that window is the zero of $H_2(\rho_n + g) - \beta_n g$, with a reserve large enough to dominate $\ell_n + O(\log n + \beta_n) + \log_2(1 + B_n^*)$. The zero-crossing description additionally requires target viability $B_n^* \ge \min\{|\Gamma_n|, n - a_n + 1\}$, and in particular $B_n^* \ge 1$.

Proof. The first assertion is Proposition 18.13. For the second, use (13.1). Away from a tangential endpoint, an agreement reserve s_n changes the entropy by $s_n |H_2'(\rho_n + g) - \beta_n| + o(s_n)$. Requiring this quantity to dominate the written logarithmic costs gives (13.2). This formulation remains valid when β_n varies; the phrase "$o(n)$ corridor" alone is not quantitative enough. $\square$

Proposition 19.2 (Exact unsafe test)**.** *Let*

$$L_n = \left\lceil \binom{n}{a_n} |\mathbb{B}_n|^{-(a_n - k_n - 1)} \right\rceil, \qquad q_n = |\mathbb{F}_n| > n.$$

If

$$\left\lceil \frac{|\Gamma_n|}{q_n} \left\lceil \frac{L_n(q_n - n)}{q_n - n + k_n(L_n - 1)} \right\rceil \right\rceil > B_n^*, \tag{13.3}$$

then agreement a_n is unsafe at the target. The same statement holds with L_n replaced by any larger identity, quotient, Chebyshev, or remainder-profile list proved for the dimension-k_n+1 code.

Proof. The identity list is Proposition 4.1, and Theorem 4.2 produces a bad-slope set $Z \subseteq \mathbb{F}_n$ of the inner-ceiling size. Replacing a line (r_0, r_1) by $(r_0 + \delta r_1, r_1)$ translates Z by $-\delta$. Averaging over $\delta \in \mathbb{F}_n$ gives $|Z||\Gamma_n|/q_n$ challenge slopes, so some translate has at least the outer-ceiling number. The profile variants use the same conversion after their list sizes have been established. □

Proof of Theorem 1.6 and Corollary 1.7. At $a_{+,n}$, the profile-envelope hypothesis is precisely (13.2), so $B^{\mathrm{MCA}}_{C_n,\Gamma_n}(a_{+,n}) \le B^*_n$. At $a_{-,n}$, the profile-list hypothesis is (13.3), with the proved list size for the selected identity, quotient, Chebyshev, or remainder profile, so $B^{\mathrm{MCA}}_{C_n,\Gamma_n}(a_{-,n}) > B^*_n$. The numerator is nonincreasing in the agreement threshold: every witness agreeing on at least $a+1$ coordinates is also a witness at threshold a. Consequently the safe set is nonempty, its least element satisfies $a_{-,n} < a^*_n \le a_{+,n}$, and substituting $\delta^*_n = 1 - a^*_n/n$ proves (1.6).

If $a_{+,n} - a_{-,n} = o(n)$, any common crossing a_n between them satisfies $a^*_n = a_n + o(n)$. Thus $a_n = k_n + 1 + g_n n + o(n)$ and $k_n/n \to \rho$ give $\delta^*_n = 1 - \rho - g_n + o(1)$. In the identity-dominant window, Stirling's formula (13.1) says

$$\frac{1}{n}\log_2 \overline{N}_1 = H_2(\rho_n + g) - \beta_n g + o(1).$$

Comparison with $n^{-1}\log_2(1+B^*_n) = \tau_n$ yields (1.7); concavity selects its rightmost crossing as in (1.7). The corollary assumes, rather than deduces from exponent notation, that quantitative reserves furnish the exact safe and unsafe tests within $o(n)$ of that crossing. The bracket therefore gives its claimed threshold. When $\log(1 + B^*_n) = o(n)$, $\tau_n = o(1)$, and continuity of the rightmost level-set endpoint gives $g_{T,n} = g^*(\rho_n, \beta_n) + o(1)$, which proves the zero-exponent specialization. □

Proof of Theorem 1.9. The numerator bound is Proposition 18.13. If agreements $a_- < a_+$ satisfy (13.3) at a_- and (13.2) at a_+, monotonicity localizes the integer threshold to $[a_-, a_+]$. Under the additional hypotheses in Theorem 1.6 and Corollary 1.7, these agreements can be chosen $o(n)$ apart around the target-adjusted entropy crossing, giving the stated radius corridor. □

20. Finite-row audit and asymptotic scope

The theorem above is designed for limits. It does not assert a finite adjacent-row certificate. For a finite challenge set Γ, the integer target is $B^* = \lfloor \varepsilon^* |\Gamma| \rfloor$, and every cell payment must be made with explicit constants. The finite Q target is row-sharp: if the entire remaining budget is assigned to Q, one must prove

$$\max_s |\Omega^\circ \cap \Phi^{-1}(s)| \le B^*$$

inside the actual first-match residual, not merely inside the raw null fiber $\Phi^{-1}(0)$ before first-match deletion. Moment arguments at fixed order are usually incapable of proving such a tight finite target unless the order is comparable with $w \log|\mathbb{B}|/\Delta$, where Δ is the remaining bit margin.

The balanced-core and split-pencil ledgers also require finite care. Raw support counts are the wrong numerator for MCA because many supports can saturate to the same slope ray. One must count saturated primitive line rays. The one-parameter moving-root pencil is paid after common roots are removed, but a finite proof must verify that every residual balanced-core chart has actually reduced to a paid pencil or has been isolated as another cell. Similarly, the Sidon/Fourier cell needs explicit Fourier-tail constants or an explicit major-arc stratification.

The same issues persist asymptotically as explicit hypotheses: a Sidon payment on the genuinely primitive residual, the full partial-occupancy profile census, and the transverse-secant ray bound must be proved with $e^{o(n)}$ losses. The unrestricted ambient identity-normalized Sidon statement is already false by Theorem 8.9. A closed ledger is therefore a substantive row theorem, not merely a matter of restoring finite constants.

21. Summary of the proof chain

The unconditional analytic chain is short. Once a primitive Boolean leaf has a logarithmic moment satisfying $\log L/q = o(N)$ and a Sidon moment payment, any positive-rate max-fiber failure produces a large high-energy fiber. BSG then gives a large subset with $e^{o(N)}$ difference doubling, whereas Boolean-cube difference growth forces $|A-A| \geq |A|^{3/2}$, a contradiction. Thus the Sidon hypothesis implies primitive Q. To reach an MCA numerator one must additionally supply the algebraic distinct-slope budgets and (RC); to reach a threshold one must then compare the full profile envelope with the target and use the exact collision-aware lower test. The quotient construction in Section 8 shows why none of these additional comparisons may be omitted.

22. Ledger interfaces and notation

For clarity we collect the interfaces between the pieces of the proof. The objects below are the quantities a row-specific verification must output.

MCA interface. Input: a received line (r_1, r_2), an agreement a, and the Reed–Solomon row $\mathrm{RS}_{\mathbb{F}}(D, k)$. Output: an ordered first-match list of slope supersets E_i covering all bad slopes. The required bound is $\sum_i |E_i| \leq e^{o(n)} \mathfrak{E}_n(a)$. The individual witnesses may overlap; slopes do not after first match.

Algebraic-cell interface. Input: a constructible stratum $\mathcal{C} \subseteq \mathfrak{W}_a$. Output: a slope bound for first-match witnesses in $\mathcal{C}$. The permitted reasons for such a bound are degree of a finite projection, entropy loss in support choices, field descent, determinantal rank loss, or ray saturation of many raw witnesses to fewer line rays. A cell is not paid merely because it has a name; the bound is part of the data.

Primitive-prefix interface. Input: a primitive profile λ. Output: a fixed-density Boolean slice, an explicitly verified prefix map, its realized image size L_λ, the image scale $\overline{N}_\lambda = |\Omega^0_\lambda|/L_\lambda$, and the moment-accessibility conditions of Theorem 17.2. An ambient scale may be substituted only after (FI) or a direct ambient theorem has been proved. A Sidon moment payment then yields the stated Q bound; merely assigning a name to a heavy leaf does not.

Sidon/Fourier interface. Input: the same primitive leaf. Output: a proof that the low-energy heavy moment $\Gamma^{\mathrm{sid}}_{q,\sigma}$ is $e^{o(Nq)}$. One sufficient route is (PF) plus the aggregate (MA); a dual major-arc label by itself is insufficient. Once this interface is supplied, the high-energy proof uses only BSG and Boolean-cube difference growth.

Support-pair and ray interfaces. Q gives the exact support-pair bound $\sum_s Q_\lambda(s)^2$. It does not saturate those pairs to slopes. That final projection is the separate ray interface (RC), or a direct distinct-slope theorem.

Entropy-frontier interface. Input: the asymptotic numerator $e^{o(n)}\mathfrak{E}_n(a)$, the target, and a target-viable agreement. Output: the safe side of the threshold. When the identity profile dominates, the leading calculation is

$$\log_2 \overline{N}_n = n(H_2(\rho + g_n) - \beta_n g_n) + o(n).$$

Crossing the relevant target-adjusted level by a quantitative reserve that dominates every error proves safety; a bare $o(n)$ displacement does not.

The conditional compiler can therefore be read as

$$\text{paid algebraic profiles} + \text{Sidon moment payment} \longrightarrow \text{primitive Q},$$

$$\text{primitive Q} + (\text{RC}) + \text{profile envelope} \longrightarrow \text{MCA numerator},$$

$$\text{numerator} + \text{target reserve} + \text{exact lower test} \longrightarrow \text{threshold corridor}.$$

The analytic implication to primitive Q and the formal compiler implications are proved here. The premises (A2), (MA), and (RC) are row-specific enumerative inputs; smooth/circle geometry organizes them but does not establish them unconditionally.

Remark 22.1 (Why the theorem is not phrased as a black-box inverse theorem). A single black-box theorem of the form "Rényi collision excess implies an approximate group" would be false for the ordinary moment, because of Sidon-heavy fibers. The correct inverse statement is ledger-valued: collision excess implies either a Sidon/Fourier cell or a high-energy fiber; the high-energy fiber is impossible in a large Boolean primitive leaf. This ledger-valued form is the reason the paper is phrased in moduli terms.

23. Coordinate form of the witness atlas

This section expands the moduli language into coordinates. It is included to make clear that the cells used above are not informal cases but constructible pieces of a single incidence problem. Fix a row $C = \mathrm{RS}_{\mathbb{F}}(D,k)$, $D = \{t_1,\dots,t_n\}$, and an agreement a. For an a-support S write

$$Q_S(X) = \prod_{t\in S}(X-t) = X^a + q_1(S)X^{a-1} + \cdots + q_a(S).$$

The prefix map is the projection $S \mapsto (q_1(S),\dots,q_w(S))$, where $w = a-k-1$ when using the dimension-$(k+1)$-to-k simple-pole conversion. The shift by one comes from passing from k to $K = k+1$: the first w coefficients are those not absorbed by a degree-$< K$ explanation.

Definition 23.1 (Support-prefix incidence). *Let* $\mathrm{Gr}_a(D)$ *denote the set of a-subsets of D. The* support-prefix incidence *is the graph*

$$\mathcal{I}_w = \{(S,z) \in \mathrm{Gr}_a(D)\times \mathbb{A}^w : z = (q_1(S),\dots,q_w(S))\}.$$

Its projection to $\mathbb{A}^w$ *is the* support-prefix image. *For* $z \in \mathbb{B}^w$*, write* $N_w(z) = |\{S \in \mathrm{Gr}_a(D) : (q_1(S),\dots,q_w(S)) = z\}|$.

For the unprofiled locator-list problem put $L_w = |\{\Phi_w(S) : S \in \binom{D}{a}\}|$. Primitive Q is exactly the assertion $\max_z N_w(z) \le e^{o(n)}\binom{n}{a}/L_w$. Under the full-image certificate $L_w \ge e^{-o(n)}|\mathbb{B}|^w$, this is equivalent at exponential accuracy to the familiar ambient bound $\max_z N_w(z) \le e^{o(n)}\binom{n}{a}|\mathbb{B}|^{-w}$. An arbitrary received-line MCA witness instead satisfies the interpolation-functional pencil (12.2); identifying all of its boundary charts with locator-prefix fibers is part of the atlas and ray hypotheses, not an automatic elimination step. The following dictionary applies to the unprofiled prefix map in large characteristic.

Lemma 23.2 (Newton dictionary). *Assume the characteristic is larger than w. For $1 \le j \le w$, the first w elementary coefficients $(q_1,\dots,q_w)$ and the first w power sums $p_j(S) = \sum_{t\in S} t^j$ determine each other by triangular polynomial automorphisms. Consequently the prefix fibers may be counted using the linear map*

$$x \longmapsto \left(\sum_{t\in D} x_t t, \sum_{t\in D} x_t t^2, \dots, \sum_{t\in D} x_t t^w\right)$$

$on the fixed-density slice$ with only bounded-complexity coordinate changes.

Proof. Newton's identities give $p_j + q_1 p_{j-1} + \cdots + q_{j-1}p_1 + jq_j = 0$. Thus q_j is determined recursively from $p_1, \ldots, p_j$ when j is invertible, and conversely. No small-characteristic linearization is claimed here; one must retain elementary coordinates or prove a suitable divided-power replacement separately. □

Definition 23.3 (MCA witness scheme)**.** *For a received line $r = (r_0, r_1)$, the exact-a MCA witness scheme means the fiber $\mathcal{W}_a(r)$ of the universal locally closed incidence in Definition 2.5. Its support coordinate belongs to the finite reduced support scheme $\mathfrak{S}_a(D) = \coprod_{S \in \binom{D}{a}} \operatorname{Spec} \mathbb{F}$, and its slope projection is $(\gamma, S, h) \mapsto \gamma$. By Lemma 2.4, its slope image is the threshold-a MCA-bad set whenever $a \geq k+1$.*

For fixed received data and S, the closed incidence equations in Definition 23.3 are linear in the coefficients of h and in γ; support-wise nontriviality is the separate open condition of Definition 2.1. Eliminating h gives the exact support-dependent equations (12.1)–(12.2); their weights involve Q'_S and the received values. A reduction of these equations to fixed top locator coefficients is an additional chart theorem and is required by (A2) or (RC).

Proposition 23.4 (Atlas criterion)**.** *Suppose a smooth or circle sequence admits a witness-exhaustive first-match atlas of the algebraic types in Section 5.1, together with its analytic Sidon component and primitive leaves, and suppose the total description entropy is $o(n)$. Then the atlas has $e^{o(n)}$ profiles and satisfies the census part of* (A2)*.*

Proof. The assumed atlas is exhaustive. Its algebraic pieces are constructible by Lemma 5.3; its Sidon component is analytic rather than constructible. The description-entropy hypothesis and Lemma 11.2 give the subexponential count. The proposition deliberately does not derive exhaustiveness from smoothness. □

Lemma 23.5 (Slope multiplicity after support fixation)**.** *Fix a support S with $|S| \geq k+1$. For a received line (r_1, r_2), either the pair is explained on S, or the set of slopes γ for which $r_1 + \gamma r_2$ is explained on S has size at most one.*

Proof. If two distinct slopes γ_1, γ_2 are explained by polynomials h_1, h_2 of degree less than k on S, then $(h_1 - h_2)/(\gamma_1 - \gamma_2)$ explains r_2 on S, and $h_1 - \gamma_1 r_2$ explains r_1 on S. Since $|S| \geq k+1$, the interpolants are unique, so the pair is explained on S. Thus in the MCA-bad locus, each support contributes at most one slope. □

This lemma says that one fixed noncommon support yields at most one slope. It does not put all bad supports into one prefix fiber and does not control how many supports represent different slopes. Consequently Q remains a support-fiber estimate, and (RC) or a direct slope projection is still required.

24. The Li–Wan–Weil calculation and residual stability

This section spells out two interfaces that are easy to obscure in a ledger proof. First, the Li–Wan sieve is used through the elementary-coefficient estimate already proved in Theorem 14.1; the finite numerical certificate is (PF), while the asymptotic smooth/circle theorem uses Theorem 14.8. Second, the primitive residual Ω° is a first-match subset of a full fixed-density profile slice; it need not be Zariski open. All prefix-flat estimates are applied to the full slice, so arbitrary first-match deletion cannot create a larger fiber than the full slice already had.

Lemma 24.1 (Cycle-index coefficient bound)**.** *Let a_t be complex numbers with $|a_t| \le 1$, indexed by a finite set T. If $|\sum_t a_t^j| \le L_j$ for $1 \le j \le m$, then*

$$|e_m(a_t : t \in T)| \le [u^m] \exp\left(\sum_{j=1}^{m} \frac{L_j}{j} u^j\right).$$

In particular, if $L_j \le L$ for all $j \le m$, then $|e_m| \le \binom{L+m-1}{m}$.

Proof. Use the identity $\sum_{r\ge0} e_r u^r = \exp(\sum_{j\ge1}(-1)^{j-1}(\sum_t a_t^j)u^j/j)$, truncate at degree m, and compare coefficients after taking absolute values. The uniform bound gives $\exp(L\sum_{j\ge1} u^j/j) = (1-u)^{-L}$. □

Lemma 24.2 (Residual monotonicity)**.** *Let Ω^0 be a full profile slice, let $\Omega^\circ \subseteq \Omega^0$, and let $\Phi : \Omega^0 \to \mathbb{B}^R$ be a syndrome map. If $|\Omega^0 \cap \Phi^{-1}(s)| \le U$ for every s, then the same holds on Ω°. Consequently every logarithmic moment of the residual, normalized at the ambient scale $\overline{N}_0^{\mathrm{amb}} = |\Omega^0||\mathbb{B}|^{-R}$, is bounded by the corresponding full-slice max-fiber bound. If $U = e^{o(N)}\overline{N}_0^{\mathrm{amb}}$, then* (FI) *follows and the same conclusion holds in image normalization.*

Proof. The fiber inclusion gives the first assertion. An ambient max-fiber bound gives the ambient moment bound by the upper half of the moment sandwich. It also proves (FI) by Remark 10.2; now apply Lemma 10.1. □

Proposition 24.3 (Smooth and circle prefix-flatness criterion)**.** *Consider a primitive smooth or circle chart with full active set T, density $m/|T| \in [\alpha, 1-\alpha]$, prefix depth R, base field $\mathbb{B}$, and minor-arc degree bound D. Suppose $m < \operatorname{char}\mathbb{B}$, the corresponding Λ satisfies* (PF)*, and the major characters satisfy* (MA)*. Then the full profile slice and every first-match residual satisfy the ambient max-fiber estimate and hence image-normalized Q.*

Proof. Weil and the cycle-index estimate bound the minor-character aggregate, while (MA) bounds the major-character aggregate. Fourier inversion gives the full-slice estimate, and residual monotonicity transfers it to the residual. □

Remark 24.4 (How to read the numerical condition)*.* Condition (PF) says exactly that the Li–Wan–Weil error term is subexponential relative to the random prefix fiber. Written exponentially, it is

$$|\mathbb{B}|^R\binom{\Lambda+m-1}{m} \le e^{o(N)}\binom{N}{m}.$$

For finite certificates this is the minor-character check. It must be paired with an exact major-character estimate. If (PF) or (MA) fails, the Fourier argument gives no flatness conclusion; one must prove the Sidon moment bound by another method.

Proposition 24.5 (From prefix flatness to Sidon payment)**.** *Under the hypotheses of Proposition 24.3, put $L = |\Phi(\Omega^0)|$ and $\overline{N}_0^{\mathrm{img}} = |\Omega^0|/L$. For every fixed $\sigma > 0$ and logarithmic q,*

$$L^{-1}\sum_{\Delta_s \le e^{-\sigma N}} \left(\frac{f_s}{\overline{N}_0^{\mathrm{img}}}\right)^q = e^{o(Nq)}.$$

Proof. This is Corollary 14.6. The low-energy restriction can only reduce the image-normalized full moment. □

25. Threshold compiler details

The asymptotic theorem uses two numerical facts repeatedly. The first converts a max-fiber failure into a logarithmic moment failure; the second converts the Q max-fiber bound into the SP second-moment bound. They are elementary, but they are the bookkeeping bridge between the moduli ledger and the MCA numerator.

Lemma 25.1 (Largest fiber and logarithmic moments). *Let $N(z)$ be nonnegative integers with $\sum_z N(z) = M$, and let L be the number of targets. Put $\bar{N} = M/L$ and $R = \max_z N(z)/\bar{N}$. For every $q \geq 2$,*

$$R^q/L \leq \sum_z (N(z)/\bar{N})^q/L \leq R^{q-1}.$$

Consequently a positive-rate max-fiber failure $R \geq e^{\eta n}$ gives a positive-rate logarithmic moment failure for every $q \to \infty$ with $\log L/q = o(n)$.

Proof. The lower bound is the contribution of the largest fiber, divided by L. For the upper bound, $N(z)^q \leq (\max N)^{q-1}N(z)$, so the normalized sum is at most R^{q-1}. Taking logarithms in the lower bound gives the stated moment-accessibility condition. □

Lemma 25.2 (Q-to-SP second moment). *If $\max_z N(z) \leq \kappa\bar{N}$, then $M^{-1}\sum_z N(z)^2 \leq \kappa\bar{N}$. In particular $\kappa = e^{o(n)}$ gives the SP scale $e^{o(n)}\bar{N}$.*

Proof. One has $\sum_z N(z)^2 \leq (\max_z N(z))\sum_z N(z) = \kappa\bar{N}M$. Dividing by M proves the claim. □

Definition 25.3 (Integer staircase). *At agreement $a = K + w$, the identity profile has scale $\overline{N}_1 = \binom{n}{a}|\mathbb{B}|^{-w}$. Its entropy exponent is $nH_2(a/n) - w\log_2|\mathbb{B}| + o(n)$. It is a safe numerator budget only after the full profile envelope has been proved to be $e^{o(n)}\overline{N}_1$ and the other compiler hypotheses hold.*

Proposition 25.4 (Target-aware crossing criterion). *Assume the full profile envelope is $e^{o(n)}\overline{N}_1$ in a window, the conditional numerator hypothesis of Theorem 1.5 holds there, and the target is viable. An agreement is certified safe by (13.2) and certified unsafe by (13.3). The zero of $H_2(\rho + g) - \beta g$ describes the leading location only when the target has subexponential logarithm and a quantitative reserve dominates all errors and pole collisions.*

Proof. This is Lemma 19.1 and Proposition 19.2. Concavity identifies the leading zero, but an unspecified $o(1)$ displacement need not dominate the error terms. The lower side uses the exact list size and the collision-aware conversion, not merely the sign of the identity exponent. □

Proposition 25.5 (First-match summation at the frontier). *Suppose a witness-exhaustive ordered ledger has $e^{o(n)}$ profiles and their distinct-slope budgets sum to $e^{o(n)}\mathfrak{E}_n(a)$, including all primitive ray budgets. Then the total bad-slope numerator is $e^{o(n)}\mathfrak{E}_n(a)$.*

Proof. This is the union bound of Lemma 2.10; the supremum in (1.4) makes it uniform over received lines. □

26. Finite-row interface and scope

The theorem in this paper is asymptotic. It deliberately separates exponent-level closure from finite adjacent-row certification. The finite problem is not a different mathematical object, but it requires the same cells with constants, exact divisor counts, and exact margins. This section records the interface so that the asymptotic proof can be specialized without changing its logic.

A finite row supplies a number B^*, the desired bad-slope ceiling. A certificate consists of a witness-exhaustive ordered atlas $\mathcal{C}_i$, exact supersets $E_i \supseteq Z_i^\circ$ of its actual first-match projections, and integer budgets $|E_i| \le U_i$ with $\sum_i U_i \le B^*$. The proof of Lemma 2.10 is already finite; only the estimates behind the payments need sharpening. Quotient cells require exact divisor-lattice recursion. Planted cells require exact binomial ratios rather than entropy inequalities. Tangent and rank-pivot cells require explicit eliminants. Fourier/Sidon cells require explicit prefix-equidistribution error terms. Split-pencil cells require exact shift-pair enumeration or a certified upper envelope.

Proposition 26.1 (Finite specialization principle)**.** *Let a finite smooth or circle row admit a witness-exhaustive first-match atlas, exact distinct-slope budgets for its algebraic profiles, an exact Sidon moment estimate yielding Q, and an exact* (RC) *or direct ray bound. If the resulting first-match sum is at most B^*, then $B_C^{\mathrm{MCA}}(a) \le B^*$.*

Proof. This is Lemma 2.10 with exact budgets. The exhaustive atlas, major-character estimate, and ray compiler are structural inputs, not constant-tracking details. □

The largest practical finite cost is the moment-to-max conversion. In asymptotics, a logarithmic moment order $q \to \infty$ eliminates a fixed positive exponential max-fiber gap. In a printed adjacent row with margin Δ bits and prefix depth $w\beta$, a direct moment hierarchy may require order comparable to $w\beta/\Delta$. This is why the finite program benefits from exact prefix equidistribution, exact split-pencil charts, and row-specific eliminants rather than a blind high-moment bound.

Remark 26.2 (Why the asymptotic proof is still useful for finite rows)*.* The framework supplies a diagnostic list, but it does not prove that the list is exhaustive for every row. A finite certificate must verify the atlas, Fourier aggregates, and ray projection explicitly.

27. Conditional proof chain

We record the logical dependencies without turning geometric labels into unproved enumerative bounds.

Theorem 27.1 (Domain geometry supplies profile maps)**.** *Multiplicative cosets and circle twin cosets have the explicit power and Chebyshev folding maps of Section 7. These maps classify natural quotient profiles and their scaled quotient coefficient fields. They do not, by themselves, prove an exhaustive first-match atlas, a major-arc aggregate, or a distinct-slope budget.*

Proof. This is the content of Definition 7.5 and Lemma 7.8; the limitations are witnessed quantitatively by Theorem 8.9. □

Theorem 27.2 (Primitive residue with Sidon payment implies Q)**.** *A fixed-density primitive Boolean leaf satisfying the moment-accessibility, image-scale, and Sidon moment hypotheses of Theorem 17.2 satisfies $\max_z N_w(z) \le e^{o(n)}\overline{N}_\lambda$.*

Proof. This is Theorem 17.2; (PF) plus (MA) is one sufficient way to verify its Sidon premise. □

Theorem 27.3 (Paid profiles, Q, and RC imply the numerator)**.** *For a ledger-admissible sequence, the paid algebraic profile budgets, primitive Q bounds, and* (RC) *imply $B_{C_n}^{\mathrm{MCA}}(a_n) \le e^{o(n)}\mathfrak{E}_n(a_n)$.*

Proof. This is Propositions 18.11 and 18.13. The second moment alone is not used as a substitute for (RC). □

Corollary 27.4 (Conditional target-adjusted frontier)**.** *Under the additional profile-envelope comparison (identity-dominance when the identity formula is invoked), target-reserve, viability, and exact lower-side hypotheses of Theorem 1.6 and Corollary 1.7, the threshold lies in the corridor determined by* (13.2)–(13.3)*. It reduces to the zero of* $H_2(\rho + g) - \beta g$ *only in the stated identity-dominant, subexponential-target regime.*

Proof. Apply Theorem 27.3 and Proposition 25.4. ☐

28. Algebraic-geometric payment criteria

This section records what a row-specific payment proof must establish. Constructibility, codimension in an ambient witness space, and support-pair counts are not by themselves distinct-slope estimates. Each criterion below therefore separates the algebraic description from the missing projection or profile comparison.

Quotient profiles. For a multiplicative coset $D = \theta H$, let $H_c \leq H$ have order c and use the normalized quotient $\pi_c(x) = (x/\theta)^c \in H^c$. If S is a union of H_c-orbits, then $S = \pi_c^{-1}(\bar{S})$ and, with $a = |S|$, $Q_S(X) = \theta^a Q_{\bar{S}}((X/\theta)^c)$. Thus its top coefficients are lacunary. A witness descends only when the received and explaining data satisfy the corresponding fiber-invariance equations; otherwise the noninvariant residues must be retained and counted, not automatically routed to another paid cell.

The quotient budget is obtained from the entropy inequality

$$\binom{n/c}{a/c} \leq \exp\{(n/c)H_2(a/n) + O(\log n)\}.$$

When $c \geq 2$ is fixed, the support count is exponentially smaller than $\binom{n}{a}$, but the quotient also has fewer prefix equations and may be defined over a smaller field. Its correct budget is therefore its own term $\overline{N}_\lambda$ in $\mathfrak{E}_n$. It need not be comparable with the identity term; Theorem 8.9 gives an explicit exponential separation.

Lemma 28.1 (Pure-power quotient lacunarity)**.** *In the normalized multiplicative quotient above, the first* w *locator coefficients satisfy exactly the zero relations in degrees not divisible by* c*; hence at least* $w - \lceil w/c \rceil$ *independent linear relations hold. No analogous count is asserted for an arbitrary degree-*c *map without an explicit triangular coefficient calculation.*

Proof. Expand $Q_S(X) = \theta^a Q_{\bar{S}}((X/\theta)^c)$. Only powers congruent to a modulo c occur, and the surviving top coefficients depend triangularly on those of $Q_{\bar{S}}$. ☐

Tangent and common-line charts. A tangent chart appears when the slope projection of the witness incidence has *ramification*, meaning that its differential has smaller than the expected rank. Write the interpolation equations on a support S as $A_S h + \gamma b_S = r_{1,S}$, where A_S is the Vandermonde evaluation matrix for degree $< k$ polynomials and $b_S = r_{2,S}$. If $|S| = k+1$, the existence of (h, γ) is the vanishing of one determinant. If the determinant vanishes to first order under a support deformation, the derivative determinant vanishes as well. These two equations are the tangent cell. At larger support size, the same condition is expressed by the rank of the augmented matrix $(A_S, b_S, r_{1,S})$.

The common-line locus contributes no bad slope on that support. A neighboring tangent locus, however, is not automatically paid: its defining minors must have the claimed independence and its slope projection must be controlled.

Lemma 28.2 (Ramification payment criterion)**.** *Suppose a tangent chart is cut out by* h *independent bounded-degree minors and the projection of that chart to slopes has fibers large enough that the resulting codimension gives a factor* $|\mathbb{B}|^{-h+o(n)}$ *in the distinct-slope* *image. Then the chart is paid at its natural profile scale. The existence of one nonzero determinant gives only codimension one and does not imply this projection estimate.*

Proof. Independent bounded-degree equations give the finite-field point loss by the usual degree estimate. The assumed projection-fiber bound converts that point loss into a slope-image loss. Both hypotheses are necessary; dependence among minors need not have a quotient explanation. □

Extension and subfield profiles. Let $\mathbb{F}/\mathbb{F}_0$ be a finite extension. A locator or slope profile is $\mathbb{F}_0$-defined when its coefficient vector is fixed by the Frobenius $x \mapsto x^{|\mathbb{F}_0|}$. Such loci are closed in affine coefficient space. Galois orbits organize their support profiles, but field of definition alone does not guarantee a favorable numerator exponent; the resulting profile must be counted at its own field and prefix depth.

Lemma 28.3 (Weil restriction interface)**.** *A bounded-degree extension-line locus becomes a bounded-complexity base-field incidence after Weil restriction. This coordinate change does not itself pay the locus; a base-field point count and a slope-projection bound at the appropriate profile scale are still required.*

Proof. Choose an $\mathbb{F}_0$-basis of $\mathbb{F}$. Coefficient variables and equations expand into bounded tuples and systems over $\mathbb{F}_0$, so complexity grows by a bounded factor. Nothing in this operation controls the number of rational slopes; that is the additional hypothesis stated above. □

Effective-image rank collapse. The prefix map on a verified primitive chart may have an expected image dimension w. An *effective-image rank-collapse locus* is described by a rank drop of its Jacobian or incidence matrix. It is distinct from ray saturation, which quotients witnesses producing the same projective explanation direction. To pay rank collapse one must exhibit independent minors and control the slope projection as in Lemma 6.3; a name such as quotient, planted, or field descent is not a substitute for that verification.

Lemma 28.4 (Vandermonde column rank)**.** *Let $U \subseteq D$ be active coordinates in a verified weighted Vandermonde leaf, and let $v_t = \rho(t)(1, t, \ldots, t^{R-1})$ with $\rho(t) \neq 0$. If the parameters $t \in U$ are distinct, every subset of at most R columns is linearly independent.*

Proof. For $r \le R$, the determinant is $(\prod_j \rho(t_j)) \prod_{i<j}(t_j - t_i) \neq 0$. This proves only the column-independence assertion. Classification and payment of a higher-dimensional image-rank locus remain separate atlas and projection problems. □

Split pencils. A split-pencil chart is most transparent in locator language. Let A and B be monic split polynomials with disjoint root sets in a residual domain D', both of degree e. A depth-w shift pair satisfies $\deg(A - B) \le e - w - 1$, which is equivalent to equality of the first w nonleading coefficients. Thus SP is the second moment of the prefix fibers. The top stratum $e = w + 1$ has $A - B = c \in \mathbb{F}^\times$. The polynomial family $A - c$ must remain split over D', and this is a one-dimensional pencil in the projective locator space.

Lemma 28.5 (Split-pencil accounting)**.** *For an actual one-dimensional locator pencil, the moving-root bound* (7.1) *controls parameters. For a general family, Q controls only the number of support pairs with common prefix. A distinct-slope conclusion requires* (RC)*, and no exhaustive decomposition of higher-dimensional balanced cores is asserted.*

Proof. The one-pencil assertion is Proposition 9.10. The exact common-prefix pair identity is Lemma 18.7. Neither statement supplies the missing higher-dimensional exhaustion or the pair-to-ray fiber bound. □

29. Smooth and circle profile examples

The two domain families below supply folding geometry. Without the remaining checks in Definition 1.4, they are not automatic instances of the conditional compiler.

Multiplicative smooth rows. Let $\mathbb{B}_n$ be a finite field and let $D_n = \theta_n H_n \subseteq \mathbb{B}_n^\times$ be a coset of a cyclic subgroup of order n. The quotient scales are divisors $a_n \mid n$ for which the quotient size $N_n = n/a_n$ tends to infinity and the first-match Fourier alternative of Theorem 14.8 is applied at the prefix depth used; the displayed Li–Wan/Weil error is the corresponding finite certificate when one wants constants. Power maps $x \mapsto x^{a_n}$ give exact uniform fibers. The stabilizer lattice is the divisor lattice of H_n, and quotient descent is exact because a polynomial of degree less than k that is constant on enough fibers must be a pullback from the quotient row.

Circle line rounds. Let $p \equiv 3 \pmod 4$, let $\mathbb{U} \leq \mathbb{F}_{p^2}^\times$ be the norm-one torus, and let $\mathcal{D} = gH \cup g^{-1}H$ be a twin coset with $g^{2a} \notin H^{(a)}$ at the scales used. The x-coordinate domain $D = \chi(\mathcal{D})$ is Chebyshev smooth by Lemma 7.8. Multiplicative quotient profiles therefore have Chebyshev analogues with X^a replaced by T_a. Minor-character sums pull back to the torus with the factor two in Theorem 7.14; major characters still require (MA).

Circle codes. The bivariate circle code of degree w on a circle-domain set is diagonally equivalent to $\mathrm{RS}(E', 2w+1)$ on the torus domain. Diagonal equivalence preserves agreement supports and slopes, so all MCA numerators are unchanged. The only visible difference is parity: $2w + 1$ is odd, while dyadic folding scales are even. This is why Lemma 7.6 is stated without assuming $a \mid k$.

Proposition 29.1 (Standard smooth/circle checklist). *A multiplicative or circle row is ledger-admissible once it verifies all of* (A1)–(A7)*. Uniform power or Chebyshev fibers verify the geometric part of* (A1)*, and a Weil parameter satisfying* (PF) *verifies only the minor-character part of* (A4)*. The atlas payments,* (MA)*, image-scale conditions,* (RC)*, and full profile envelope remain separate checks.*

Proof. This is a restatement of Definition 1.4, with the two automatic geometric observations just proved. The counterexample shows that no stronger conclusion follows from smoothness alone. □

30. Discussion of proof robustness

The proof deliberately avoids a false inverse statement. Ordinary Rényi excess splits into a Sidon-heavy part and a high-energy part. The former is an explicit analytic hypothesis, verified by Fourier flatness only under (PF) and (MA); the latter is killed by the Boolean support slice. This analytic implication is domain-independent.

The algebraic-geometric language also prevents double counting. Quotient, tangent, and saturation loci overlap heavily. A support may be periodic and planted; a tangent chart may also lie in an extension locus; a split pencil may be Chebyshev-invariant. The first-match assignment converts an overlapping constructible cover into a disjoint counting cover without requiring the strata themselves to be disjoint. This is essential for finite certification and harmless asymptotically.

Finally, finite certification replaces each asymptotic input by an exact integer bound. The paper closes the high-energy Boolean branch, but it does not close the Sidon, algebraic-projection, or higher-dimensional ray branches for every smooth/circle row.

31. Smooth/circle ingredients as theorem packages

For reference, and to make the integration with the smooth-domain and circle-domain work explicit, we isolate the exact theorem packages used by the proof. Each package has already appeared in the body of the argument, but this section records them in the order in which a reader would verify an unfamiliar row.

Theorem 31.1 (Stabilizer partition package)**.** *Let D be either a multiplicative coset or a Chebyshev twin-coset x-domain. Every support $S \subseteq D$ has a maximal periodicity scale $c(S)$. If $c(S) = c > 1$, then S is the pullback of a unique support on the quotient domain of size n/c. If $c(S) = 1$, then no nontrivial quotient map in the domain tower preserves S.*

Proof. In the multiplicative case, the acting group is cyclic, so stabilizers are exactly the subgroups H_c. A set fixed by H_c is a union of H_c-orbits and hence a pullback under $x \mapsto x^c$; maximality of $c(S)$ is maximality of the stabilizer. In the circle case, Lemma 7.8 identifies the Chebyshev fibers with the images of the same subgroup tower on the norm-one torus. The x-coordinate map is two-to-one in a way compatible with inversion, so complete torus fibers descend to complete Chebyshev fibers and the same stabilizer argument applies. □

Theorem 31.2 (Invariant quotient-descent package)**.** *Let S be periodic of scale c, and suppose the two received words and the explaining polynomial are pullbacks through the same quotient map. Then the witness descends exactly; agreement is divided by c, and pair-explanation commutes with pullback.*

Proof. Write each datum as a composition with the quotient coordinate. Equality on a union of complete fibers is then exactly equality on the quotient support, and pullback preserves simultaneous explanation. Support periodicity without data invariance does not imply this theorem. □

Theorem 31.3 (Fully primitive residual criterion)**.** *If an exhaustive atlas has separately identified all quotient, planted, tangent, extension, rank-defect, saturation, and pencil loci, its* fully primitive residual *has the negated properties by definition. Smoothness alone does not prove that these loci form such an exhaustive subexponential atlas.*

Proof. The conclusion is the definition of the complement. Constructibility of the algebraic loci follows from Lemma 5.3; exhaustiveness and description entropy are the hypotheses. □

Theorem 31.4 (Prefix flatness package)**.** *On a primitive smooth/circle leaf satisfying* (PF) *and* (MA)*, every full-slice and residual prefix fiber is at most $e^{o(n)}\overline{N}_\lambda$, and the Sidon moment is $e^{o(nq)}$.*

Proof. This is Theorems 7.13 and 7.14 and Proposition 24.5. □

Theorem 31.5 (Primitive Boolean package)**.** *Let $\Omega^\circ \subseteq \{0,1\}^T$ be a fixed-density leaf satisfying the moment-accessibility, Sidon moment, and image-scale hypotheses of Theorem 17.2. Then $\max_s |\Omega^\circ \cap \Phi^{-1}(s)| \le e^{o(n)}\overline{N}$.*

Proof. This is Theorem 17.2. □

The packages isolate exactly what is proved and what remains an input. The stabilizer and invariant quotient-descent statements organize quotient profiles; (PF) plus (MA) pays a verified prefix leaf; and the Boolean theorem closes the high-energy branch. Exhaustive algebraic slope budgets and (RC) remain conditions of the compiler.

32. A row-by-row verification template

A new row is checked against (A1)–(A7). Besides the folding tower and prefix depths, this requires an exhaustive subexponential atlas with direct slope budgets, (PF) and (MA) or a direct Sidon estimate, the primitive image-scale and moment conditions, (RC) or direct ray bounds, and the complete profile envelope.

Proposition 32.1 (Verification template)**.** *Suppose a row verifies* (A1)–(A7) *of Definition 1.4, with every asymptotic bound uniform over received lines. Then Theorem 1.5 applies.*

Proof. This is the conditional compiler theorem. □

The finite version replaces every input by an exact inequality. In particular, a second-moment identity is not a finite ray certificate without an exact pair-to-ray fiber bound.

33. Cell-budget audit

This section lists the estimates a closure proof would have to certify. Every profile is budgeted at its own $1+\overline{N}_\lambda$ and retained in $\mathfrak{E}_n$; the identity scale is not used as a universal budget.

Definition 33.1 (Profile complexity)**.** *A profile type and a realized profile are the objects of Definition 2.6. Their description data include the type λ and every realized parameter ξ needed to specify the incidence, such as a quotient map, planted core, tangent multiplicity vector, field of definition, rank-pivot shape, saturation image, or split-pencil partition. A profile class has subexponential complexity if the total number of nonempty realized pairs (λ, ξ) in a row is $e^{o(n)}$, uniformly in the received line, and the algebraic description complexity is subexponential as in Definition 5.1.*

The following observation is used repeatedly. If a cell with a fixed profile has cost e^{-cn} relative to the uncut prefix scale, then any subexponential family of such cells is paid. If the saving is only $e^{-o(n)}$, it is still paid provided the exact comparison with $1+\overline{N}$ is nonpositive at the frontier. This is the source of the integer-staircase checks in finite-row certificates.

Lemma 33.2 (Subexponential profile summation)**.** *Let $\{E_\alpha\}_{\alpha\in I_n}$ be slope sets with $|I_n| = e^{o(n)}$ and $|E_\alpha| \le e^{o(n)}(1+\overline{N})$ uniformly. Then $|\bigcup_\alpha E_\alpha| \le e^{o(n)}(1+\overline{N})$.*

Proof. The union is bounded by $\sum_\alpha |E_\alpha| \le e^{o(n)}e^{o(n)}(1+\overline{N}) = e^{o(n)}(1+\overline{N})$. □

Quotient cells. A quotient cell is attached to a finite map $\pi : D \to D'$ with uniform fiber size $d > 1$. In the multiplicative case $\pi(x) = x^d$ on a coset of a cyclic group. In the circle case π is the Chebyshev map $x = z + z^{-1} \mapsto z^d + z^{-d}$ after passing to the norm-one torus. A witness lies in the quotient cell if, after deleting a remainder support of size r, its locator is constant on the fibers of π. Equivalently, its support is the full inverse image of a support $S' \subseteq D'$, up to r marked exceptions.

For a fixed quotient profile (d, r), the support count is at most $n^r\binom{n/d}{(a-r)/d}$, and a candidate quotient prefix scale is

$$\overline{N}^{\mathrm{amb}}_{d,r} = \binom{n/d}{(a-r)/d} |\mathbb{B}_{d,r}|^{-R_{d,r}}.$$

The row must prove the correct residual depth $R_{d,r}$, field of definition $\mathbb{B}_{d,r}$, lift multiplicity, and distinct-slope projection. If $\Phi_{d,r}$ is the proved profile map, the term retained in $\mathfrak{E}_n$ is

$$\overline{N}^{\mathrm{img}}_{d,r} = \frac{\binom{n/d}{(a-r)/d}}{|\Phi_{d,r}(\Omega^0_{d,r})|},$$

unless (FI) has certified its equivalence to the ambient candidate. In particular, the inequality

$$\overline{N}^{\mathrm{img}}_{d,r} \le e^{o(n)}\overline{N}_1$$

is not automatic and is false in Theorem 8.9.

Planted cells. A planted cell must range over an algebraically controlled family $\mathcal{P}_b$, not all $\binom{n}{b}$ blocks. For a fixed block its residual image scale is computed from $\binom{n-b}{a-b}$ divided by the realized image size of the proved residual prefix map. The ambient prefix depth may be used only under (FI). Payment is exactly the criterion of Proposition 37.2; neither sublinearity nor constructibility alone supplies it.

Tangent and common-line cells. A tangent cell is a determinantal locus in the witness incidence. Rank loss h need not have codimension h; independent minors and a slope-projection estimate must be exhibited. The common-line locus itself has no bad slopes on that support. These are the hypotheses of Lemma 28.2, not automatic consequences of smoothness.

Extension and field-descent cells. An extension profile may be defined over a proper intermediate field $\mathbb{B}'$, in which case its image-normalized profile scale can be larger than the identity scale. Frobenius coincidence equations pay it only if their independence and their effect on the slope image are proved. Otherwise the profile is retained at its natural field in $\mathfrak{E}_n$.

Differential-locator and effective-image rank-collapse cells. Differential-locator and effective-image rank-collapse loci are rank-defect loci. A drop of h does not imply codimension h, and a smaller effective image can make fibers larger. Payment therefore requires the independent-minor and slope-projection hypotheses of Lemma 6.3. Ray saturation is separate and is paid by the image–fiber count for the witness-to-ray map. Ordinary Vandermonde column independence does not classify every persistent higher-dimensional defect.

Split-pencil cells. A one-dimensional split pencil is controlled by the moving-root incidence bound after common roots are removed. In higher dimension the exact second moment counts support pairs but gives no inequality of the displayed frontier size and no ray count. Such profiles require a direct balanced-core exhaustion or (RC).

Proposition 33.3 (Conditional cell-budget theorem)**.** *Suppose an exhaustive atlas has $e^{o(n)}$ profiles and each profile satisfies a direct distinct-slope estimate at most $e^{o(n)}\overline{N}_\lambda$. Then the total algebraic contribution is at most $e^{o(n)}\mathfrak{E}_n(a)$.*

Proof. Sum the assumed slope estimates over the profile atlas and use (1.4). The preceding paragraphs explain sufficient criteria and the places where a direct proof is still needed. ☐

34. Smooth-domain ledger closure in detail

Let $D = \theta H$, where $H \subseteq \mathbb{B}^\times$ is cyclic of order n. A smooth-domain quotient is the map $x \mapsto x^d$, $d \mid n$. The fiber size is exactly d, and the quotient domain is $\theta^d H^d$. A support is quotient-saturated if it is a union of fibers. A locator $Q_S(X)$ is quotient-saturated precisely when it lies in $\mathbb{B}[X^d]$ after the scalar change $X \mapsto \theta X$, up to the planted remainder support.

Lemma 34.1 (Exact invariant multiplicative quotient descent). *Let $D = \theta H$, $d \mid |H|$, and $\pi(x) = x^d$. Suppose that after deleting a marked set R, the support is a union of complete π-fibers and the restrictions of both received words and the explaining polynomial are pullbacks through π. Then the witness descends exactly to $\pi(D)$, with agreement divided by d. Conversely, pullback gives lifts; a subexponential lift count requires a separate $o(n)$-bit profile bound.*

Proof. All three data are constant on the same complete fibers, so their equality on the original support is exactly the quotient equality. Pullback proves the converse. Without the invariance hypothesis, nontrivial kernel characters remain and must be counted rather than discarded. □

Lemma 34.2 (Smooth support stabilizers). *If a support in $D = \theta H$ is invariant under a nontrivial subgroup of H, it is a union of quotient fibers. This classifies the support; it does not imply that the received or explaining data descend, nor does it classify arbitrary positive-density partial fibers.*

Proof. The stabilizer is a subgroup, and an invariant set is a union of its orbits, which are precisely the quotient fibers. □

Proposition 34.3 (Conditional smooth-domain algebraic budget). *If a multiplicative smooth row verifies the exhaustive atlas, invariant quotient-descent/lift counts, planted and determinantal projection criteria, and one-pencil or direct balanced-core ray bounds, then its algebraic cells contribute at most $e^{o(n)}\mathfrak{E}_n$.*

Proof. Apply the verified profile estimates and Proposition 33.3. Smoothness supplies the quotient maps, not the omitted projection estimates. □

For the analytic component, let ψ be a nontrivial additive character of $\mathbb{B}$. A primitive prefix Fourier coefficient has the form

$$\binom{N}{a}^{-1} e_a\Big(\psi(P(t)) : t \in T\Big),$$

where P is a nonzero polynomial of degree less than the prefix equation count R. Li–Wan and Weil give the minor-character estimate when (PF) holds. The full conclusion additionally requires (MA); a major phase label does not provide it.

Corollary 34.4 (Conditional smooth-domain compiler). *A multiplicative smooth row satisfying the algebraic hypotheses above,* (PF) *plus* (MA) *or a direct Sidon payment, the primitive-Q conditions, and* (RC) *has a closed asymptotic ledger.*

Proof. Combine the stated hypotheses with Theorem 17.2 and Propositions 18.11 and 33.3. □

35. Circle-domain ledger closure in detail

Let $\mathcal{T}$ be the norm-one torus over $\mathbb{B}$, and write $x = z + z^{-1}$. A twin coset is a union $\xi H \cup \xi^{-1}H^{-1}$ stable under $z \mapsto z^{-1}$, projected to the x-line. The Chebyshev polynomial T_d satisfies $T_d(z + z^{-1}) = z^d + z^{-d}$. Hence the quotient map on the x-line is the projection of the multiplicative quotient $z \mapsto z^d$ on the torus. The only additional issue is the two-to-one ramification at $z = \pm 1$, which contributes only bounded exceptional fibers in the asymptotic rows and is assigned to a planted cell.

Lemma 35.1 (Invariant Chebyshev quotient descent). *Let D be a circle twin-coset domain and let d divide the torus coset order. If the support and all received and explaining data are pullbacks through T_d, the witness descends exactly to the Chebyshev quotient. A lift census requires its own description-entropy bound.*

Proof. Lift to the torus, apply invariant multiplicative quotient descent, and take invariants under $z \mapsto z^{-1}$. The identity $T_d(z+z^{-1}) = z^d + z^{-d}$ identifies the quotient coordinate. □

Lemma 35.2 (Circle diagonal uniformization)**.** *For the standard circle code and evaluation set of Lemma 7.9, the diagonal multiplier gives an exact equivalence to the torus Reed–Solomon row and preserves MCA numerators.*

Proof. This is Lemma 7.9. □

Proposition 35.3 (Conditional circle-domain compiler)**.** *A circle twin-coset row with a proved exhaustive subexponential atlas and complete profile envelope, satisfying the circle analogues of the algebraic profile criteria,* (PF) *plus* (MA) *or a direct Sidon payment, the primitive image-scale conditions, and* (RC)*, has a closed asymptotic ledger. The same holds for a standard circle-code row under the exact diagonal equivalence.*

Proof. Pull the verified algebraic and analytic estimates to the torus, apply the conditional smooth compiler equivariantly under inversion, and use Lemma 35.2. No unverified major character or balanced-core chart is removed by this reduction. □

36. Why ledger closure remains an input

Smooth and circle geometry supplies uniform folding maps, but it does not prove the enumerative statements that drive the upper bound: exhaustion of all partial-occupancy profiles, their algebraic slope projections, the primitive major-character aggregate, and the higher-dimensional transverse-secant bound. The following theorem records the exact conditional status.

Theorem 36.1 (Smooth/circle compiler, expanded form)**.** *Let* (C_n, a_n) *be a ledger-admissible smooth/circle sequence. Then*

$$B^{\mathrm{MCA}}_{C_n}(a_n) \le e^{o(n)} \mathfrak{E}_n(a_n).$$

Proof. This is Theorem 1.5. The smooth and circle propositions above explain how the required verified estimates transport through power, Chebyshev, and diagonal maps. □

Corollary 36.2 (Identity-dominant specialization)**.** *If, in addition,*

$$\mathfrak{E}_n(a_n) = e^{o(n)} \binom{n}{a_n} |\mathbb{B}_n|^{-(a_n-k_n-1)},$$

then the numerator has the corresponding identity-scale upper bound. The quotient counterexample proves that this additional comparison cannot be deleted.

Proof. Substitute the profile comparison into the preceding theorem. □

37. Algebraic ledger criteria: planted, determinantal, and circle edge cases

This section records the precise forms of three payments that are often misstated if one counts supports rather than moduli. They are included to ensure that the first-match ledger is not hiding exponential candidate families.

Definition 37.1 (Algebraically planted block)**.** *A planted block of size* b *is algebraically planted if it is the zero set on* D *of a polynomial in a constructible family* $\mathcal{P}_b$ *of support locators, common factors, ramification polynomials, polynomials cutting out quotient fibers, or received-line resultants. Here a* received-line resultant *is the polynomial obtained by eliminating the explaining-polynomial variables from the received-line interpolation equations; its zeros are the coordinates forced by that elimination. We require* $|\mathcal{P}_b(\mathbb{B})| \le e^{o(n)}$

for every profile size occurring in the profiled asymptotic row datum. Arbitrary choices of b-subsets of D are not planted cells.

Proposition 37.2 (Planted payment without arbitrary-block overcount)**.** *Let P range over an algebraically planted family $\mathcal{P}_b$, and suppose the residual chart after fixing P has image-normalized profile scale $\overline{N}(P)$. If the total description entropy is $o(n)$, $\sum_P \overline{N}(P) \le e^{o(n)}\mathfrak{E}_n$, and the projection from each chart to distinct slopes has cost $e^{o(n)}\overline{N}(P)$, then the planted family is paid. These hypotheses must also be checked for positive-density blocks; arbitrary blocks are excluded.*

Proof. Sum the assumed distinct-slope budgets over P. The statement is deliberately phrased in terms of description entropy and natural profile scales, since $b = o(n)$ alone does not imply $b \log|\mathbb{B}| = o(n)$. □

Proposition 37.3 (Determinantal payment with a nonzero pivot minor)**.** *Let X be a bounded-complexity affine chart over $\mathbb{B}$, and let $M : X \to \mathrm{Mat}_{u\times v}$ be polynomial of degree $\exp(o(n))$. A nonzero $r \times r$ minor makes $\{\operatorname{rank} M < r\}$ a proper constructible locus. If h independent pivot equations give codimension h and a separate slope-projection estimate converts the point loss into image loss, then the cell gains $|\mathbb{B}|^{-h+o(n)}$. Dependence among the minors yields no automatic quotient or tangent classification.*

Proof. The minors define the rank locus. Under the stated independence, bounded-degree point counting gives $|\mathbb{B}|^{\dim X-h}\exp(o(n))$ chart points. The separately assumed projection estimate gives the slope-image bound. No conclusion is drawn when either hypothesis fails. □

Lemma 37.4 (Circle ramification and twin-coset collapse)**.** *For a circle twin coset $gH \cup g^{-1}H$, the exceptional cases $u = \pm 1$, $gH = g^{-1}H$, and $g^{2a} \in H^{(a)}$ are respectively a bounded ramification profile, a dihedral collapse, and a Chebyshev quotient profile. They are constructible and must be budgeted at their natural profile scales.*

Proof. The points $u = \pm 1$ are fixed by the involution and contribute at most two ramification points. The equality $gH = g^{-1}H$ is equivalent to $g^2 \in H$, so the twin coset is dihedrally collapsed. If $g^{2a} \in H^{(a)}$, the two image cosets under $u \mapsto u^a$ coincide. These observations classify the profiles; their quotient terms are retained in $\mathfrak{E}_n$. □

Proposition 37.5 (Conditional algebraic ledger from verified criteria)**.** *For a smooth or circle row with an exhaustive subexponential atlas, suppose every quotient, planted, tangent, extension, differential, saturation, and one-pencil profile satisfies the corresponding direct slope criterion above, while every higher-dimensional balanced core has a direct ray bound. Then the algebraic contribution is $e^{o(n)}\mathfrak{E}_n$.*

Proof. Sum the verified profile budgets. The one-pencil estimate is Proposition 9.10; no subdivision theorem for higher-dimensional pencils is used or claimed. □

38. Finite source theorems and certificates

The finite constructions in [Cho26a, Cho26b] supply the uniform-fiber viewpoint, locator-prefix coordinates, and finite ledger interfaces. The unconditional ingredients used here are the syndrome description and deep theorem, the locator-prefix list, the collision-aware pole line, the map-smooth list construction, the bounded-remainder normal form, the polynomial-field counterexample, the support-pair identity, and the one-pencil moving-root bound. General partial-occupancy exhaustion, quotient-descent, determinantal, and balanced-core payments occur only under the explicit conditional criteria proved above.

For circle rows, Chebyshev smoothness supplies the uniform-fiber maps, and the exact diagonal equivalence Lemma 7.9 transfers the quotient obstruction and all support-wise numerators to the standard circle code.

The primitive analytic residue splits Q into a Sidon-heavy branch and a high-energy Boolean branch. The high-energy branch is impossible by BSG and Boolean-cube difference growth. The Sidon moment remains an explicit input, verified by (PF) plus (MA) when those estimates hold on the residual. The polynomial-field construction proves that no such ambient identity-normalized payment can hold before quotient, subfield, planted, and remainder profiles are routed. Thus the primitive theorem resolves the high-energy branch, while the ambient identity-normalized claim is false without profile routing.

A finite certificate must additionally tabulate the major-character aggregate, direct algebraic slope projections, and the ray compiler; these are not recovered by inserting constants into an asymptotic support count.

39. A compact dependency graph

For clarity we record the logical dependencies in one place. The domain-independent analytic step is

$$\text{Sidon moment payment} + \Gamma_q^{\mathrm{ord}} \text{ large} \Rightarrow E(F_s) \geq |F_s|^3 e^{-o(n)},$$
$$E(F_s) \geq |F_s|^3 e^{-o(n)} \Rightarrow \text{BSG subset } A \Rightarrow |A - A| \leq e^{o(n)}|A|.$$

Since $A \subseteq \{0,1\}^T$, Boolean-cube difference growth gives $|A - A| \geq |A|^{3/2}$, contradicting $|A| = e^{\Omega(n)}$. Hence primitive Q holds. The compiler is

$$\text{paid algebraic profiles} + \text{primitive Q} + (\text{RC}) \Rightarrow \text{MCA numerator},$$

followed, under the required profile-envelope comparison and target/lower hypotheses, by the threshold comparison. The implications are proved; their displayed premises are not claimed for every smooth/circle row. Indeed the unrestricted ambient identity-normalized Sidon premise and identity-profile dominance are disproved in Theorem 8.9, while Proposition 18.8 shows that Q/SP statistics alone cannot provide the missing ray premise.

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